# Statistics 2 lecture notes 2026–2027
# Part II

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# Part I

# Testing

## 1 Introduction to hypothesis tests

Hypothesis testing (or just testing) is one of the most common statistical procedures employed throughout the experimental sciences. This is because tests allow us to do something new: instead of just estimating or providing plausible ranges of values for a parameter $\theta$ we can (additionally) **make decisions**. Testing increasingly plays a part in everyday life, the decisions made shape the world around us, sometimes even in an automated fashion.

Testing is a procedure designed to decide which of two competing statements (*hypotheses*), $H_0$ (*null*) and $H_1$ (*alternative*), is more consistent with an observation $\mathbf{x} = (x_1, x_2, ..., x_n)$ of $\mathbf{X} = (X_1, X_2, ..., X_n) \sim f(\cdot; \theta)$.

In Statistics 1, there was a first introduction to testing with focus in tests for the mean of realizations of normally distributed random variables (with known or unknown variance). The null hypothesis was assumed to be of the form $H_0 : \mu = \mu_0$.

In this unit we will look at the more general case:

1. $\mathbf{X} \sim f(\cdot; \theta)$ where $\theta \in \Theta$ is unknown;
2. we have an observation $\mathbf{x}$ of $\mathbf{X}$;
3. we have formulated a null hypothesis concerning possible values of $\theta$, e.g. $H_0 : \theta \in \Theta_0$
4. we have an alternative hypothesis, $H_1 : \theta \in \Theta_1 = \Theta \setminus \Theta_0$.

Testing involves considering the distribution of $\mathbf{X} \sim f(\cdot; \theta)$ for $\theta \in \Theta_0$ and $\theta \in \Theta_1$. The goal is to discriminate between these two cases.

Before we look at the procedure of performing a hypothesis test it is useful to define the different types of tests depending on the form of the two hypotheses.

**Definition 1.1.** Types of hypotheses and tests:

1. A hypothesis of the form $\theta = \theta_i$ (or $\Theta_i = \{\theta_i\}$) for $i = 0$ or $i = 1$ where $\theta_i$ is a specific value (e.g. $\theta_i = 0.37$) is called a *simple* hypothesis.
2. A hypothesis such that $\Theta_i$ is not a singleton is a *composite* hypothesis (e.g. for $\theta$ a scalar, $\theta \leq c$ and $\theta > c$ specify composite hypotheses).
3. When $\theta$ is a scalar, for a given value $\theta_0 \in \Theta$, a *one-sided test* is of the form
$$H_0 : \theta \leq \theta_0 \text{ versus } H_1 : \theta > \theta_0$$
or
$$H_0 : \theta \geq \theta_0 \text{ versus } H_1 : \theta < \theta_0.$$
4. Again in the scalar case, a *two-sided test* is of the form
$$H_0 : \theta = \theta_0 \text{ versus } H_1 : \theta \neq \theta_0$$
(there are more general forms for two-sided tests, but we will not consider them here).

Lets see now an example of how we form a hypothesis test in practice.

**Example 1.1.** A new training program for heptathletes has been proposed and it is of interest to determine whether this new program should improve general performance. We split the space of possibilities in two:

- Null hypothesis ($H_0$): the new training program does not improve performance
- Alternative hypothesis ($H_1$): the new training program improves performance.

The question now is how we convert this into a hypothesis test. In order to test $H_0$ statistically, we need a model and some data. We decide to assign the new program to $n$ athletes, and measure the changes in their heptathlon scores (before rounding) $x_1, \ldots, x_n$. We assume that these are observed realizations of i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables $X_1, \ldots, X_n$ with $\sigma^2 > 0$ known. The translation of the hypotheses is :

If you are more motivated by different application domains, there are many analogues:

| **Testing change to** | **Subjects** | **Response** |
|---|---|---|
| training program | athletes | change in score |
| webpage layout | webpages | change in ad revenue |
| high-frequency trading strategy | trading agents | money gained |
| medical treatment | patients | change in measure of health |

Table 1.1: Different application domains

In the setting of example 1.1, one can imagine that we are more willing to believe that $H_1$ is true if $\bar{x} > 0$. This does not by itself provide a satisfactory mechanism for testing. Should we "reject $H_0$" whenever $\bar{x} > 0$ or whenever $\bar{x} > c$ for some $c$? The second option is more reasonable, but how can we choose $c$ in a principled way? We should keep the following in mind:

1. A statistical approach involves designing the test by considering the behaviour of $\mathbf{X} \sim \mathcal{N}(\mu, \sigma^2)$ for different values of $\mu$.
2. Hypotheses are associated with the decision that we are trying to make; they reflect the interests of the statistician/investigator.
3. As we have seen, in order to formally define a test and its corresponding hypotheses we partition the parameter space $\Theta = \Theta_0 \cup \Theta_1$. In order to take a decision we are going to partition the set of observations $\mathcal{X}^n$ into two sets, one where we retain $H_0$ and another (called the critical region $R$ ) where we reject $H_0$. That is, in order to make a decision in light of the data we need to determine which values of the data correspond to retaining/rejecting $H_0$.

We shall see that the primary focus is on rejecting $H_0$ when it is "too implausible" in light of the data. For this reason, it is conventional never to say "accept $H_1$" but rather "reject $H_0$". It is also conventional never to say "accept $H_0$" but rather "retain $H_0$" since typically retaining $H_0$ indicates some support for $H_0$ as a model, but one must acknowledge that in most situations the statistical model is only an approximation of reality.

For the example above, we are going to partition the set of possible observations (the differences in heptathlon score) in such a way that we can distinguish between observations that lend support to the null hypothesis, and those that do not. We are going to focus here on a particular and natural way of partitioning $\mathcal{X}^n$, which relies on defining a test statistic, i.e. a statistic $T(\mathbf{x})$ that facilitates testing.

**Definition 1.2** (*Test statistic, critical region and critical value*)**.** Let $c \in \mathbb{R}$ (the "critical value") and $T : \mathcal{X}^n \to \mathbb{R}$ (the test statistic) be such that large values of $T(\mathbf{x})$ indicate rejection of the null hypothesis $H_0$. The critical region associated to $c$ and $T$ is the set

$$R := \{\mathbf{x} : T(\mathbf{x}) \geq c\} \subseteq \mathcal{X}^n = R \cup R^c.$$

*Remark* 1.1. In practice, one may find a natural statistic $T'$ such that the critical region is $R = \{\mathbf{x} : T'(\mathbf{x}) \leq c'\}$. In this case, one can always define the test statistic to be $T(\mathbf{x}) = -T'(\mathbf{x})$ and the critical value to be $c = -c'$ to meet the definition above. However, this is often needlessly complicated and it is fine to call $T'$ a test statistic even though small (rather than large) values of $T'(\mathbf{x})$ indicate rejection of the $H_0$, as long as it is very clear what the critical region is.

**Example 1.2** (Testing the mean of a normal sample, continued)**.** Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and that we want to test

$$H_0 : \mu \leq 0, \qquad H_1 : \mu > 0.$$



The two main points addressed in this unit are going to be

1. how does one choose the test statistic $T(\cdot)$?
2. how does one choose the critical value $c$?

In the motivating example, $T(\mathbf{x}) = \bar{x}$ is intuitively sensible, but one may wonder if there are better choices? I.e. is this choice optimal in some sense? How might we even define optimal? Moreover, how can we choose $c$? Choosing $c = 0$ is not sensible if $\mu$ is "very close" to 0 since we would reject $H_0$ with probability approximately $1/2$: we might as well flip a coin.

We have seen earlier that in the context of estimation (of $\theta$), quantifying uncertainty was crucial in order to formulate sensible (and honest) answers to scientific problems. Similarly, when testing hypotheses, errors are unavoidable due to uncertainty, and quantifying test errors is going to be central.

# 2 Hypothesis testing: significance and power

When performing a hypothesis test, there are two possible errors one can commit. These are displayed in the table below:

| **Truth\Decision** | **Retain $H_0$** | **Reject $H_0$** |
|---|---|---|
| **$H_0$ true** | ✓ | |
| **$H_1$ true** | | ✓ |

It is obvious that our goal should be to minimize the probabilities of commiting either of the two errors, however in practice this is not straightforward. As we will see, the two errors are not treated in a identical ('fair') way. Our main focus is to reject $H_0$ only if there is strong evidence from the data against it. Thus, we primarly want to secure that the probability of Type I error is small. In the next two sections we will see analytically how we achieve this and what is the effect on Type II error.

## 2.1 Significance

Type I errror concerns the null hypothesis, and allows us to talk about *significance*. This can be phrased as

> If $H_0$ is actually true, rejecting $H_0$ is a Type I error.

Similar to what we have seen with confidence intervals, it is typically impossible to avoid making Type I errors entirely. Instead, we set a level at which we tolerate such errors. The principle is

> *If $H_0$ is actually true, we should design a test that rejects $H_0$ with probability less than or equal to $\alpha \in (0, 1)$.*

The value of $\alpha$ is referred to as the **significance level** of the test. We will define this formally a little later, but this is sufficient to begin seeing what the ramifications are. In fact, the significance level is sufficient to define a test in many situations.

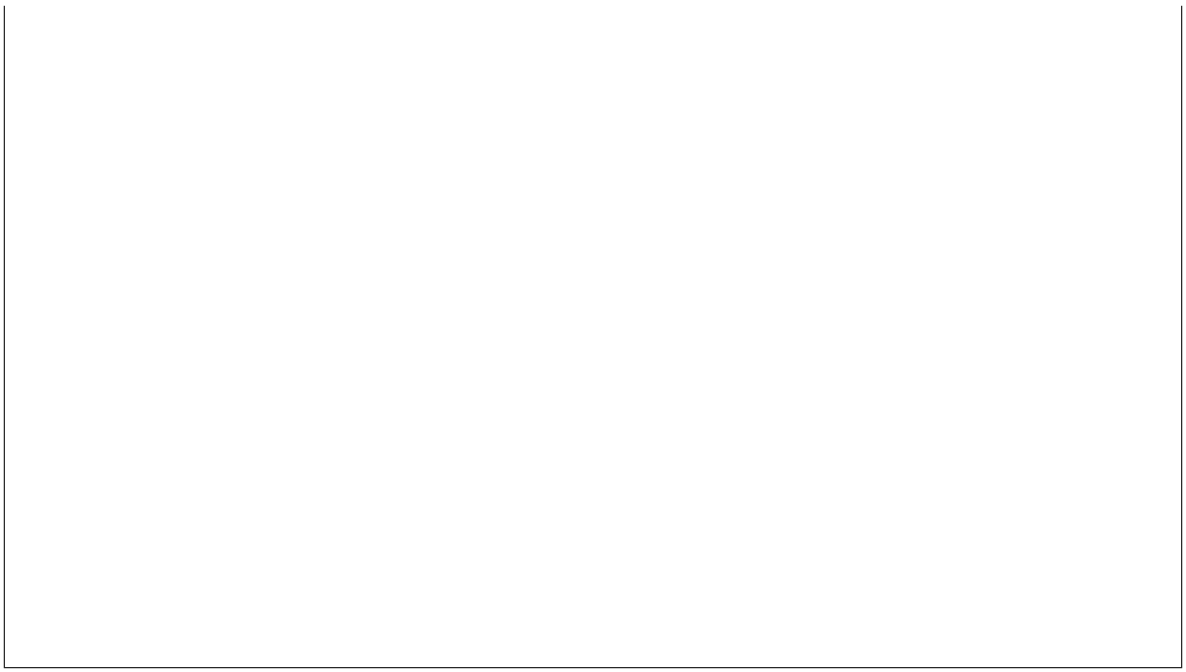
**Example 2.1** (Testing the mean of a normal sample, continued)**.** Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and that we want to test

$$H_0 : \mu \leq 0, \qquad H_1 : \mu > 0.$$

We consider the test statistic $T(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$ with critical region $R$ (for some $c \in \mathbb{R}$)

$$R = \{\mathbf{x} \in \mathcal{X}^n : \bar{x} \geq c\}.$$

How can we find $c$ so that the significance level is $\alpha$?



In the example, we have found the *smallest* $c$ so that the test has significance level $\alpha$. In fact, we have chosen $c$ so that the test does not have significance level $\alpha - \epsilon$ for any $\epsilon > 0$. This is because we want the test to have a reasonable chance of rejecting $H_0$ when it does not describe the system under study, although you may have noticed that we have made absolutely no reference to the alternative $H_1$ when developing the test so far.

You should also notice that the threshold $c$ becomes smaller as $n$ grows (and increases with $\sigma$).

*Remark* 2.1. The significance level $\alpha$ is directly related to the phrase "statistical significance". This is particularly true of cases where one is trying to establish empirically that the models in the null hypothesis do not explain the data well. In this example, we might want to say that the analysis showed "a statistically significant positive effect of the training program".

Statistical significance relates only to the Type I error rate. You should have some healthy skepticism about what we have covered so far. For example,

1. one can perform the hypothesis test with $n = 1$ or $n = 10^6$; clearly one should be persuaded more by a result in the latter case.
2. one could even treat the case $n = 0$, and flip coins to produce tests with significance level $1 - \alpha$; clearly this accomplishes absolutely nothing.

We need to consider the alternative hypothesis to make more interesting statements about tests.

## 2.2 Power

The second type of test error concerns the alternative hypothesis also, and allows us to talk about *power*. This can be phrased as

> If $H_1$ is actually true, retaining $H_0$ is a Type II error.

The definition of a Type II error leads naturally to the following definition for the power function associated with a test.

**Definition 2.1** (Power function)**.** Let $\mathbf{X} \sim f(\cdot;\theta^*)$. The power function, $\pi(\cdot;T,c) : \Theta \to [0,1]$ of a test $T(\cdot)$ with critical value $c \in \mathbb{R}$ is the probability of rejecting $H_0$ when the true value of the parameter is $\theta \in \Theta$, i.e.

$$\pi(\theta;T,c) := \mathbb{P}(\mathbf{X} \in R;\theta) = \mathbb{P}(T(\mathbf{X}) \geq c;\theta).$$

*Remark* 2.2. For a given $\theta \in \Theta_1$, the Type II error probability is $1-\pi(\theta;T,c)$. In particular, if $\Theta_1 = \theta_1$ the power function, $\pi(\theta_1;T,c)$ is called the *power* of the test.

Ideally a test should satisfy $\pi(\theta;T,c) = 0$ for $\theta \in \Theta_0$ (that is when $H_0$ is correct we do not want reject $H_0$) and $\pi(\theta;T,c) = 1$ for $\theta \in \Theta_1$ (because when $H_1$ is correct we want to reject $H_0$). This will not be possible in most statistical scenarios, as there is an unavoidable tradeoff between the probabilities of Type I and Type II errors.

1. The power function $\mathbb{P}(T(\mathbf{X}) \geq c;\theta)$ is non-increasing in $c$ regardless of whether $\theta \in \Theta_0$ or $\theta \in \Theta_1$.
2. To make the probability of Type I error go to 0 we should make $c$ very large so we very rarely reject $H_0$.
3. If $c$ is really very large, we will also very rarely reject $H_0$ even if $\theta \in \Theta_1$, so the power is low and the Type II error probability is high.

In practice, we manage this tradeoff by specifying at least implicitly a significance level $\alpha$ and then trying to find a powerful test, at least for relevant values of $\theta \in \Theta_1$.

*Remark* 2.3. When it is clear from the context that the test is $(T,c)$, we may write $\pi(\theta)$ instead of $\pi(\theta;T,c)$.

**Example 2.2** (Testing the mean of a normal sample, continued)**.** Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu,\sigma^2)$ and that we want to test

$$H_0 : \mu \leq 0, \qquad H_1 : \mu > 0.$$

We consider the test statistic $T(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^n x_i = \bar{x}$ with critical region $R$ (for some $c \in \mathbb{R}$)

$$R = \{\mathbf{x} \in \mathcal{X}^n : \bar{x} \geq c\}.$$

What is the power function for the test?

We can draw this as a function of $\mu$ for a value of $c = 0$, or as a function of $c$ for $\mu = 0$.

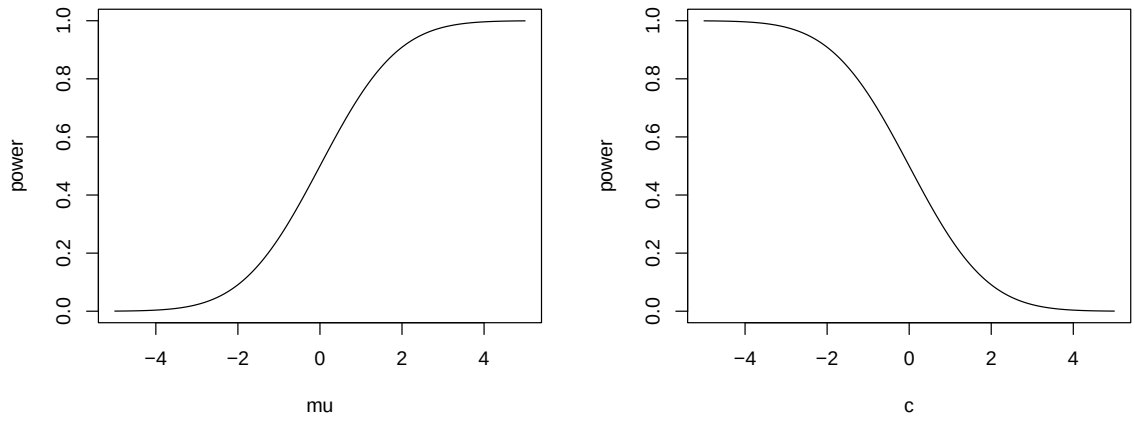



Figure 2.1: Power function for $c = 0$, $\sigma = 3.0$, $n = 4$ (left) and power function for $\mu = 0$, $\sigma = 3.0$, $n = 4$ (right)

## 2.3 Summary and definitions

Now consider the simple case where we test $H_0 : \mu = \mu_0 = 0$ against $H_1 : \mu = \mu_1 = 2$. One can see that $\pi(\mu_0; T, c) = \mathbb{P}(T(\mathbf{X}) \geq c; \mu_0)$ (i.e. the probability of rejecting $H_0$ when it is actually true) is decreasing as $c$ increases but that the type II error $\mathbb{P}(T(\mathbf{X}) < c; \mu_1) = 1 - \pi(\mu_1; T, c)$ (i.e the probability of deciding that $H_0$ is true when $H_1$ is actually true) is increasing at the same time. This illustrates the necessity of a trade-off between the type I and II errors which cannot be set to zero. One could naturally, in the present case, choose $c$ to minimize the maximum of the two errors, or their sum or a linear combination in order to give different weights to the two types of errors. We illustrate these curves for two different values of $\sqrt{n}/\sigma$ and draw the line corresponding to a significance level of $\alpha = 0.05$. The value of $c$ at the point at which the black curve intersects this significance level line also indicates the Type II error probability.

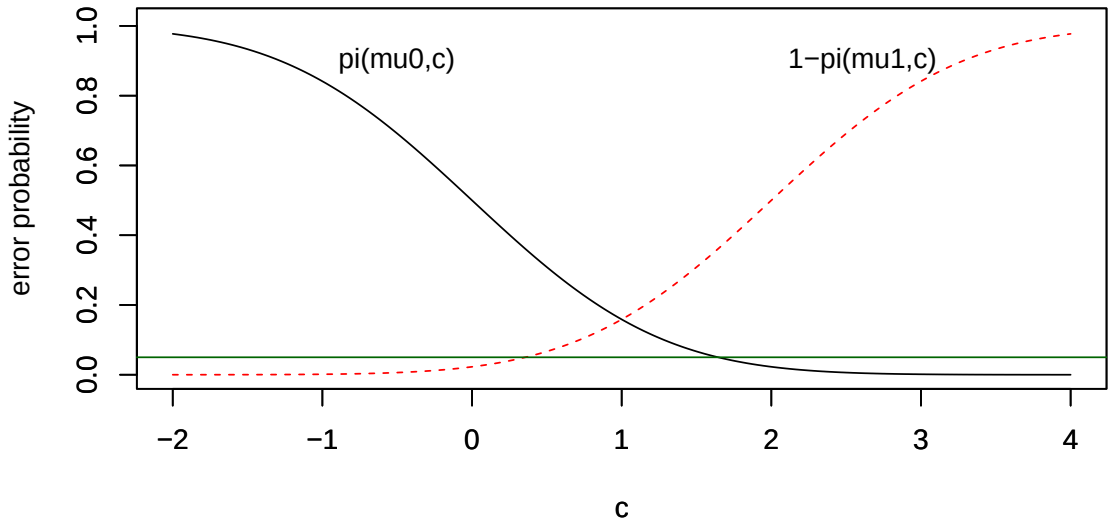



Figure 2.2: Type I (black, solid) and Type II (red, dashed) errors against $c$ for $\mu_0 = 0$, $\mu_1 = 2$, $\sqrt{n}/\sigma = 1$

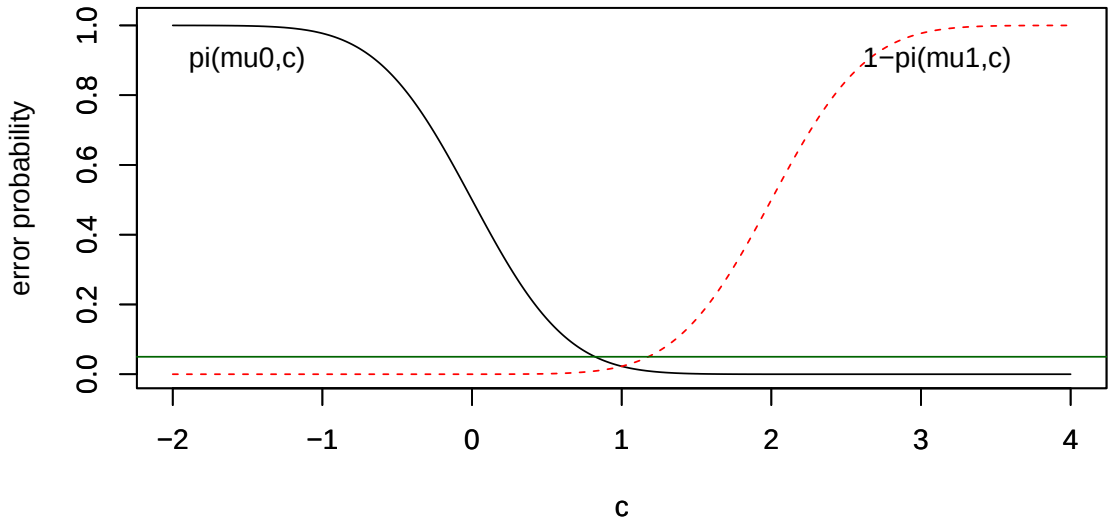



Figure 2.3: Type I (black, solid) and Type II (red, dashed) errors against $c$ for $\mu_0 = 0$, $\mu_1 = 2$, $\sqrt{n}/\sigma = 2$

**Definition 2.2** (Size of a test)**.** The size of a test is the largest possible probability of making an error of type I

$$\alpha = \sup_{\theta \in \Theta_0} \pi\left(\theta; T, c\right),$$

i.e. it is the maximum power under the null hypothesis.

*Remark* 2.4. If you look back at how we chose $c$ for our example, we chose $c$ so that the test has size $\alpha$. This is a general recipe for identifying $c$.

**Definition 2.3** (Significance level of a test)**.** A test has level $\alpha$ if its size is less than or equal to $\alpha$. The corresponding test is then called a level $\alpha$ test.

*Remark* 2.5. In many situations the size and level of a test are equal.

When a hypothesis is simple it is easy to talk about a probability of error under that hypothesis.

**Definition 2.4.** When $\Theta_0 = \{\theta_0\}$ then $\alpha = \pi(\theta_0, T, c)$ is the significance level, or Type I error rate (the probability of rejecting $H_0$ when it is actually true).

**Definition 2.5.** When $\Theta_1 = \{\theta_1\}$ then $1 - \pi(\theta_1, T, c)$ is the Type II error rate (the probability of failing to reject $H_0$ when it is false, and hence $\theta = \theta_1$ is correct).

**Example 2.3** (Testing the mean of a normal sample, continued)**.** Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and that we want to test

$$H_0 : \mu \leq 0, \qquad H_1 : \mu > 0.$$

We consider the test statistic $T(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$ with critical region $R$. A test of size $\alpha$ is obtained by choosing

$$c = \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1-\alpha) = \frac{\sigma}{\sqrt{n}} z_\alpha.$$

So we consider the fact that $c = z_\alpha \sigma/\sqrt{n}$, and we obtain

$$\mathbb{P}\left(\bar{X} \geq \frac{\sigma}{\sqrt{n}} z_\alpha; \mu\right) = 1 - \Phi\left(z_\alpha - \frac{\sqrt{n}\mu}{\sigma}\right).$$

This gives the power for all $\mu \in \mathbb{R}$ and we are interested in particular in it for $\mu > 0$.

**Exercise.** Plot this power function at home as a function of $\mu \geq 0$. You can see that the test is more powerful for larger values of $\mu$. You can also determine what value of $n$ is required to discriminate with low error probability between $\mu = \mu_0 = 0$ and different values of $\mu = \mu_1 \in (0, \infty)$. This is one way that people determine how much data is required to answer a particular question (or to determine that it is not feasible to answer the question with reasonable resources).

*Remark* 2.6. In many cases, rejecting $H_0$ corresponds to evidence of an "effect". For a significance level $\alpha$ test, we could say that the effect is "statistically significant" (at that level). Naturally, this does not mean the effect is large. In our motivating example, for large enough $n$, one might reject $H_0$ with data corresponding to $\bar{x} = 0.00001$ and $\alpha = 10^{-9}$. Clearly the evidence for the effect is very significant, but the effect itself is apparently very small.

## 2.4 Testing as a legal trial

Many people find that thinking of testing in a legal context helps clarify some critical concepts.

Consider a case where forensic evidence has been obtained. A suspect has been apprehended and they are tested for a match to the forensic evidence. Let $H_0$ be the null hypothesis that they are innocent and $H_1$ the alternative that they are guilty. The nature of the test is that 1 in 10000 innocent people will match, and the actual perpetrator will be matched with probability 1.

| | Retain $H_0$ | Reject $H_0$ |
|---|---|---|
| $H_0$ true | Correct: _____ | $\mathbb{P}$(Type I error): _____ |
| $H_1$ true | $\mathbb{P}$(Type II error): _____ | Correct:_____ |

That is $\alpha = \mathbb{P}_{H_0}(\text{reject } H_0) =$ _____ and $\mathbb{P}_{H_1}(\text{reject } H_0) =$ _____. If the match is negative, then they are definitely innocent.

If the match is positive, then this does not mean that the probability they are innocent is $\alpha = \frac{1}{10000}$ [this claim is known as the "prosecutor's fallacy"]. Obviously, we cannot make any probabilistic statements about their guilt in this framework: the hypotheses (more generally, the statistical parameter) are not random variables. What we can say is that if convictions are made on the basis of this type of test, then the frequency with which innocent people are convicted is 0.0001, and guilty people will always be convicted. This is the nature of the testing procedure.

We can make the rates different if we want to. Imagine that instead of just looking at matches, a procedure is to convict if there is a match and the result of 10 coin tosses are all heads. Then we get

| | Retain $H_0$ | Reject $H_0$ |
|---|---|---|
| $H_0$ true | Correct, $1 - \alpha \approx 1 - 10^{-7}$ | $\mathbb{P}$(Type I error) is $\alpha = \frac{1}{10000 \times 2^{10}} \approx 10^{-7}$. |
| $H_1$ true | $\mathbb{P}$(Type II error) is $1 - \frac{1}{2^{10}} \approx 0.999$ | Correct, $\frac{1}{2^{10}} \approx 0.001$ |

We would then convict innocent people at a much lower rate (1 in 10 million) but we also often let guilty people go free (1023 times out of 1024). Clearly it is not satisfactory to try legal cases on the basis of coin flips: the result of the coin flips should not alter the assessment of innocence or guilt. In practice, if we required a much lower rate of innocents being convicted, we would need to devise a better matching process / obtain higher quality forensic evidence.

This rate-adjustment mechanism should also indicate that considering only the significance level of a test is not satisfactory; we want tests to be powerful so that we are not led to believe all kinds of things on the basis of random chance alone: if 10 million experiments test null hypotheses at a significance level of $\alpha = 0.05$, then 500000 of them will result in rejecting the null even if all of the null hypotheses are true. This becomes more important when decisions are not made in isolation. For more important decisions with greater consequences, smaller values of $\alpha$ are often taken ($\approx 3 \times 10^{-7}$ for testing the Higgs boson at CERN), and independent reproduction of the results of an experiment is often a key step in the experimental sciences.

# 3 Designing tests: the Neyman-Pearson approach

When testing at significance level $\alpha$, we recall that what we plan to do is

1. Define a model $f(\cdot;\theta)$ for $\theta \in \Theta$.
2. Define a null hypothesis $H_0 : \theta \in \Theta_0$ and an alternative hypothesis $H_1 : \theta \in \Theta_1 = \Theta \setminus \Theta_0$.
3. Define a test statistic $T(\mathbf{x})$.
4. Choose a critical value $c$ such that $\sup_{\theta\in\Theta_0} \mathbb{P}(T(\mathbf{X}) \geq c;\theta) \leq \alpha$, i.e. we reject $H_0$ with probability at most $\alpha$ when $H_0$ is true.

We notice that once the model, hypotheses and test statistic are chosen, one can choose $c$ to be the smallest value satisfying the inequality above. In this case the critical value is determined by the significance level $\alpha$.

So far so good, but how can we choose a good test statistic? There are very many possibilities in general.

The Neyman–Pearson lemma tells us what the optimal test statistic is when considering two simple hypotheses. This gives rise to a general Neyman–Pearson approach to testing.

**Lemma 3.1** (The Neyman--Pearson Lemma)**.** *Suppose we test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ and use the likelihood ratio test statistic*

$$T_{\mathrm{NP}}(\mathbf{x}) := \frac{f_n(\mathbf{x};\theta_1)}{f_n(\mathbf{x};\theta_0)} = \frac{L(\theta_1;\mathbf{x})}{L(\theta_0;\mathbf{x})}.$$

*Let the critical value $c_{\mathrm{NP}} \geq 0$ be such that the test has size $\alpha$,*

$$\mathbb{P}\left(T_{\mathrm{NP}}(\mathbf{X}) \geq c_{\mathrm{NP}};\theta_0\right) = \alpha.$$

*Then this test is the **most powerful** level $\alpha$ test. That is, among all tests with level $\alpha$, this test maximizes the power function $(T, c) \mapsto \pi(\theta_1; T, c)$ (or, equivalently, minimizes the Type II error probability $\beta(\theta_1; T, c) = 1 - \pi(\theta_1; T, c)$).*

*Remark* 3.1. The likelihood ratio test statistic, a.k.a. the Neyman–Pearson test statistic, is intuitively meaningful since when the observations are much more likely under $H_1$ than $H_0$, we reject the latter. For example, let $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ and $\mathbf{x} = (-1.1, -2.3, -3.5, -4.1, -2.87)$, $\Theta_0 = \{10\}$ and $\Theta_1 = \{-3\}$. We expect $f_n(\mathbf{x};\theta_1)/f_n(\mathbf{x};\theta_0) \gg 1$.

*Remark* 3.2. Note the order in which we have chosen the constants to define the test. The critical value $c_{\mathrm{NP}}$ is chosen first and determines the significance level $\alpha$. We may however be interested in choosing $\alpha$ first and then determining what critical value $c_{\mathrm{NP}}$ leads to this significance level.

*Remark* 3.3. If $T(\mathbf{X})$ is not a continuous random variable, then one can have the issue that no such $c_{\mathrm{NP}}$ exists: in this situation one can perform an appropriate randomized test, and this will also be the most powerful size $\alpha$ test. We will not cover the details here: the idea is to find an appropriate $c$ and reject $H_0$ when $T_{\mathrm{NP}}(\mathbf{x}) > c$, retain $H_0$ when $T_{\mathrm{NP}}(\mathbf{x}) < c$ but when $T_{\mathrm{NP}}(\mathbf{x}) = c$ one should reject/retain with a probability that makes the test have size $\alpha$. This is really a mathematical, rather than a practical, issue: one would need special justification to randomize a test.

*Proof.* (of Lemma 3.1) Consider for an arbitrary level $\alpha$ test $(T, c)$, the linear combination of Type I and Type II errors

$$\phi\big(T, c\big) := c_{\mathrm{NP}}\alpha\big(T, c\big) + \beta(T, c),$$

where $\alpha(T, c) = \mathbb{P}(T(\mathbf{X}) \geq c;\theta_0)$ and $\beta(T, c) = \mathbb{P}(T(\mathbf{X}) < c;\theta_1) = 1 - \mathbb{P}(T(\mathbf{X}) \geq c;\theta_1)$.



Now consider the difference

$$\begin{aligned}&\phi(T,c)-\phi(T_{\mathrm{NP}},c_{\mathrm{NP}})\\&=\int\left[\mathbb{I}\{T(\mathbf{x})\geq c\}-\mathbb{I}\{T_{\mathrm{NP}}(\mathbf{x})\geq c_{\mathrm{NP}}\}\right]\{c_{\mathrm{NP}}-T_{\mathrm{NP}}(\mathbf{x})\}\,f(\mathbf{x};\theta_0)\mathrm{d}\mathbf{x}.\end{aligned}$$

We observe that

$$\mathbb{I}\{T_{\mathrm{NP}}(\mathbf{x})\geq c_{\mathrm{NP}}\}=1\iff c_{\mathrm{NP}}-T_{\mathrm{NP}}(\mathbf{x})\leq 0$$

and

$$\mathbb{I}\{T_{\mathrm{NP}}(\mathbf{x})\geq c_{\mathrm{NP}}\}=0\iff c_{\mathrm{NP}}-T_{\mathrm{NP}}(\mathbf{x})>0,$$

so for any $\mathbf{x}\in\mathcal{X}^n$,

$$\left[\mathbb{I}\{T(\mathbf{x})\geq c\}-\mathbb{I}\{T_{\mathrm{NP}}(\mathbf{x})\geq c_{\mathrm{NP}}\}\right]\{c_{\mathrm{NP}}-T_{\mathrm{NP}}(\mathbf{x})\}\geq 0,$$

and hence as the integral of a non-negative function,

$$\phi(T,c)-\phi(T_{\mathrm{NP}},c_{\mathrm{NP}})\geq 0.$$

We have established

$$\begin{aligned}0&\leq\phi(T,c)-\phi(T_{\mathrm{NP}},c_{\mathrm{NP}})\\&=c_{\mathrm{NP}}\alpha(T,c)+\beta(T,c)-c_{\mathrm{NP}}\alpha(T_{\mathrm{NP}},c_{\mathrm{NP}})-\beta(T_{\mathrm{NP}},c_{\mathrm{NP}})\\&=c_{\mathrm{NP}}\left[\alpha(T,c)-\alpha(T_{\mathrm{NP}},c_{\mathrm{NP}})\right]+\beta(T,c)-\beta(T_{\mathrm{NP}},c_{\mathrm{NP}}).\end{aligned}$$

Since $(T,c)$ specifies a level $\alpha$ test, we know $\alpha(T,c)\leq\alpha$ while $(T_{\mathrm{NP}},c_{\mathrm{NP}})$ specifies a size $\alpha$ test so $\alpha(T_{\mathrm{NP}},c_{\mathrm{NP}})=\alpha$. It follows that

$$\alpha(T,c)-\alpha(T_{\mathrm{NP}},c_{\mathrm{NP}})\leq 0$$

so we have

$$\beta(T, c) - \beta(T_{\mathrm{NP}}, c_{\mathrm{NP}}) \geq 0,$$

which means $(T_{\mathrm{NP}}, c_{\mathrm{NP}})$'s Type II error rate is no higher, or equivalently $(T_{\mathrm{NP}}, c_{\mathrm{NP}})$ is at least as powerful as $(T, c)$. Since $(T, c)$ is an arbitrary level $\alpha$ test, we conclude that $(T_{\mathrm{NP}}, c_{\mathrm{NP}})$ is the most powerful test with level $\alpha$.

□

From this lemma one can deduce the Neyman–Pearson procedure for testing two simple hypotheses:

1. Choose the likelihood ratio as test statistic
$$T(\mathbf{x}) = \frac{f_n(\mathbf{x}; \theta_1)}{f_n(\mathbf{x}; \theta_0)} = \frac{L(\theta_1; \mathbf{x})}{L(\theta_0; \mathbf{x})},$$
2. Choose a critical value $c$ in order to target a particular significance level (say $\alpha = 5\% = 0.05$) such that
$$\alpha = \pi(\theta_0) = \mathbb{P}(T(\mathbf{X}) \geq c; \theta_0) = 0.05$$
3. Compute the power
$$\pi(\theta_1, T, c) = \mathbb{P}(T(\mathbf{X}) \geq c; \theta_1)$$
4. Compute $T(\mathbf{x})$ and report whether $T(\mathbf{x}) \geq c$, as well as the power $\pi(\theta_1, T, c)$ or the type II error rate $1 - \pi(\theta_1, T, c)$.

There are objections/extensions to the Neyman–Pearson approach. The main issues are the following:

1. Reporting rejection/retention of $H_0$ does not show strength of evidence against the null hypothesis [e.g. for $c > 0$, $T(\mathbf{x}) = (1 + \epsilon)c$ or $T(\mathbf{x}) = 10c$ both lead to a rejection],
2. We might prefer to set $\alpha$ and $\beta = 1 - \pi(\theta_1, T, c)$ together, or optimize them. We might also consider choosing $n$ in order to ensure both error rates are low enough.

**Example 3.1** (Normal example.)**.** The model is $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ and we want to test $H_0 : \mu = 0$ against $H_1 : \mu = 1$. It can be shown (see the next section and Homework sheet 5) that for any $n \in \mathbb{N}$, $c_{\mathrm{NP}} = \exp\left(2.326n^{1/2} - 0.5n\right)$ leads to a type I error of 0.01, and type II error of the form $\Phi\left(2.326 - n^{1/2}\right)$. The latter naturally vanishes rapidly, but so does $c_{\mathrm{NP}}$. In particular, for $n = 100$ this means that we reject the null hypothesis whenever

$$\frac{f_n(\mathbf{x}; \theta_1)}{f_n(\mathbf{x}; \theta_0)} \geq c_{\mathrm{NP}} \approx 2.5 \times 10^{-12},$$

that is including situations when the likelihood for $\theta_0$ (the null) is almost a trillion times the likelihood for $\theta_1$ (the alternative)! One can check that the type two error probability in this case is about $8 \times 10^{-15}$, which is perhaps too conservative. This may seem paradoxical, but it is generally not feasible to think intuitively about arbitrary ratios of likelihoods; as we can seen here. Despite the tiny figure for $c_{\mathrm{NP}}$, which is decreasing very rapidly as $n$ increases, the Type I error rate is indeed 0.01.

In addition, the different error rates are strikingly different. In general, one should probably consider managing the tradeoff between them better: in this scenario, one could reduce the Type I error rate without sacrificing much power at all. We will see later, in the Bayesian approach to testing, that the relative importance of $\alpha(T, c)$ and $\beta(T, c)$ is explicitly used to specify a critical value.

# 4 Testing: p-values, equivalent test statistics and computing the power function

The Neyman–Pearson approach is mathematically very elegant, and also emphasizes clearly the decision-making nature of testing. There are two issues we will focus on in this section.

1. In many situations, just rejecting $H_0$ or retaining $H_0$ is not satisfactory; we may want more information.
2. It is not obvious how to calibrate a likelihood ratio test; e.g. to find the critical value or to compute the power function.

## 4.1 p-values

In practice, many studies prefer not to select in advance just one significance level $\alpha$, or they may wish to report something more informative than a binary decision. In such cases, they can report the $p$-value associated with the observed test statistic.

**Definition 4.1** (p-value)**.** Assume $\mathbf{X} \sim f_n(\cdot;\theta^*)$ for some $\theta^* \in \Theta$. The $p-$value for a test with test statistic $T(\mathbf{x})$ is the probability of seeing a test statistic $T(\mathbf{X})$ at least as extreme as $T(\mathbf{x})$, that is

$$p(\mathbf{x}) := \sup_{\theta_0 \in \Theta_0} \mathbb{P}(T(\mathbf{X}) \geq T(\mathbf{x}); \theta_0).$$

Equivalently, $p(\mathbf{x})$ *is the smallest significance level at which we would reject* $H_0$.

In many situations, $p(\mathbf{x})$ can be calculated by considering a single $\theta_0$ [e.g., the value in $\Theta_0$ associated with the highest probability of producing extreme test statistics.

Informally the $p-$value is a measure of the evidence against $H_0$. The smaller it is the less likely it is that $\mathbf{x}$ is a realization of $\mathbf{X} \sim f(\cdot;\theta_0)$, resulting in strong evidence against $H_0$. One should however note that a large $p-$value is not evidence in favour of $H_0$, nor is it necessarily evidence in favour of $H_1$ as $H_1$ is not involved at all when computing the $p$-value.

*Remark* 4.1. The standard caution: $p(\mathbf{x})$ is NOT the probability that $H_0$ is true. Apart from the reasons above, the most important reason is that we are not able to say anything interesting about the probability of $H_0$ being true since $\theta$ is not a random variable (at least in the frequentist setting that we study now).

*Remark* 4.2. In the case of a simple null hypothesis $\Theta_0 = \{\theta_0\}$, and assuming $T(\mathbf{X})$ is a continuous random variable when $\mathbf{X} \sim f(\cdot;\theta_0)$, the distribution of $p(\mathbf{X})$ is in fact uniform under the null hypothesis.

**Example 4.1** (Normal example)**.** The model is $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ and we want to test $H_0 : \mu = \mu_0 < 0$ against $H_1 : \mu = \mu_1 > 0$. The $p-$value for $T(\mathbf{x}) = \bar{x} = n^{-1}\sum_{i=1}^n x_i$ is

$$p(\mathbf{x}) := \sup_{\mu \in \Theta_0} \mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n X_i \geq T(\mathbf{x}) = \bar{x}; \mu\right) = \mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n X_i \geq T(\mathbf{x}) = \bar{x}; \mu_0\right).$$

A very large positive value of the empirical mean leads to a small $p-$value, and is an indication of how unlikely it is to have observed $\mathbf{x}$ if it was a realization of $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_0, 1)$. A large $p-$value is not an argument in favour of $H_0$ : in fact it could suggest that $T(\mathbf{x})$ is an unlikely realization (lower values are unlikely) under $H_0$.

We have already calculated this kind of expression: under the null, $\bar{X} \sim \mathcal{N}(\mu_0, 1/n)$ so

$$\begin{aligned}\mathbb{P}(\bar{X} \geq c; \mu) &= \mathbb{P}\left[\sqrt{n}(\bar{X} - \mu_0) \geq \sqrt{n}(c - \mu_0); \mu_0\right] \\ &= \mathbb{P}\left[Z \geq \sqrt{n}(c - \mu_0)\right] \\ &= 1 - \Phi\left[\sqrt{n}(c - \mu_0)\right].\end{aligned}$$

It follows that

$$p(\mathbf{x}) = 1 - \Phi\left[\sqrt{n}(\bar{x} - \mu_0)\right].$$

## 4.2 Equivalent test statistics and computing the power function

**Definition 4.2** (Equivalent statistics)**.** A statistic $T'(\mathbf{x})$ is *equivalent* to $T(\mathbf{x})$ if for every critical value $c \in \mathbb{R}$ of $T(\cdot)$ we can find $c' \in \mathbb{R}$ such that for any $\mathbf{x} \in \mathcal{X}^n$

$$T(\mathbf{x}) \geq c \iff T'(\mathbf{x}) \geq c'$$

or in other words for any $c \in \mathbb{R}$ there exist $c' \in \mathbb{R}$ such that the corresponding critical regions of $T(\cdot)$ and $T'(\cdot)$ respectively coincide, i.e.

$$\{\mathbf{x} \in \mathcal{X}^n : T(\mathbf{x}) \geq c\} = \{\mathbf{x} \in \mathcal{X}^n : T'(\mathbf{x}) \geq c'\} \quad .$$

This can be very practically helpful when trying compute the power function.

*Remark* 4.3. To verify that $T'(\mathbf{x})$ is an equivalent statistic in practice it is sufficient to factorize $T(\mathbf{x})$ as

$$T(\mathbf{x}) = Mf(T'(\mathbf{x})),$$

where $M$ is independent of $\mathbf{x}$ and $f$ is increasing and bijective. This is because

$$\begin{aligned}T(\mathbf{x}) \geq c &\iff Mf(T'(\mathbf{x})) \geq c \\ &\iff f(T'(\mathbf{x})) \geq c/M \\ &\iff T'(\mathbf{x}) \geq f^{-1}(c/M).\end{aligned}$$

In the next two examples we will see how (simpler) equivalent test statistics can help us compute the critical value or the power of a test.

**Example 4.2** (Geometric example)**.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \text{Geometric}(p)$, so that $f(x; p) = (1-p)^{x-1}p\mathbb{I}\{x \in \mathbb{N} \setminus \{0\}\}$ and assume that we want to test $H_0 : p = p_0$ versus $H_1 : p = p_1$, with $p_0 > p_1$. Compute the Neyman–Pearson test statistic and give a simple equivalent test statistic which could be used to compute the power function.

**Example 4.3** (Normal example.)**.** The model is $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ and we want to test $H_0 : \mu = 0$ against $H_1 : \mu = 1$. Find the power function for the NP test (using an equivalent test statistic) and provide a justification for the expressions of the critical value and type I and II error probabilities in Example 3.1



# 5 Uniformly most powerful tests

The Neyman-Pearson test allows one to address in a principled manner the scenario where the two hypotheses to test are simple, i.e. we are interested in $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$. But we are often interested in more general situations involving composite hypotheses: we focus here on the the situation where we are interested in testing

$$H_0 : \theta \leq \theta_0 \text{ (or } \theta = \theta_0) \qquad \text{vs.} \qquad H_1 : \theta > \theta_0.$$

We would like to find a test $(T, c)$ such that

$$\sup_{\theta \in \Theta_0} \pi(\theta; T, c) \leq \alpha$$

for a significance level $\alpha$ of our choice, in order to control the probability of type I errors. Subject to this constraint, we would like to choose $(T, c)$ such that $\pi(\theta; T, c)$ is maximized for any $\theta \in \Theta_1$ as this corresponds to minimizing the probability of type II errors. That is, we would like to choose $(T, c)$ such that for *any* $\theta \in \Theta_1$ and any test statistic $(T', c')$

$$\pi(\theta; T, c) \geq \pi(\theta; T', c').$$

If it exists, such a test can be called a ***Uniformly Most Powerful (UMP)*** test, since it has the largest power $\pi(\theta; T, c)$ among all possible tests, *uniformly* in $\theta \in \Theta_1$.

*Remark* 5.1. The Type II error probability/rate depends on a specific value of $\theta \in \Theta_1$. Typically, the Type II error probability is close to $1-\alpha$ for values of $\theta \in \Theta_1$ "very close to being in" $\Theta_0$. For example, in the scenario above, $\pi(\theta; T, c) \approx \alpha$ for $\theta = \theta_0 + \epsilon$ with $\epsilon$ very small. In general, when we talk about the Type II error rate for composite alternative hypotheses, this should be stated in terms of a given $\theta \in \Theta_1$.

The main idea here is to ascertain whether there is a test statistic $T$ such that

1. For any $c \in \mathbb{R}$, the power function $\pi(\theta, T, c)$ is monotonically increasing in $\theta$.
2. For any $\theta_2 > \theta_1$, the test statistic $T$ is equivalent to a likelihood ratio test statistic for $H_0 : \theta = \theta_1$ vs. $H_1 : \theta = \theta_2$.

Our main result is the following.

**Theorem 5.1.** *Let $\Theta_1 = \{\theta : \theta > \theta_0\}$ for some $\theta_0 \in \Theta$. Assume that for the simple hypotheses*

$$H_0' : \theta = \theta_1 \quad \textit{vs.} \quad H_1' : \theta = \theta_2,$$

*the Neyman–Pearson test statistic*

$$T_{\mathrm{NP}}(\mathbf{x}) = \frac{f_n(\mathbf{x}; \theta_2)}{f_n(\mathbf{x}; \theta_1)},$$

*is equivalent to the same test statistic $T(\mathbf{x})$ for any $\theta_1 < \theta_2$, and $T(\mathbf{x})$ does not depend on $\theta_1$ or $\theta_2$. Then $T(\mathbf{x})$ is the uniformly most powerful test statistic for*

$$H_0 : \theta \leq \theta_0 \ (\textit{or}\ \theta = \theta_0) \quad \textit{vs.} \quad H_1 : \theta > \theta_0,$$

*and the associated $p-$value is*

$$p(\mathbf{x}) = \mathbb{P}(T(\mathbf{X}) \geq T(\mathbf{x}); \theta_0) \quad .$$

The proof is in Section 5.1.

In practice this suggests the following simple strategy in situations where we wish to test such hypotheses.

1. Find the equivalent statistic $T(\mathbf{x})$, if it exists, starting from the likelihood ratio.
2. Compute the $p-$value using the distribution of $T(\mathbf{X})$ when the true value of the parameter is assumed to be $\theta = \theta_0$.

**Example 5.1** (Poisson)**.** For $\mathbf{X} \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$ for some $\lambda > 0$ and we want to test $H_0 : \lambda \leq \lambda_0$ vs. $H_1 : \lambda > \lambda_0$. Compute the p-value associated with this test.

Another familiar example.

**Example 5.2.** In the case where $\mathbf{X} \overset{\text{iid}}{\sim} \text{Geometric}(p)$ we have already shown that for two simple hypotheses $\bar{X}$ is equivalent to the likelihood ratio test statistic when $p_0 > p_1$. It follows that $T(\mathbf{x}) = S_n(\mathbf{x})$ is also equivalent. We have also noticed that $S_n(\mathbf{X}) \sim \text{NegBinomial}(n, p)$. Compute the $p-$value associated to

$$H_0 : p \leq p_0 \quad vs \quad H_1 : p > p_0,$$

*Remark* 5.2. UMP tests need not exist in general. It might be the case that $(T_1, c_1)$ is best for, say, $\theta_{1,1} \in \Theta_1$ i.e. for any test $(T', c')$,

$$\pi(\theta_{1,1}; T_1, c_1) \geq \pi(\theta_{1,1}; T', c')$$

but that there exists another test $(T_2, c_2)$ better than $(T_1, c_1)$ for some other $\theta_{1,2} \in \Theta_1$, that is

$$\pi(\theta_{1,2}; T_2, c_2) > \pi(\theta_{1,2}; T_1, c_1).$$

At home, you may consider the test $H_0 : \theta = 0$ vs. $H_1 : \theta \neq 0$ for $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, and you can find that no UMP test exists [the most powerful test statistic when considering $\theta_1 > 0$ is not the same as the most powerful test statistic for $\theta_1 < 0$].

## 5.1 Why a test can be UMP (not examinable)

We need a notion closely related to the notion of equivalence presented earlier.

**Definition 5.1.** Let $\mathbf{X} \sim f_n(\cdot; \theta)$. One says that $f_n(\cdot; \theta)$ has a monotone increasing likelihood ratio in the statistic $T(\mathbf{x})$ if for any $\theta_1 < \theta_2$ the likelihood ratio satisfies

$$\frac{f_n(\mathbf{x}; \theta_2)}{f_n(\mathbf{x}; \theta_1)} = \phi_{\theta_1, \theta_2}(T(\mathbf{x})),$$

where $\phi_{\theta_1, \theta_2}$ is a monotone increasing function of $T(\mathbf{x})$ for every $\theta_1 < \theta_2$. That is, the likelihood ratio depends on value of the data $\mathbf{x}$ only through the value of the statistic $T(\mathbf{x})$, and the ratio is a monotone increasing function of $T(\mathbf{x})$.

*Remark* 5.3. The difference to the factorization for equivalent statistics is the additional requirement that $T(\mathbf{x})$ be the same statistic for any $\theta_1 < \theta_2$.

**Example 5.3.** For a Geometric model, testing $H_0 : p \leq p_0$ vs. $H_1 : p > p_0$, we obtain

$$T_{\text{NP}}(\mathbf{x}) = \left(\frac{p_1}{p_0}\right)^n \left(\frac{1 - p_1}{1 - p_0}\right)^{n(\bar{x}-1)},$$

which is increasing in the statistic $T(\mathbf{x}) = -\bar{x}$ for any $p_1 > p_0$.

**Example 5.4.** For a Normal$(\mu, 1)$ model we can compare $H_0 : \mu \leq 0$ with $H_1 : \mu > 0$.

For $H_0' : \mu = \mu_1$ vs. $H_1' : \mu = \mu_2$ with $\mu_1 < \mu_2$ we obtain

$$T_{\text{NP}}(\mathbf{x}) = \exp\left\{n\bar{x}(\mu_2 - \mu_1) + \frac{n}{2}(\mu_1^2 - \mu_2^2)\right\},$$

which is increasing in the statistic $T(\mathbf{x}) = \bar{x}$.

*Remark* 5.4. We require the following technical lemma.

**Lemma 5.1.** *Assume that $f_n(\cdot; \theta)$ has an increasing monotone likelihood ratio in the statistic $T(\mathbf{x})$, and let $\psi : \mathbb{R} \to \mathbb{R}$ be a non-decreasing function, then*

$$g(\theta) := \mathbb{E}\big[\psi(T(\mathbf{X})); \theta\big],$$

*is a non-decreasing function of $\theta$.*

*Proof.* Let $\theta_1 < \theta_2$ and introduce the two following sets $A, B \subset \mathcal{X}^n$,

$$A := \{\mathbf{x} \in \mathcal{X}^n : f_n(\mathbf{x}; \theta_1) > f_n(\mathbf{x}; \theta_2)\}, \qquad B := \mathcal{X}^n \setminus A = \{\mathbf{x} \in \mathcal{X}^n : f_n(\mathbf{x}; \theta_1) \leq f_n(\mathbf{x}; \theta_2)\}.$$

Define $a := \sup_{\mathbf{x} \in A} \psi(T(\mathbf{x}))$ and $b := \inf_{\mathbf{x} \in B} \psi(T(\mathbf{x}))$. It follows that $b \geq a$ since $\sup_{\mathbf{x} \in A} T(\mathbf{x}) \leq \inf_{\mathbf{x} \in B} T(\mathbf{x})$ and $\psi$ is non-decreasing. Now consider

$$
\begin{aligned}
g(\theta_2) - g(\theta_1) &= \int \psi\left(T(\mathbf{x})\right) \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} \\
&\geq a \int_A \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} + b \int_B \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} \\
&= (b-a) \int_B \left\{ f_n(\mathbf{x};\theta_1) - f_n(\mathbf{x};\theta_2) \right\} \mathrm{d}\mathbf{x} \geq 0,
\end{aligned}
$$

where we have used that $f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \geq 0$ for $\mathbf{x} \in B$, and

$$
\begin{aligned}
0 = 1 - 1 &= \int_{\mathcal{X}^n} \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} \\
&= \int_A \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} + \int_B \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x},
\end{aligned}
$$

so $\int_B \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x} = -\int_A \left\{ f_n(\mathbf{x};\theta_2) - f_n(\mathbf{x};\theta_1) \right\} \mathrm{d}\mathbf{x}$. Hence $g$ is non-decreasing in $\theta$. □

**Corollary 5.1.** *If $f_n(\cdot;\theta)$ has an increasing monotone likelihood ratio in the statistic $T(\mathbf{x})$, the power function*

$$
\theta \mapsto \pi(\theta; T, c) = \mathbb{P}\left[ T(\mathbf{X}) \geq c; \theta \right],
$$

*is non-decreasing in $\theta$.*

*Proof.* Let $\psi(z) = \mathbb{I}(z \geq c)$ which is indeed non-decreasing and observe that

$$
\mathbb{P}\left[ T(\mathbf{X}) \geq c; \theta \right] = \mathbb{E}\left[ \mathbb{I}\{T(\mathbf{X}) \geq c\}; \theta \right] = \mathbb{E}\left[ \psi(T(\mathbf{X})); \theta \right].
$$

□

**Theorem 5.2** (Restatement of Theorem 5.1)**.** *Assume that $f_n(\cdot;\theta)$ has an increasing monotone likelihood ratio in the statistic $T(\mathbf{x})$. Let $\alpha \in [0,1]$ and $c \in \mathbb{R}$ be such that*

$$
\mathbb{P}\big(T(\mathbf{X}) \geq c; \theta_0\big) = \alpha.
$$

*Then $(T, c)$ is a uniformly most powerful level $\alpha$ test for the one-sided alternative hypothesis*

$$
H_0 : \theta \leq \theta_0 \text{ and } H_1 : \theta > \theta_0.
$$

*Proof.* Consider the hypothesis test, for some $\theta_1 > \theta_0$,

$$
H_0' : \theta = \theta_0 \text{ vs. } H_1' : \theta = \theta_1.
$$

We know from the Neyman–Pearson lemma that $T_{\mathrm{NP}}(\mathbf{x}) = f_n(\mathbf{x};\theta_1)/f_n(\mathbf{x};\theta_0) = \phi_{\theta_0,\theta_1}\left(T(\mathbf{x})\right)$ with $c_{\mathrm{NP}}(\theta_0,\theta_1)$ such that $\mathbb{P}\left[T_{\mathrm{NP}}(\mathbf{X}) \geq c_{\mathrm{NP}}(\theta_0,\theta_1); \theta_0\right] = \alpha$ is most powerful for testing $H_0'$ vs. $H_1'$, and equivalent to $T(\mathbf{x})$ from the monotone likelihood ratio assumption. Then $(T, c)$ also maximizes $\pi(\theta_1; T', c')$ among all possible level $\alpha$ tests $(T', c')$, where $c$ is taken to solve

$$
\mathbb{P}\big(T(\mathbf{X}) \geq c; \theta_0\big) = \alpha.
$$

We notice that $c$ is independent of $\theta_1$. So $(T, c)$ is in fact a most powerful test for

$$
H_0' : \theta = \theta_0 \text{ vs. } H_1' : \theta = \theta_1 > \theta_0,
$$

no matter what the value of $\theta_1$ is. So we know that $(T, c)$ is a UMP test for $\theta = \theta_0$ vs. $\theta > \theta_0$. It remains only to show that $(T, c)$ is a size $\alpha$ test for $H_0 : \theta \leq \theta_0$. From Corollary 5.1, we deduce that $\pi(\theta; T, c)$ is non-decreasing in $\theta$ so in particular $\pi(\theta_0; T, c) = \alpha$ implies $\pi(\theta; T, c) \leq \alpha$ for all $\theta < \theta_0$, and we conclude that indeed $\sup_{\theta \leq \theta_0} \pi(\theta, T, c) = \pi(\theta_0, T, c) = \alpha$. □

*Remark* 5.5. It is natural to ask whether a result similar to the above can be applied to the one-sided alternative hypothesis

$$H_0 : \theta \geq \theta_0 \text{ vs. } H_1 : \theta < \theta_0.$$

The answer is yes: one can rework the statements and proofs appropriately.

# 6 More general hypothesis tests

## 6.1 Generalized likelihood ratio tests

In the most general case we would like to test

$$H_0 : \theta \in \Theta_0 \quad \text{vs.} \quad H_1 : \theta \in \Theta_1.$$

It might be difficult or impossible to find a UMP test and there is no guarantee of the existence of an optimal test statistic. One useful approach is to generalize the likelihood ratio test for simple hypotheses (that is $\Theta_0 = \{\theta_0\}$ and $\Theta_1 = \{\theta_1\}$ for $\theta_0, \theta_1 \in \Theta$) from

$$T_{\mathrm{NP}}(\mathbf{x}) = \frac{f_n(\mathbf{x};\theta_1)}{f_n(\mathbf{x};\theta_0)},$$

to

$$T_{\text{suggested}}(\mathbf{x}) := \frac{\sup_{\theta\in\Theta_1} f_n(\mathbf{x};\theta)}{\sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)}.$$

That is we could consider the best fit to the data for both situations $\theta \in \Theta_0$ and $\theta \in \Theta_1$ and compare the corresponding likelihoods. However for technical reasons we will instead use the likelihood ratio

$$\begin{aligned}\Lambda_n(\mathbf{x}) : &= \frac{\sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)}{\sup_{\theta\in\Theta} f_n(\mathbf{x};\theta)} = \frac{\sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)}{f_n(\mathbf{x};\hat{\theta}_n)} \\ &= \min\left\{1, \frac{\sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)}{\sup_{\theta\in\Theta_1} f_n(\mathbf{x};\theta)}\right\} \in [0,1].\end{aligned}$$

We observe that

- the denominator corresponds to plugging in the ML estimate in the likelihood (assuming it exists and is unique),
- the last equality follows from the fact that $\sup_{\theta\in\Theta} f_n(\mathbf{x};\theta) \geq \sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)$; if the inequality is strict then
$$\sup_{\theta\in\Theta} f_n(\mathbf{x};\theta) = \sup_{\theta\in\Theta_1} f_n(\mathbf{x};\theta) > \sup_{\theta\in\Theta_0} f_n(\mathbf{x};\theta)$$
and if it is an equality then $\Lambda_n(\mathbf{x}) = 1$.

We have seen a similar quantity before, when considering Wilks confidence intervals. In fact, we will consider the statistic

$$T_n(\mathbf{x}) = -2\log\Lambda_n(\mathbf{x}),$$

which is large when the maximum likelihood estimate lies outside $\Theta_0$, in other words when there is a better (statistical) explanation of the data in $\Theta_1$ than in $\Theta_0$. This is coherent with the way we have defined critical regions (i.e. reject $H_0$ when the test statistic is large). In order to quantify the properties of the test (i.e. evaluate the power function $\pi(\theta, T, c)$ of the test in order to compute $\alpha$, $\beta$ and its $p-$value) we investigate the properties of the distribution of the random variable $T_n(\mathbf{X})$. There is no tractable expression for this distribution for any $n \in \mathbb{N}$, but again a convenient asymptotic approximation is available.

In order to make sense of the asymptotic distribution of $T_n(\mathbf{X})$, we first consider the degrees of freedom in $\Theta_0$ and $\Theta$. We will denote by $\dim(\Theta_0)$ the associated dimension of $\Theta_0$.

**Example 6.1.** Say $d \geq 1$ and that we are interested in the set

$$\Theta_0 = \Big\{\theta \in \Theta : \theta = \theta_0\Big\}$$

for some specific value $\theta_0 \in \Theta$. Then $\Theta_0$ consists of only a single point and $\dim(\Theta_0) = 0$.

**Example 6.2.** Say $d = 2$, write $\theta = (\theta_1, \theta_2)$ for the two components of this vector and introduce $\Theta_1$ and $\Theta_2$ such that $\Theta = \Theta_1 \times \Theta_2$, $\theta_1 \in \Theta_1$ and $\theta_2 \in \Theta_2$. Assume we are interested in the set

$$\Theta_0 := \Big\{\theta \in \Theta : \theta_1 = \theta_{0,1}\Big\},$$

for $\theta_{0,1} \in \mathbb{R}$, a specific value of the first component of $\theta$. Then $\dim(\Theta_0) = 1$ since only $\theta_2$ is free.

**Example 6.3.** Assume that $d = 2$ and with the notation from the previous example, $\Theta_1 = \Theta_2$ and

$$\Theta_0 := \Big\{\theta \in \Theta : \theta_1 = \theta_2\Big\}.$$

Then $\dim(\Theta_0) = 1$ since there is only one free parameter.

**Definition 6.1.** Assume $\Theta \subseteq \mathbb{R}^d$ for some $d \geq 1$. Let $\phi = (\phi_1, \phi_2) : \Theta \to \Phi_1 \times \Phi_2$ be a continuously differentiable bijection with $\Phi_1 \subseteq \mathbb{R}^r$ and $\Phi_2 \subseteq \mathbb{R}^{d-r}$, and for some $c \in \Phi_1$,

$$\Theta_0 = \{\theta \in \Theta : \phi_1(\theta) = c\},$$

i.e. $\Theta_0$ contains all $\theta$ satisfying a particular equation involving $\phi_1$. Then $\Theta_0$ is *nested* within $\Theta$ and the dimension of $\Theta_0$ is $\dim(\Theta_0) = d - r$.

Alternatively, one can write

$$\Theta_0 = \big\{\theta \in \Theta : \text{ for some } c_2 \in \Phi_2, \theta = \phi^{-1}(c, c_2)\big\}.$$

*Remark* 6.1. We will not use the full generality of this definition in this unit. The main point is that there are some restrictions on which types of $H_0$ one can consider in the following theorem.

In the following theorem $\theta$ is not assumed to be a scalar, and $\Theta \subset \mathbb{R}^d$.

**Theorem 6.1.** *Assume* $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta)$ *for some* $\theta \in \Theta_0$ *(i.e.* $H_0$ *is true), that* $f$ *defines a regular statistical model, and that* $\Theta_0$ *is nested within* $\Theta$*. Then*

$$T_n(\mathbf{X}) = -2\log\Lambda_n(\mathbf{X}) \to_{\mathcal{D}(\cdot;\theta)} W$$

*where* $W \sim \chi^2(r)$ *with* $r = \dim(\Theta) - \dim(\Theta_0)$.

The result relies again on a Taylor expansion of the likelihood similar to that used earlier to prove the result underpinning the construction of Wilks confidence intervals. However, the proof will be omitted as it requires one to consider the case where $\theta$ is multidimensional, which is not in the scope of this unit.

*Remark* 6.2. Although the result is similar to that proved in the lectures concerned with the Wilks confidence intervals, it should be emphasized that the number of degrees of freedoms $r$ of the asymptotic distribution depends on the dimension of both $\Theta$ and $\Theta_0$. For example, assume that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $\mu$ and $\sigma^2$ unknown and that we wish to test

$$H_0 : \theta = (0, \sigma^2) \quad \text{vs.} \quad H_1 : \mu \neq 0, \sigma^2 \geq 0$$

In this case $\dim(\Theta) = 2$ while $\dim(\Theta_0) = 1$ as we have

$$\Theta_0 = \Big\{\theta \in \Theta = \mathbb{R} \times \mathbb{R}_+ : \theta_1 = 0\Big\}.$$

Therefore $r = 2 - 1 = 1$ and

$$\Lambda_n(\mathbf{x}) = \frac{\sup_{\sigma^2 \in \mathbb{R}_+} f_n\big(\mathbf{x}; (0, \sigma^2)\big)}{\sup_{(\mu,\sigma^2) \in \mathbb{R} \times \mathbb{R}_+} f_n\big(\mathbf{x}; (\mu, \sigma^2)\big)}.$$

This general result suggests the following simple procedure to compute an approximate $p-$value for this problem.

1. Compute $T_n(\mathbf{x}) = -2\log\Lambda_n(\mathbf{x})$.
2. Determine $r = \dim(\Theta) - \dim(\Theta_0)$.
3. Compute the approximate $p-$value

$$p(\mathbf{x}) = \mathbb{P}(W \geq -2\log\Lambda_n(\mathbf{x})),$$

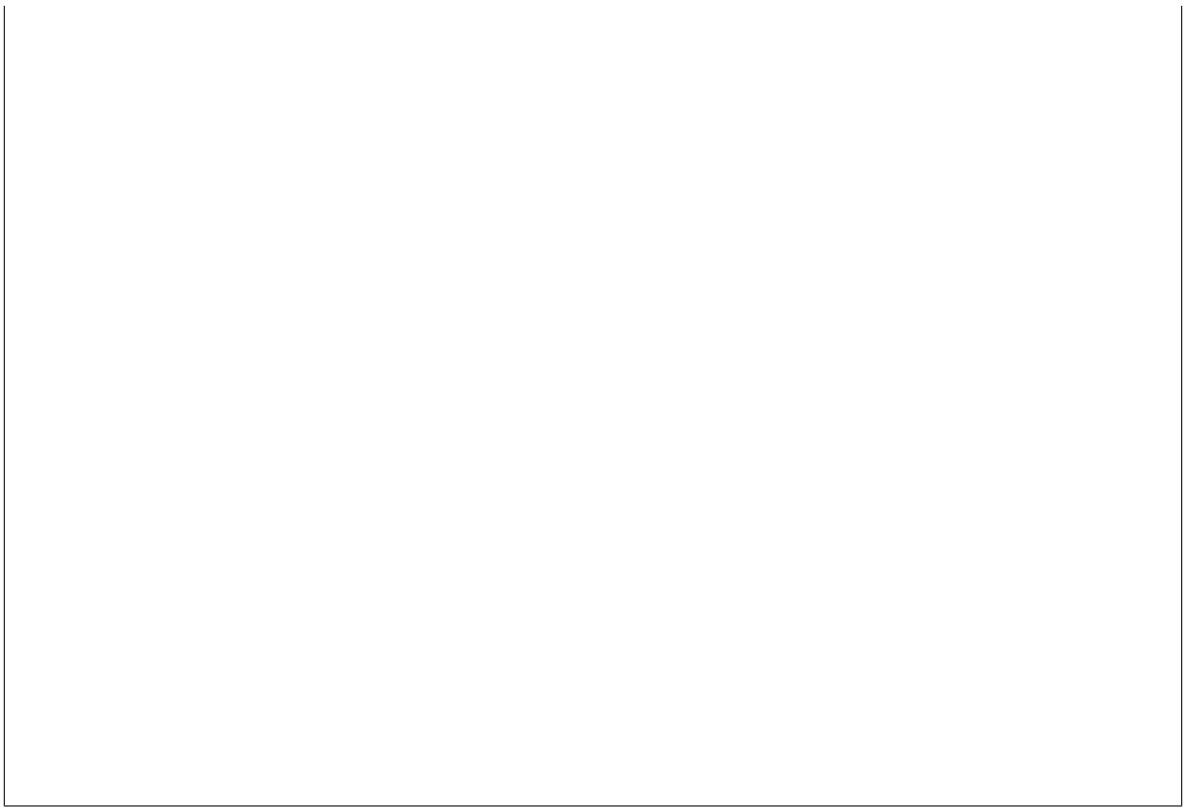
where $W \sim \chi^2(r)$.

If instead we want to make a decision and the size of the test to be approximately $\alpha$, we would reject $H_0$ if $T_n(\mathbf{x}) \geq \chi^2_{r,\alpha}$, with $\chi_{r,\beta}$ the value satisfying $\mathbb{P}(W \geq \chi_{r,\beta}) = \beta$ for $W \sim \chi^2(r)$. This is because

$$\mathbb{P}(T_n(\mathbf{X}) \geq \chi^2_{r,\alpha}; \lambda_0) \approx \alpha.$$

**Example 6.4.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \text{Poisson}(\lambda^*)$ for some $\lambda^* > 0$ and that we wish to test

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda \neq \lambda_0.$$



The scenario considered in the example above is a particular instance of what is called a "two sided" test. Such tests are of the form

$$H_0 : \theta = \theta_0 \quad vs. \quad H_1 : \theta \neq \theta_0$$

and are called "two sided" in contrast with "one sided" scenarios, such as

$$H_0 : \theta = \theta_0 \quad vs. \quad H_1 : \theta \geq \theta_0 \quad .$$

When the theorem above applies, we can use $T_n$ as defined earlier and exploit the fact that

$$T_n(\mathbf{X}) = -2 \log \Lambda_n(\mathbf{X}) \rightarrow_{\mathcal{D}(\cdot;\theta_0)} \chi^2(r),$$

where $r = \dim(\Theta)$, since $\dim(\Theta_0) = 0$ for a point null hypothesis.

## 6.2 Connection to confidence intervals

We can see that the convergence in distribution mentioned above

$$T_n(\mathbf{X}) = -2 \log \Lambda_n(\mathbf{X}) \rightarrow_{\mathcal{D}(\cdot;\theta_0)} \chi^2(r),$$

is a generalization of the result motivating Wilks confidence sets. Indeed, in the case where $\Theta_0 = \{\theta_0\}$, we have $r = \dim(\Theta)$ and when $\Theta \subseteq \mathbb{R}$ we recover Theorem **??**.

In fact, there is a connection between testing point null hypotheses and confidence intervals. Recall that a test of size approximately $\alpha$ is obtained by rejecting $H_0$ if

$$T_n(\mathbf{x}) = -2 \left[ \ell_n(\theta_0; \mathbf{x}) - \ell_n(\hat{\theta}_n; \mathbf{x}) \right] \geq \chi^2_{r,\alpha},$$

or equivalently we retain $H_0$ if

$$T_n(\mathbf{x}) = -2\left[\ell_n(\theta_0;\mathbf{x}) - \ell_n(\hat{\theta}_n;\mathbf{x})\right] < \chi^2_{r,\alpha}.$$

We can consider those values of $\theta_0$ that would lead to retention of $H_0$,

$$\begin{aligned} C(\mathbf{x}) &= \left\{\theta : -2\left[\ell_n(\theta;\mathbf{x}) - \ell_n(\hat{\theta}_n;\mathbf{x})\right] < \chi^2_{r,\alpha}\right\} \\ &= \left\{\theta : \ell_n(\theta;\mathbf{x}) > \ell_n(\hat{\theta}_n;\mathbf{x}) - \chi^2_{r,\alpha}/2\right\}. \end{aligned}$$

This corresponds to an asymptotically exact $1-\alpha$ observed confidence set for $\theta$. In summary, an asymptotically exact $1-\alpha$ observed confidence set can be constructed by considering all values of $\theta_0$ for which we would retain $H_0$ in a size $\alpha$ test.

Alternatively, testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ at a significance level $\alpha$ can be performed by constructing a $1-\alpha$ observed confidence set $C(\mathbf{x})$ for $\theta$ using Wilks' approach and retaining $H_0$ if $\theta_0$ lies inside it.

More generally, if $\mathcal{I}_\alpha(\mathbf{X})$ is a $1-\alpha$ confidence interval, i.e.

$$\mathbb{P}(\theta \in \mathcal{I}_\alpha(\mathbf{X});\theta) = 1-\alpha,$$

then a size $\alpha$ test corresponds to rejecting $H_0$ if $\theta_0 \notin \mathcal{I}_\alpha(\mathbf{x})$.

# 7 Categorical distributions and Pearson's chi-squared test

## 7.1 Categorical & Multinomial random variables

One very common application of tests is to test "goodness-of-fit". In this scenario, the null hypothesis typically corresponds to a model that is believed to be appropriate and one wishes to test whether the data observed is consistent with the model.

The most flexible distribution for a random variable $Y$ taking values in a finite set $\{1, \dots, m\}$ is

$$Y \sim \text{Categorical}(\mathbf{p}), \qquad \mathbf{p} = (p_1, \dots, p_m),$$

where $\mathbf{p}$ is a vector of probabilities: non-negative and summing to 1. This distribution is flexible because it has a parameter associated with each probability

$$p_k = \mathbb{P}(Y = k),$$

meaning that $\mathbf{p}$ is essentially a vector of $m - 1$ free parameters [the last is given by the first $m - 1$, e.g., since $\sum_{k=1}^{m} p_k = 1$.]

**Example 7.1.** A Bernoulli($p$) random variable is a translation of a Categorical($1 - p, p$) random variable. This is the case $m = 2$.

**Example 7.2.** The outcome of throwing a 6-sided die can be modelled as a Categorical($p_1, \dots, p_6$) random variable. If the die is fair then $p_1 = \cdots = p_6 = 1/6$.

If we observe $n$ i.i.d. Categorical($\mathbf{p}$) random variables $Y_1, \dots, Y_n$, in most situations it is convenient to represent their counts by

$$N_k := \sum_{i=1}^{n} \mathbb{I}(Y_i = k),$$

which gives rise to the random vector

$$\mathbf{X} = (N_1, \dots, N_m) \sim \text{Multinomial}(n, \mathbf{p}),$$

satisfying $\sum_{i=1}^{m} N_i = n$, and with $\mathbf{x} = (n_1, \dots, n_m)$ has PMF

$$\begin{aligned} f_n(\mathbf{x}; \mathbf{p}) &= \frac{n!}{n_1! n_2! \cdots n_m!} p_1^{n_1} p_2^{n_2} \cdots p_m^{n_m} \mathbb{I}\left\{ \sum_{i=1}^{m} n_i = n \right\} \\ &= \mathbb{I}\left\{ \sum_{i=1}^{m} n_i = n \right\} \left\{ \frac{n!}{\prod_{i=1}^{m} n_i!} \right\} \prod_{i=1}^{n} p_i^{n_i}. \end{aligned}$$

This is the distribution of the numbers of each outcome in $n$ independent trials where each trial can have one of $m$ outcomes indexed by $\{1, \dots, m\}$ and the probabilities of each outcome are $p_1, p_2, \dots, p_m$, respectively. For $j \in \{1, \dots, m\}$ the random variable $N_j$ represents the number of outcomes of type $j$.

**Example 7.3.** When $m = 2$, $(N_1, N_2) \sim \text{Multinomial}(n, (1-p, p))$ can be related to $S \sim \text{Binomial}(n, p)$ via $(N_1, N_2) = (n - S, S)$.

**Example 7.4.** For a 6-sided die, $N_i$ is the number of times the die shows $i$ after being thrown $n$ times.

## 7.2 GLR test statistic

Let $\mathcal{S}_m$ be the simplex of probability vectors of length $m$, so

$$\mathcal{S}_m = \left\{ (p_1, \ldots, p_m) \in [0,1]^m \ : \ \sum_{i=1}^{m} p_i = 1 \right\}.$$

Now imagine that we want to test, for some

$$(p_{0,1}, \ldots, p_{0,m}) = \mathbf{p}_0 \in \mathcal{S}_m,$$

the hypotheses

$$H_0 : \quad \mathbf{p} = \mathbf{p}_0 \quad \text{vs.} \quad H_1 : \mathbf{p} \in \mathcal{S}_m \setminus \{\mathbf{p}_0\} \ .$$

We can use a generalized likelihood ratio test. First we consider the generalized likelihood ratio

$$\Lambda_n(\mathbf{x}) = \frac{f_n(\mathbf{x}; \mathbf{p}_0)}{\sup_{\mathbf{p} \in \mathcal{S}_m} f_n(\mathbf{x}; \mathbf{p})}.$$

To proceed, we need to find an expression for the ML estimate of $\mathbf{p}$.

**Lemma 7.1.** *Assume* $\mathbf{X} = (N_1, \ldots, N_m) \sim \text{Multinomial}(n, \mathbf{p}^*)$ *for some* $\mathbf{p}^* \in \mathcal{S}_m$. *Given a realization* $\mathbf{x}$ *of* $\mathbf{X}$, $\mathbf{x} = (n_1, \ldots, n_m)$, *the ML estimate of* $\mathbf{p}^*$ *is*

$$\hat{\mathbf{p}}(\mathbf{x}) = (\hat{p}_1(\mathbf{x}), \ldots, \hat{p}_m(\mathbf{x})) = \left( \frac{n_1}{n}, \ldots, \frac{n_m}{n} \right) \quad .$$

*Proof.* One could use the Lagrange multipliers technique in order to take into account the constraint. However one can reparameterize the problem in terms of $p_1, p_2, \ldots, p_{m-1}$ only, which yields the log-likelihood



It follows then that

$$\Lambda_n(\mathbf{x}) = \frac{f_n(\mathbf{x};\mathbf{p}_0)}{\sup_{\mathbf{p}\in\mathcal{S}_m} f_n(\mathbf{x};\mathbf{p})} = \prod_{i=1}^{m} \frac{p_{0,i}^{n_i}}{\hat{p}_i^{n_i}} = \prod_{i=1}^{m} \frac{p_{0,i}^{n_i}}{(n_i/n)^{n_i}},$$

so that

$$T_n(\mathbf{x}) = -2\log\Lambda_n(\mathbf{x}) = -2\sum_{i=1}^{m} n_i \left\{\log(p_{0,i}) - \log(n_i/n)\right\},$$

is the GLR test statistic. From the general theorem,

$$T_n(\mathbf{X}) \to_{\mathcal{D}(\cdot;\mathbf{p}_0)} \chi^2(m-1),$$

since $\dim(\mathcal{S}_m) = m-1$.

Many people rewrite this statistic as

$$T_n(\mathbf{x}) = 2\sum_{j=1}^{m} o_j \log\left(\frac{o_j}{e_j}\right),$$

where $o_j = n_j$ is the observed number in category $j$ and $e_j = np_{0,j}$ the expected number in category $j$. □

## 7.3 Pearson's chi-squared test statistic

The GLR test statistic is closely related to the popular and classical Pearson chi-squared test statistic,

$$T_{\text{Pearson}}(\mathbf{x}) := \sum_{j=1}^{m} \frac{(n_j - np_{0,j})^2}{np_{0,j}} = \sum_{j=1}^{m} \frac{(o_j - e_j)^2}{e_j},$$

which can be thought of as an approximation of $T_n(\mathbf{x})$.

We can use a second order Taylor series expansion of

$$g(x) = x\log\left(\frac{x}{x_0}\right),$$

to find that for some $\xi \in (x, x_0)$,

$$\begin{aligned} g(x) &= g(x_0) + (x-x_0)g'(x_0) + \frac{1}{2}(x-x_0)^2 g''(x_0) + \frac{1}{6}(x-x_0)^3 g'''(\xi) \\ &= (x-x_0) + \frac{1}{2x_0}(x-x_0)^2 - \frac{1}{6\xi^2}(x-x_0)^3, \end{aligned}$$

from which we deduce that

$$\begin{aligned} -2\log\Lambda_n(\mathbf{x}) &= 2\sum_{j=1}^{m} o_j \log\left(\frac{o_j}{e_j}\right) \\ &= \sum_{j=1}^{m} 2(o_j - e_j) + \frac{(o_j-e_j)^2}{e_j} - \frac{(o_j-e_j)^3}{3\xi_j^2} \\ &= \sum_{j=1}^{m} \frac{(o_j-e_j)^2}{e_j} - \frac{(o_j-e_j)^3}{3\xi_j^2}, \end{aligned}$$

where $\xi_j \in (o_j, e_j)$ for $j \in \{1, \ldots, m\}$ and we have used the fact that $\sum_{i=1}^m o_j - e_j = n - n = 0$. Heuristically, to see that

$$-2 \log \Lambda_n(\mathbf{X}) \to_{\mathbb{P}(\cdot;\mathbf{p}_0)} \sum_{j=1}^m \frac{(O_j - E_j)^2}{E_j},$$

we note that for each $j \in \{1, \ldots, m\}$,

$$\begin{aligned}\frac{(O_j - E_j)^3}{3\xi_j^2} &= \frac{n^3(O_j/n - E_j/n)^3}{3\xi_j^2} \\ &= \frac{n(\hat{p}_j - p_{0,j})^3}{3\left(\xi_j/n\right)^2} \to_{\mathbb{P}(\cdot;\mathbf{p}_0)} 0,\end{aligned}$$

since $\sqrt{n}(\hat{p}_j - p_{0,j})$ converges in distribution to a normal random variable with mean 0, while $\xi_j/n$ converges in probability to $p_{0,j}$.

In fact, $T_{\text{Pearson}}(\mathbf{X})$ and $T_n(\mathbf{X})$ have the same asymptotic distribution under $H_0$, i.e.

$$T_{\text{Pearson}}(\mathbf{X}) \to_{\mathcal{D}(\cdot,\mathbf{p}_0)} \chi^2(m-1).$$

*Remark* 7.1. For some direct intuition, let $Y_{i,j} = 1$ if the $i$th draw is of type $j$ and $Y_{i,j} = 0$ otherwise. We have $N_j = \sum_{i=1}^n Y_{i,j}$, and $\hat{p}_j(\mathbf{X}) = N_j/n$ so

$$\frac{(n\hat{p}_j(\mathbf{X}) - np_{0,j})^2}{np_{0,j}} = \left(\sqrt{n}\frac{\hat{p}_j(\mathbf{X}) - p_{0,j}}{\sqrt{p_{0,j}}}\right)^2 = \left(\sqrt{n}\frac{\frac{1}{n}\sum_{i=1}^n Y_{i,j} - p_{0,j}}{\sqrt{p_{0,j}}}\right)^2.$$

The inner term should satisfy a CLT, and taking into account the dependence structure (i.e. the $N_1, \ldots, N_m$ are not independent) leads ultimately to $T_{\text{Pearson}}(\mathbf{X})$, the sum of all $m$ of these terms, being asymptotically distributed as $\chi^2(m-1)$.

In fact, the asymptotic approximation is quite good whenever $e_j = np_{0,j} \geq 5$ for all $j = 1, \ldots, m$, which motivates the aggregation of categories with low expected counts in order to achieve this condition.

To summarize, with $\mathbf{X} \sim \text{Multinomial}(\mathbf{p}^*)$ and $\mathbf{p}_0 \in \mathcal{S}$, if we want to test

$$H_0: \quad \mathbf{p} = \mathbf{p}_0 \quad vs. \quad H_1: \mathbf{p} \neq \mathbf{p}_0 \quad .$$

then, given $n_1, n_2, \ldots, n_m$, we can

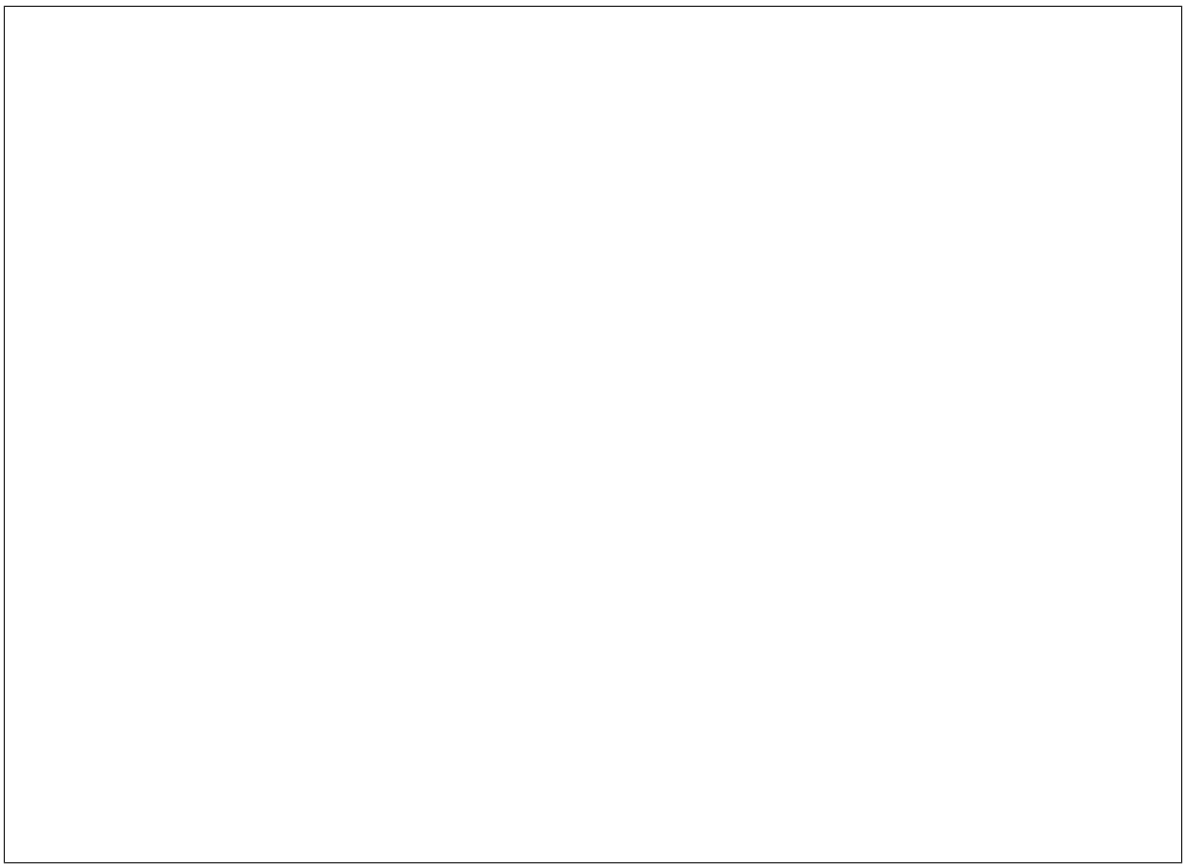
1. compute $e_j = np_{0,j}$ for $j = 1, \ldots, m$,
2. amalgamate the groups where $e_j < 5$ (effectively this means changing the model to a Multinomial with $m' < m$ categories),
3. compute the test statistic $T_{\text{Pearson}}(\mathbf{x})$,
4. compute the $p-$value using the $\chi^2(m'-1)$ distribution.

**Example 7.5** (The Montana Business Outlook Survey)**.** This is a poll of Montana residents where respondents were asked whether their personal financial status was worse/same/better than a year earlier. The data is as follows

| Worse | Same | Better | Total |
|---|---|---|---|
| 58 | 64 | 67 | 189 |

One may wonder whether the answers are uniformly distributed? Formally we would like to test

$$H_0: \mathbf{p} = \mathbf{p}_0 = (1/3, 1/3, 1/3) \quad \text{vs.} \quad H_1: \mathbf{p} \neq \mathbf{p}_0$$



## 7.4 Goodness-of-fit for continuous distributions

The goodness-of-fit approach to model testing is very useful. However the approach is, as such, limited to categorical observations and one may wonder whether this can be extended to the situations where the observations take continuous values?

More precisely, assuming that we observe $\mathbf{x}$ and decide to model this dataset by assuming that $\mathbf{x}$ is a realization of the random variable $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta^*)$ for some density $f(x;\theta)$ and some $\theta^* \in \Theta$, how does one test whether this model is plausible for $\mathbf{x}$? A simple way of achieving this consists of "quantizing" or "binning" the continuous space $\mathcal{X}$ of observations into a discrete set.

**Example 7.6.** Let $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta)$ and we want to test the goodness-of-fit for a particular, fixed value $\theta_0 \in \Theta$ given data $\mathbf{x} = (x_1, \ldots, x_n)$.

A solution is to partition $\mathcal{X} = \mathbb{R}$ into intervals $I_1, \ldots, I_m$, i.e. we require $I_k \cap I_l = \emptyset$ for $k \neq l$ and $\bigcup_{j=1}^m I_j = \mathcal{X}$. For example, $I_1, \ldots, I_m$ can be specified as

$$(-\infty, v_1], (v_1, v_2], \ldots, (v_{m-2}, v_{m-1}], (v_{m-1}, \infty),$$

where $-\infty < v_1 \cdots < v_{m-1} < \infty$. Then we consider the random variables

$$Y_{i,j} = \mathbb{I}\{X_i \in I_j\}, \qquad i \in \{1, \ldots, n\},\ j \in \{1, \ldots, m\},$$

which satisfy $Y_{i,j} = 1$ if $X_i$ is in interval $I_j$. We define

$$p_j(\theta_0) = \mathbb{P}\left(Y_{i,j} = 1; \theta_0\right) = \mathbb{P}\left(X_i \in I_j; \theta_0\right) = \int_{I_j} f(x;\theta_0)dx,$$

and $\mathbf{p}(\theta_0) = (p_1(\theta_0), \ldots, p_m(\theta_0))$. We can then proceed by considering $(N_1, \ldots, N_m) \sim \text{Multinomial}(n, \mathbf{p}(\theta_0))$, where

$$N_j = \sum_{i=1}^{n} Y_{i,j} = \sum_{i=1}^{n} \mathbb{I}\{X_i \in I_j\},$$

and its realized values $(n_1, \ldots, n_m)$ satisfying $n_j = \sum_{i=1}^{n} \mathbb{I}\{x_i \in I_j\}$. That is, we test whether the counts of values in each "bin" are consistent with being a realization of a $\text{Multinomial}(n, \mathbf{p}(\theta_0))$ random variable using a GLR test or Pearson's $\chi^2$-test.

Notice that we essentially test as if we had only observed $(n_1, \ldots, n_m)$ rather than $(x_1, \ldots, x_n)$, so we have "thrown away" some of the information in the data.

**Example 7.7.** Let $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot; \theta)$ and we want to test the goodness-of-fit given data $\mathbf{x} = (x_1, \ldots, x_n)$, without specifying any particular $\theta$. We can again partition $\mathcal{X} = \mathbb{R}$ into intervals $I_1, \ldots, I_m$, and define

$$Y_{i,j} = \mathbb{I}\{X_i \in I_j\}, \qquad i \in \{1, \ldots, n\},\ j \in \{1, \ldots, m\}.$$

We define the vector $\mathbf{p}(\theta)$ by

$$p_j(\theta) = \mathbb{P}\left(Y_{i,j} = 1; \theta\right) = \mathbb{P}\left(X_i \in I_j; \theta\right) = \int_{I_j} f(x; \theta) dx,$$

for each $\theta \in \Theta$, and consider the counts $N_j := \sum_{i=1}^{n} Y_{i,j}$ and their realizations $n_j = \sum_{i=1}^{n} \mathbb{I}\{x_i \in I_j\}$, for $j \in \{1, \ldots, m\}$.

We now want to test whether $(n_1, \ldots, n_m)$ are consistent with

$$(N_1, \ldots, N_m) \sim \text{Multinomial}(n, \mathbf{p}(\theta)),$$

for some $\theta \in \Theta$. Notice that $\theta$ is the parameter for $f$ but $\mathbf{p}(\theta)$ is the parameter of the Multinomial distribution.

What we can do now is consider the model $(N_1, \ldots, N_m) \sim \text{Multinomial}(n, \mathbf{q})$, and test the hypotheses

$$H_0 : \mathbf{q} \in \{\mathbf{p}(\theta) \,:\, \theta \in \Theta\}, \qquad H_1 : \mathbf{q} \notin \{\mathbf{p}(\theta) \,:\, \theta \in \Theta\}.$$

Notice that we are *not* testing whether $\theta$ is in some $\Theta_0$ but instead whether $\mathbf{q} \in \{\mathbf{p}(\theta) \,:\, \theta \in \Theta\}$

The likelihood ratio is

$$\Lambda_n(\mathbf{x}) = \frac{\sup_{\theta \in \Theta} \prod_{j=1}^{m} p_j(\theta)^{n_j}}{\prod_{i=1}^{m} (n_i/n)^{n_i}},$$

and so we must optimize $\prod_{j=1}^{m} p_j(\theta)^{n_j}$ w.r.t. $\theta \in \Theta$ to evaluate it. In practice the resulting $\theta^{\text{opt}}$ will often need to be found by numerical methods. Notice that the denominator corresponds to the ML estimate for $\mathbf{q}$, and we are essentially comparing the original continuous model $f$ with an arbitrary model for the quantized data. The asymptotic distribution of the test statistic is given by

$$T_n(\mathbf{X}) = -2 \log \Lambda_n(\mathbf{X}) \to_{\mathcal{D}(\cdot;\theta)} \chi^2(m - 1 - n_\theta),$$

where $\theta \in \Theta$ (i.e. $H_0$ is true) and $n_\theta$ is the dimension of $\Theta$.

*Remark* 7.2. In some situations, finding $\theta^{\text{opt}}$ is difficult. A natural question is whether one can replace $\theta^{\text{opt}}$ with $\hat{\theta}_{\text{ML}}$ in the GLR test statistic to avoid the constrained optimization. One can do this, but the the modified test statistic

$$T_{\text{modified}}(\mathbf{x}) = -2 \log \frac{\prod_{j=1}^{m} p_j(\hat{\theta}_{\text{ML}})^{n_j}}{\prod_{i=1}^{m} (n_i/n)^{n_i}},$$

then satisfies asymptotically [under the null], for any $c \in \mathbb{R}_+$,

$$\mathbb{P}(W_{m-1-n_\theta} \geq c) \leq \mathbb{P}\left[T_{\text{modified}}(\mathbf{X}) \geq c; \theta\right] \leq \mathbb{P}(W_{m-1} \geq c)$$

where $W_{m-1-n_\theta} \sim \chi^2(m - 1 - n_\theta)$ and $W_{m-1} \sim \chi^2(m-1)$. This allows one to compute approximate upper and lower bounds on the p-value of data $\mathbf{x}$.

# 8 Two goodness-of-fit examples from genetics

## 8.1 Mendel's peas

Gregor Mendel, the father of modern genetics, performed experiments to test his theory of inheritance in the 1850s and 1860s.

There are $k = 4$ types of peas: round yellow, wrinkled yellow, round green and wrinkled green. According to his theory of inheritance, the number of peas of each type in the second generation of a crossing experiment should be an observation of a Multinomial$(n, \mathbf{p}_0)$ random vector, where

$$\mathbf{p}_0 = \left(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16}\right).$$

This constitutes the null hypothesis. In his experiment, with $n = 556$, he recorded $\mathbf{x} = (n_1, \ldots, n_4) = (315, 101, 108, 32)$. The test statistic is

$$\begin{aligned} w &= \sum_{i=1}^{k} \frac{(n_i - np_{0,i})^2}{np_{0,i}} \\ &= \frac{(315 - 312.75)^2}{312.75} + \frac{(101 - 104.25)^2}{104.25} + \frac{(108 - 104.25)^2}{104.25} + \frac{(32 - 34.75)^2}{34.75} \\ &= 0.47. \end{aligned}$$

The (approximate) $p$-value is therefore the probability that a $\chi^2(k-1) = \chi^2(3)$ random variable exceeds 0.47, which is 0.925. We would not reject the null hypothesis for any reasonable $\alpha$, so the data do not contradict his theory. In fact, the $p$-value is quite large, and there is some controversy regarding whether Mendel's results are "too good".

In R one could perform these calculations by running the following commands.

```
p0 <- c(9/16, 3/16, 3/16, 1/16)
x <- c(315, 108, 101, 32)
n <- sum(x)
T <- sum((x-n*p0)^2/(n*p0))
r <- 3
T

## [1] 0.470024

p_value <- pchisq(T, r, lower.tail=FALSE)
p_value

## [1] 0.9254259
```

One can also approximate the power of the test for a given $(\alpha, \mathbf{p})$, say $\alpha = 0.05$ and $\mathbf{p} = (8/16, 3/16, 3/16, 2/16)$.

```
p0 <- c(9/16, 3/16, 3/16, 1/16)
p1 <- c(8/16, 3/16, 3/16, 2/16)
alpha <- 0.05
r <- 3
n <- sum(c(315,108,101,32)) # use the same n
```

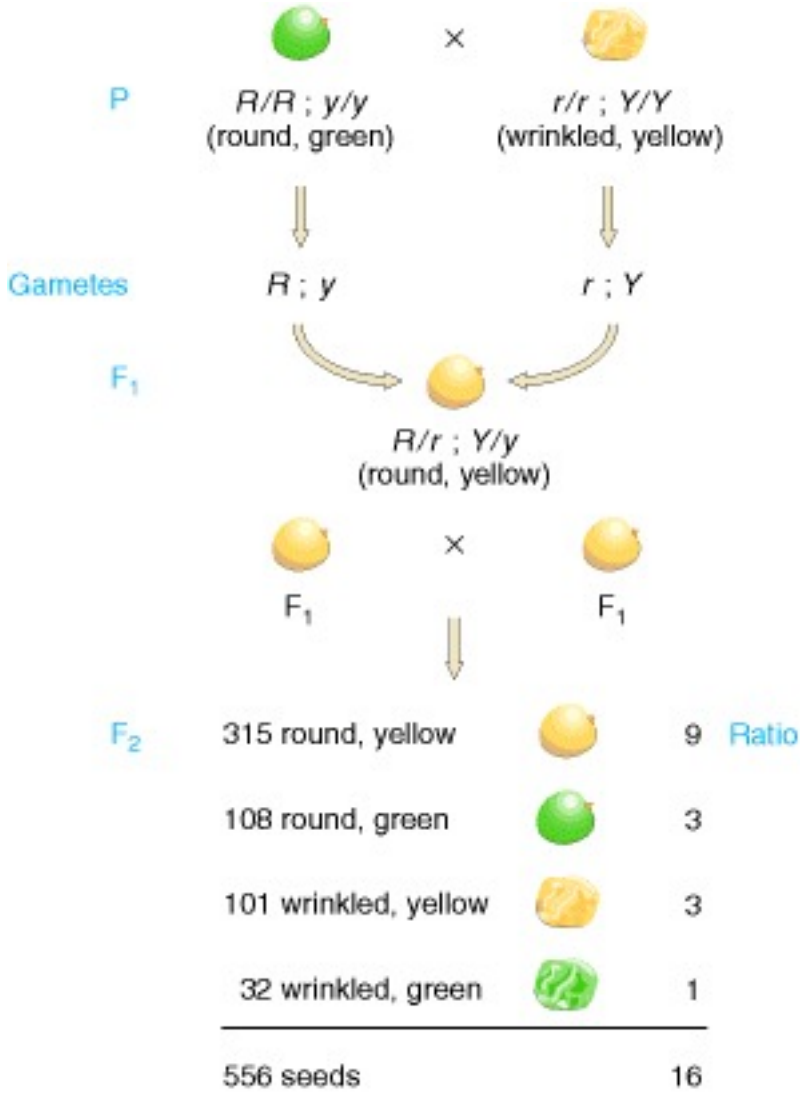



Figure 8.1: From "An Introduction to Genetic Analysis.", 7th edition. Griffiths AJF, Miller JH, Suzuki DT, et al. New York: W. H. Freeman; 2000.

```r
chisq.val <- qchisq(1-alpha, r)
trials <- 1e3
rejections <- 0
for (i in 1:trials) {
  x <- rmultinom(1,n,p1)
  T <- sum((x-n*p0)^2/(n*p0))
  rejections <- rejections + (T >= chisq.val)
}
power.est <- rejections/trials
power.est

## [1] 0.997
```

Ronald Fisher looked at Mendel's data for a sequence of $m$ similar experiments, and discovered that they seemed all to obtain this kind of unusually good agreement. Each test statistic $w_i$ is modelled (approximately) as an observation of an independent $\chi^2(k_i-1)$ random variable, where $k_i$ is the number of possible outcomes in experiment $i$. Since the sum of independent $\chi^2(k_i-1)$ random variables is a $\chi^2\left(\sum_{i=1}^m (k_i-1)\right)$ random variable, one can use the sum of the test statistics $\sum_{i=1}^m w_i$ as a test statistic and compute the p-value.

$$p(\mathbf{x}) = \mathbb{P}(Y \geq \sum_{i=1}^m w_i), \qquad Y \sim \chi^2\left(\sum_{i=1}^m (k_i-1)\right).$$

In fact, he obtained a "pooled $\chi^2$" test statistic of $\sum_{i=1}^n w_i = 42$ and a value of $\sum_{i=1}^n (k_i-1) = 84$. The $p$-value for a goodness-of-fit test is

$$\mathbb{P}\left[Y \geq 42\right] \approx 1 - 3.5 \times 10^{-5} = 0.999965, \qquad Y \sim \chi^2(84).$$

But Fisher's $H_0$ was that Mendel's data was "gathered honestly" and assumed that Mendel's theory is true, so in fact his $p$-value was

$$\mathbb{P}\left[-Y \geq -42\right] = \mathbb{P}\left[Y \leq 42\right] = 3.5 \times 10^{-5} = 0.000035,$$

so Fisher's $H_0$ is rejected with a very low $\alpha$. There are many who believe that neither Mendel nor his assistant(s) did anything untoward in their experiments, but it is an interesting application nonetheless.

## 8.2 Hardy–Weinberg model

We now consider an example where $H_0$ is not a simple hypothesis. A scientific theory, the Hardy–Weinberg model, provides a probabilistic model for the way genotypes are passed on from one generation to the next. If there are two alleles, $A$ and $a$, in a population, then each individual in the population has one of the following genotypes: $A/A$, $A/a$ or $a/a$. The model predicts that the proportions of each genotype are given by

$$\mathbf{p}(\delta) = \left([1-\delta]^2, 2\delta[1-\delta], \delta^2\right),$$

for some $\delta \in [0,1]$. We want to test whether the model fits some collected data, but without specifying any particular value of $\delta$.

In a real population of 1029 individuals it was observed that the respective genotype counts were 342, 500 and 187. One can model these counts as being a realization of a multinomial random vector

$$\mathbf{X} = (N_1, N_2, N_3) \sim \text{Multinomial}\left(n = 1029, \mathbf{p}^*\right)$$

for some vector $\mathbf{p}^* = (p_1, p_2, p_3) \in \mathcal{S}_3$. In the light of what we said above, we would like to test whether $\mathbf{p}^* = \mathbf{p}(\delta)$ for some $\delta \in [0, 1]$. In other words, we let

$$\Theta_0 = \{\mathbf{p}(\delta) = \left([1-\delta]^2, 2\delta[1-\delta], \delta^2\right) \ : \ \delta \in [0,1]\},$$

$$\Theta_1 = \mathcal{S}_3 \setminus \Theta_0.$$

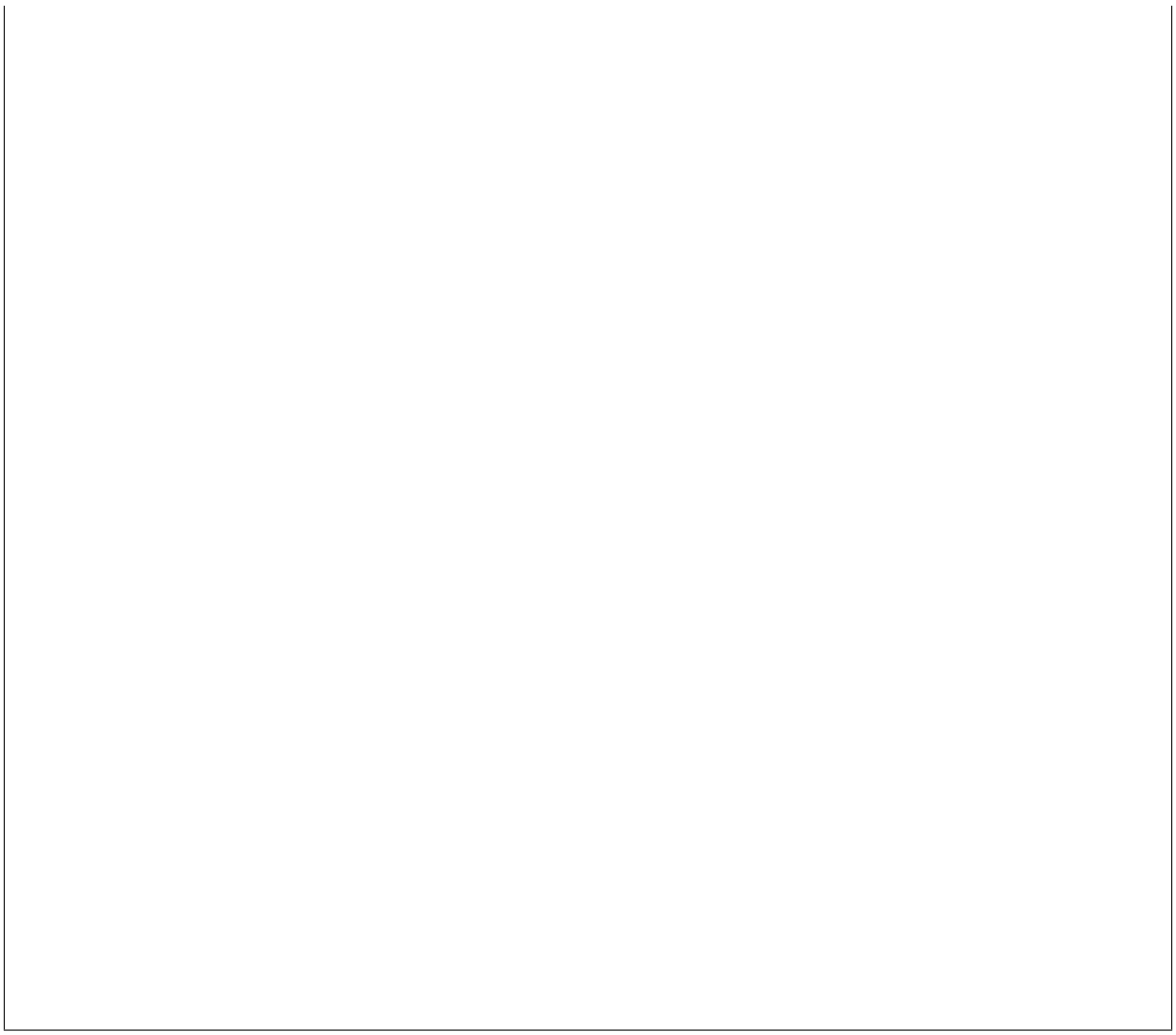
And we construct the hypotheses

$$H_0 : \mathbf{p} \in \Theta_0 \quad \text{vs.} \quad H_1 : \mathbf{p} \in \Theta_1 \quad .$$

We can then use the generalized likelihood ratio approach, with $f_n(\mathbf{x}; \mathbf{p})$ the probability mass function of the multinomial distribution Multinomial$(n, \mathbf{p})$ and consider first the following quantity,

$$\Lambda_n(\mathbf{x}) = \frac{\sup_{\mathbf{p}\in\Theta_0} f_n(\mathbf{x}; \mathbf{p})}{\sup_{\mathbf{p}\in\Theta} f_n(\mathbf{x}; \mathbf{p})} = \prod_{i=1}^{m} \frac{\check{p}_i^{n_i}}{\hat{p}_i^{n_i}},$$

where $\check{\mathbf{p}}$ is the maximizing $\mathbf{p} \in \Theta_0$ of $f_n(\mathbf{x}; \cdot)$ and $\hat{\mathbf{p}}$ is the ML estimate, i.e. it maximizes $f_n(\mathbf{x}; \cdot)$ and can be any $\mathbf{p} \in \Theta$.



# Part II

# Bayesian inference

## 9 Bayesian inference

Remember what the frequentist approach to statistical inference postulates:

1. Probability refers to the limiting relative frequencies of events. Probabilities are objective properties of the world.
2. The parameter $\theta$ of the model is fixed, even though it is unknown. Since $\theta$ is not random, we cannot make probabilistic statements about it.
3. Statistical procedures should be designed to have well-defined long run properties. For example,
   - At least 95% of all 95% confidence intervals should contain $\theta$.
   - We should make Type I errors at most 5% of the time if we test at significance level 0.05.

The Bayesian approach in contrast postulates that

1. Probabilities can also describe degrees of belief, i.e. subjective statements. Therefore we can use probability to model things beyond just data. For example, if I flip a coin and hold it concealed in my hand I could say "the probability that the coin has turned up heads is 1/2".
2. We can make subjective statements about a model parameter $\theta$. In particular, we can model it as a random variable with a particular *prior* distribution, before receiving the data.
3. Inferences about $\theta$ can be conducted on the basis of the conditional distribution of the random variable $\theta$ given the data $\mathbf{x}$. This conditional, or *posterior*, distribution will be the object through which we define point estimates and interval estimates.

*Remark* 9.1. The third postulate has an important consequence. We do not need to consider "hypothetical repeated experiments" when conducting Bayesian inference: the impact of the data and the model will be solely through the posterior distribution, which in turn will be defined in terms of the prior and the observed likelihood function.

Let us now examine a limitation of the classical approach. Often in applications, the nature of the data and the background of the problem suggest that certain values of $\theta$ are more "likely" than others, even before any measurement is made. The model describes possible probabilistic explanations of the data and, in a sense, the statistician believes more strongly in certain explanations than in others. This is illustrated by the following example, which is due to Leonard Savage [Letter to Cornfield on 22nd February 1962.].

**Example 9.1.** Consider the following three statistical experiments:

1. A lady who drinks milk in her tea claims to be able to tell which was poured first, the tea or the milk. In ten trials, she determines correctly whether it was tea or milk which was poured in the cups first.
2. A music expert claims to be able to tell whether a page of music was written by Haydn or by Mozart. In ten trials conducted, he correctly determines the composer every time.

3. A drunken friend says that he can predict the outcome of a fair coin-flip. In ten trials, he is right every time.

In all three experiments, $\theta \in \Theta = [0,1]$ is the probability that the person gives the right answer. Assuming that the trials form an i.i.d.-sequence of Bernoulli measurements with success-rate $\theta$, we test their respective claims with frequentist methods, posing the hypotheses:

$$H_0 : \theta = \theta_0 = 1/2 \quad \text{vs.} \quad H_1 : \theta > 1/2.$$

That is, the answers of the lady, the music expert and drunken friend are purely random or indeed well-informed.

To test we use the total number of successes out of ten trials as our test-statistics, noting that its distribution is binomial with $n = 10$. Given the data $\mathbf{x} = (1, \ldots, 1)$ when $\mathbf{X} \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ [which is the same in all three examples], we reject $H_0$ with a



So there is strong evidence to support the claims made in all three cases.

Note that all three cases are treated exactly in the same way in the frequentist approach. Yet intuitively one would be inclined to treat the three claims on different footings: in the second experiment, we have no reason to doubt the expert's claim, whereas in the third case the friend's condition makes their claim even less plausible than it would be if they were sober, if that's even possible. In the first experiment, the validity of the lady's claim is hard to guess beforehand. The outcome of the experiments would be as expected in the second case and remarkable in the first. In the third case, one would likely consider the friend to be extremely lucky or somehow to have manipulated the experiment.

The example convincingly makes the point that in our intuitive approach to statistical issues, we include all knowledge we have, even potentially resorting to strongly biased estimators. The Bayesian approach allows us to choose the prior in order to reflect this subjectivity: from the outset, we attach more prior mass to parameter-values that we deem more likely, or that we believe in more strongly. In the above example, we would choose a prior that gives higher probability to high values of $\theta$ in the second case unlike the third case where one would consider higher probabilities around $1/2$. In the first case, we may use a fairly weak prior, e.g. a Uniform$(0,1)$ prior in order to hedge our bets.

## 9.1 From prior to posterior

We have seen that we can formalize our prior beliefs by choosing a prior distribution for $\theta$, essentially treating the unknown $\theta$ as a realization of a random variable $\vartheta$. In the example above we could use a Beta$(\alpha, \beta)$ distribution whose density is, for $\alpha, \beta > 0$,

$$p_\vartheta(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}\mathbb{I}\{0 \leq \theta \leq 1\}.$$

The mean and variance of $\vartheta$, a priori, would then be

$$\mathbb{E}(\vartheta) = \frac{\alpha}{\alpha+\beta} \quad \text{and} \quad \mathbb{V}\text{ar}(\vartheta) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)},$$

and we could use these expressions to adjust the parameters in order to take into account our prior beliefs.

How can we combine these prior beliefs to obtain "updated" beliefs once we have evidence, i.e. data? We do so by using Bayes' rule from probability. Note that $\alpha$ and $\beta$ are traditionally referred to as hyperparameters.

**Example 9.2.** Consider a bag containing a ball of unknown colour which may be either black or white (we will refer to this unknown colour as $\theta \in \{w, b\}$). A white ball is added to the bag, then a ball is drawn at random, the ball we pick is white. What is the probability that the remaining ball is white, or in other words the probability that $\vartheta = w$?

So the question is whether it is possible to generalize this idea to the general situation where we have some prior belief about $\theta$ in the form of a prior distribution and some observations from a model $\mathbf{X} \sim f_n(\cdot;\theta)$ for any $\theta \in \Theta$? The answer is straightforward in the situation where both $\theta$ and $\mathbf{x}$ are discrete valued, since then the identity

$$\mathbb{P}(\mathbf{X} = \mathbf{x}, \vartheta = \theta) = \mathbb{P}(\vartheta = \theta \mid \mathbf{X} = \mathbf{x})\mathbb{P}(\mathbf{X} = \mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x} \mid \vartheta = \theta)\mathbb{P}(\vartheta = \theta)$$

yields Bayes' rule

$$\begin{aligned}\mathbb{P}(\vartheta = \theta \mid \mathbf{X} = \mathbf{x}) &= \frac{\mathbb{P}(\mathbf{X} = \mathbf{x} \mid \vartheta = \theta)\mathbb{P}(\vartheta = \theta)}{\mathbb{P}(\mathbf{X} = \mathbf{x})} \\ &= \frac{\mathbb{P}(\mathbf{X} = \mathbf{x}, \theta = \theta)}{\mathbb{P}(\mathbf{X} = \mathbf{x})}.\end{aligned}$$

We observe that

$$\mathbb{P}(\mathbf{X} = \mathbf{x}) = \sum_{\theta \in \Theta} \mathbb{P}(\mathbf{X} = \mathbf{x}, \vartheta = \theta).$$

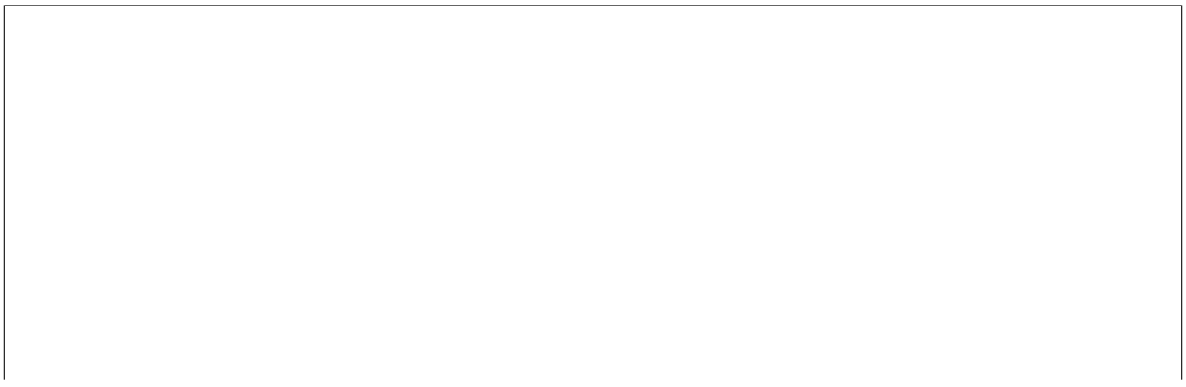
In the case where either $\vartheta$ is a continuous random variable, one resorts to densities. In fact given a prior density $p(\theta)$ one can always write the following generalization of the formula above

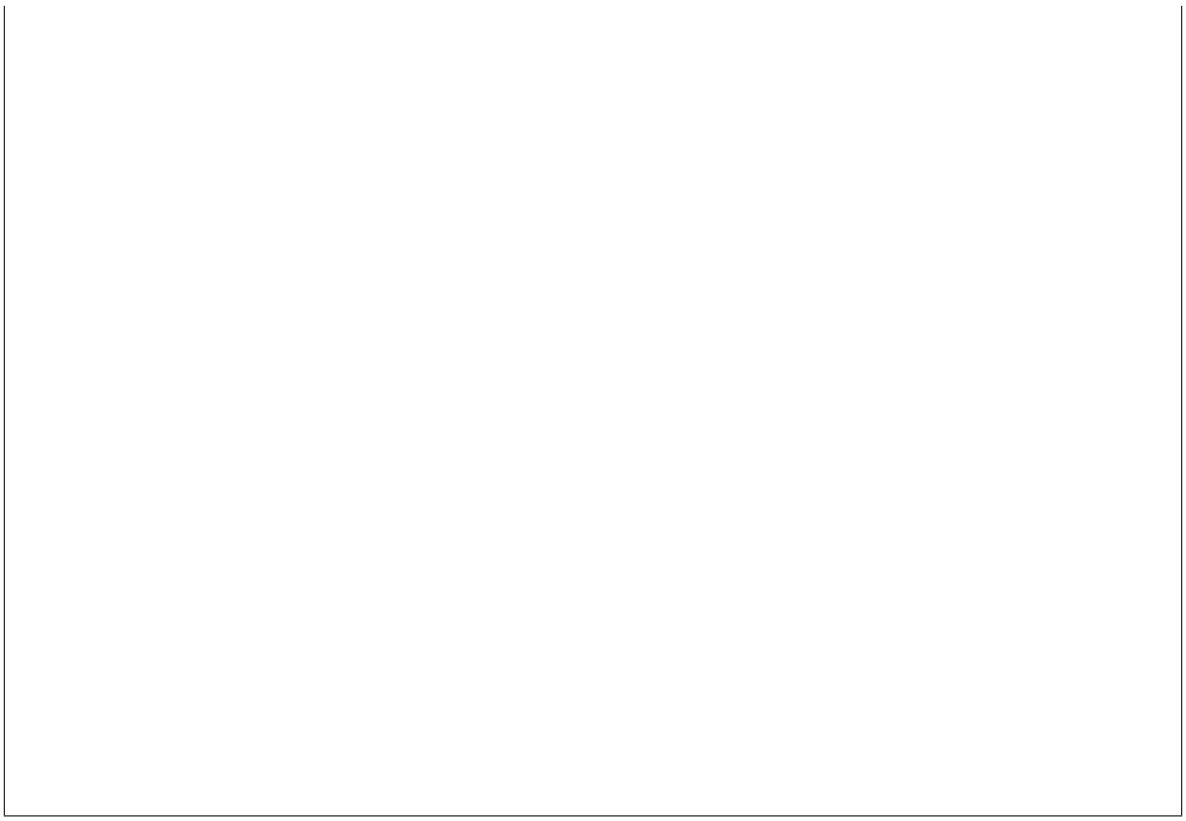
$$p(\theta \mid \mathbf{x}) = \frac{f_n(\mathbf{x};\theta)p(\theta)}{\int_\Theta f_n(\mathbf{x};\theta')p(\theta')d\theta'} = \frac{L(\theta;\mathbf{x})p(\theta)}{\int_\Theta L(\theta';\mathbf{x})p(\theta')d\theta'} \propto L(\theta;\mathbf{x})p(\theta),$$

which you can assume to hold for the content of this unit.

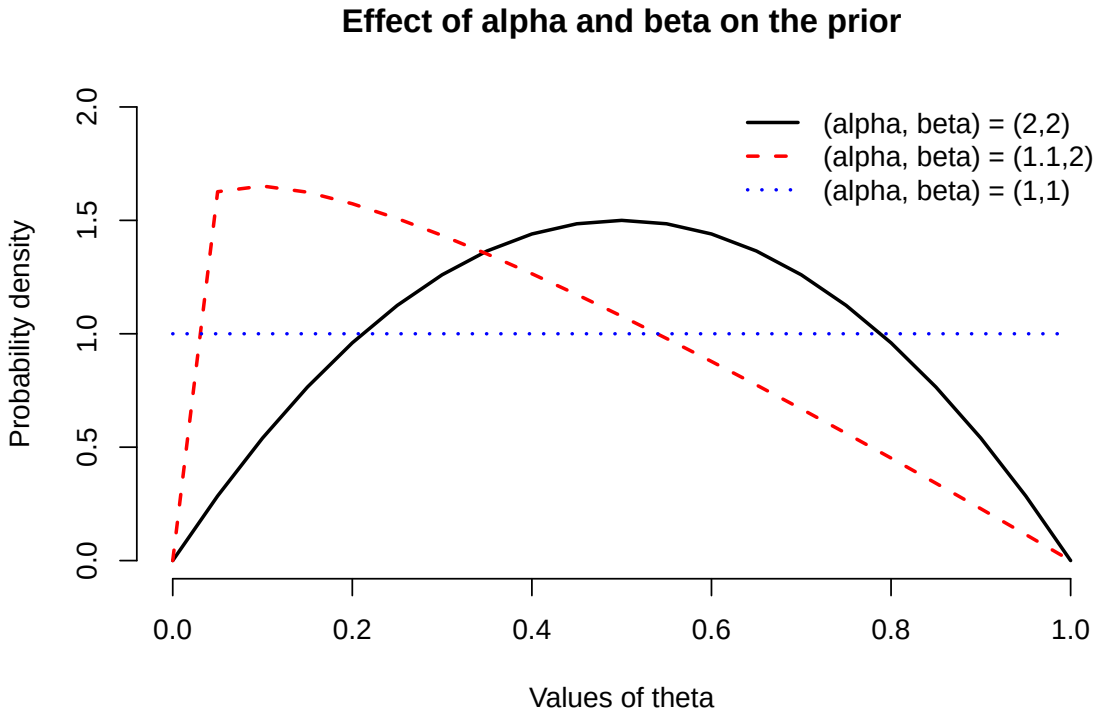
All inference in a Bayesian framework is based on the posterior distribution with PDF or PMF $p(\theta|\mathbf{x})$. The posterior density/distribution combines the prior information and the information carried by the data [given the model]. The combination is not arbitrary: it is determined by the rules of probability.

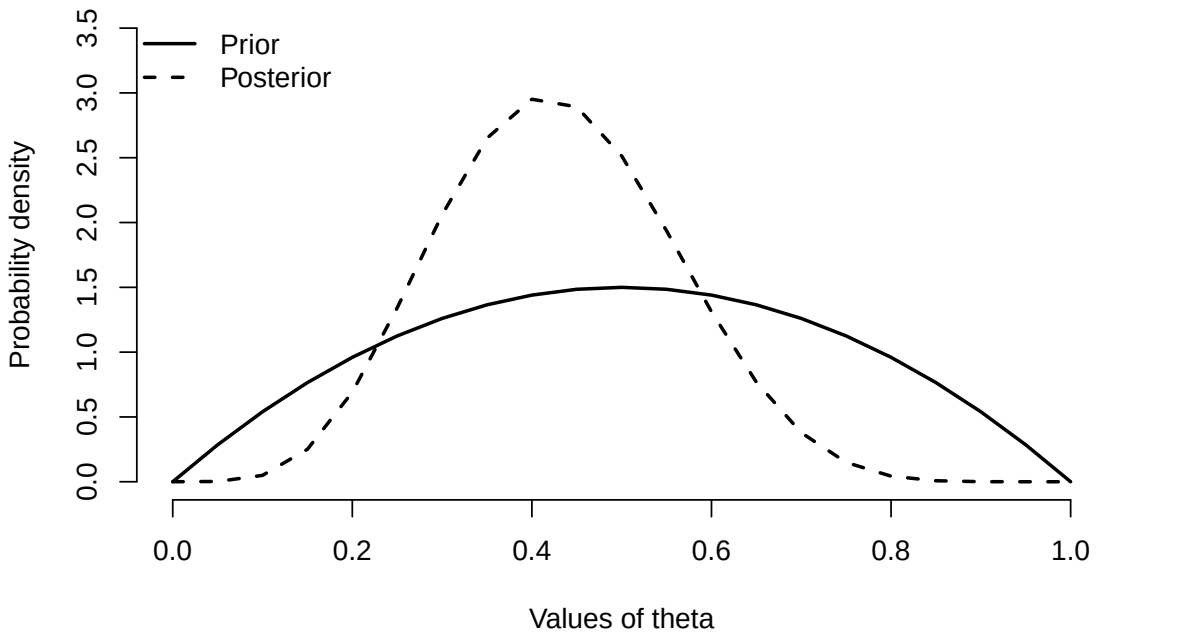
**Example 9.3** (Bernoulli likelihood/Beta prior)**.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ and that as suggested earlier we express our belief in the various values of $\theta$ by using a $\text{Beta}(\alpha, \beta)$ distribution as a prior distribution for $\theta$. We will derive the posterior distribution $p(\theta|\mathbf{x})$.





One can draw the prior density for various values of the parameters, and also the prior and posterior densities. The posterior density reflects both the prior and the data that has been recorded.


Effect of alpha and beta on the prior
(alpha, beta) = (2,2)
(alpha, beta) = (1.1,2)
(alpha, beta) = (1,1)
Probability density
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Values of theta


Prior and posterior probability density functions for (alpha, beta) = (2,2)
Prior
Posterior
Probability density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0.0
0.2
0.4
0.6
0.8
1.0
Values of theta

# 10 Posterior distributions: Bayes estimates and credible intervals

In example 9.3 we observe that the posterior distribution is the same distribution (Beta) as the prior up to the parameters. This property is called conjugacy, and one refers to the prior as being a "conjugate prior". In other words, a prior is called *conjugate* if its distribution is of the same family of distributions as the posterior.

This scenario has historically played an important role, however, as when the prior is not conjugate, one must typically use a computer to conduct inference. Not all models (i.e. likelihoods) have conjugate priors.

First conclusions:

1. compared to the ML approach where the estimator of $\theta$ is usually a point, in a Bayesian framework one considers a posterior distribution, which means that the information we get is potentially richer, but also more difficult to handle,
2. the practical selection of a prior distribution may be delicate and subjective. However, it should be noted that at least in the example considered above, as $n$ becomes large the effect of the prior distribution vanishes. This holds somewhat more generally, but not always.

One may wonder how to use the possibly overwhelming amount of information provided by the posterior distribution - one solution consists of using standard characteristics of probability distributions in order to help summarize its properties. One may consider:

- the posterior mean of the distribution, $\int_\Theta \theta p(\theta \mid \mathbf{x})\, d\theta$ (useful if the posterior distribution is unimodal),
- the maximum a posteriori estimator, $\arg\max_{\theta\in\Theta} p(\theta \mid \mathbf{x})$ (might be misleading in some contrived situations),
- the median or more generally quantiles,
- posterior variance, etc.

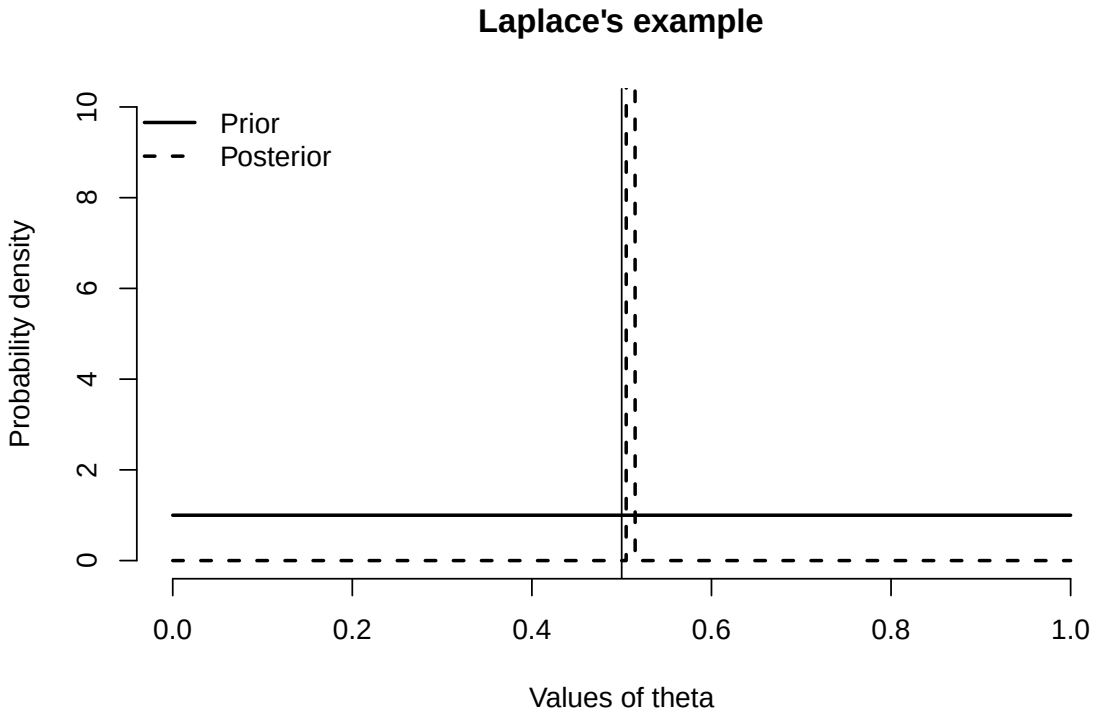
**Example 10.1** (Laplace's example)**.** Laplace proposed in 1786 one of the very first examples of the use of the Bayesian principle (which he referred to as "inverse probability"). He considered births in Paris : $x = 251,527$ male births and $n - x = 241,945$ female births and wanted to decide whether the probability of a male birth, $\theta$, was above $1/2$ or not. He considered a Binomial$(n, \theta)$ model, with $\theta$ the probability of male birth. He decided that based on his lack of prior information about the true value of $\theta$ he would choose a uniform prior distribution for $\theta$, which corresponds to $\alpha = \beta = 1$ in our previous example. This yields a posterior distribution

$$\vartheta \mid (X = x) \sim \text{Beta}\,(251528, 241946)\,,$$

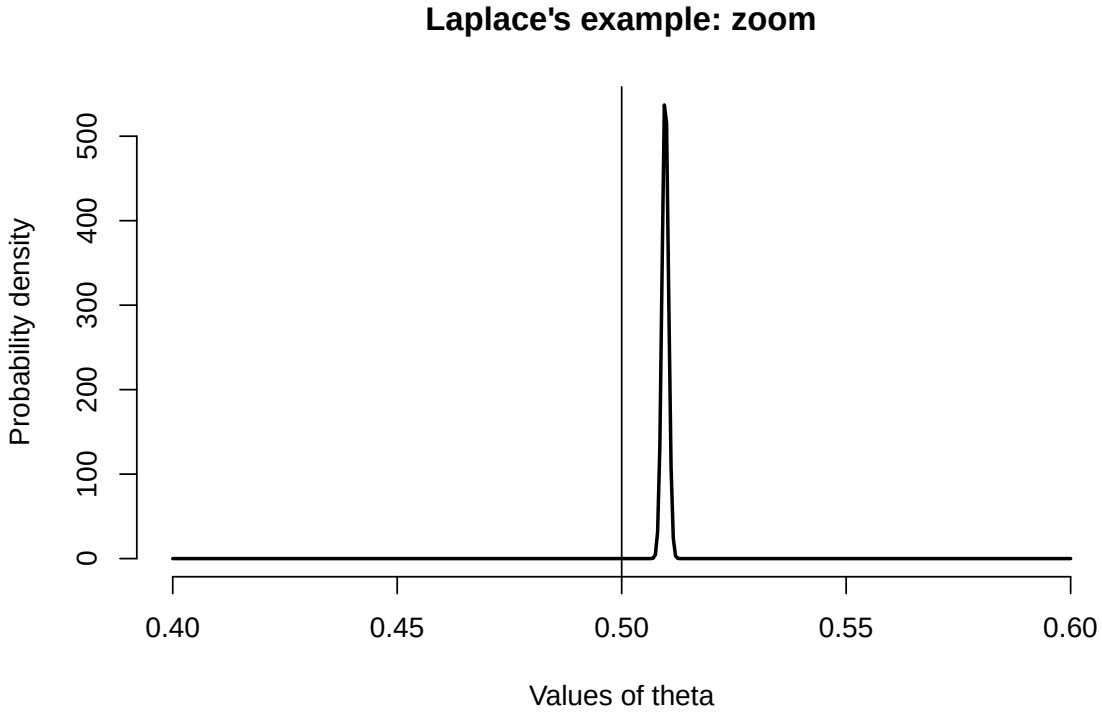
and from this he computed

$$\mathbb{P}\left(\vartheta > \frac{1}{2} \mid X = x\right) \approx 1 - 1.15 \times 10^{-42},$$

and concluded that $\theta$ was most likely to be above $1/2$.




We now zoom in




More generally one may wonder how to deliver point and interval estimates using the Bayesian approach. The most important point is that we do not consider explicitly "hypothetical repeated experiments", i.e. the distribution of a potential estimator for different values of $\theta$.

## 10.1 Bayes estimators

**Definition 10.1** (Posterior expected loss and Bayes estimators.)**.** Let $p(\theta \mid x)$ denote the posterior density. Then the *posterior expected loss*, or posterior risk, of an estimate $\hat{\theta}$ of $\theta$ is

$$R(\hat{\theta} \mid \mathbf{x}) = \int L(\theta, \hat{\theta}) p(\theta \mid \mathbf{x}) \, d\theta,$$

where $L(\theta, \hat{\theta})$ is a non-negative loss function. Notice that $\hat{\theta}$ is a fixed value. A *Bayes estimate* is obtained by minimizing the posterior risk, i.e.

$$\hat{\theta}_{\text{Bayes}} = \arg\min_{\hat{\theta}} R(\hat{\theta} \mid \mathbf{x}).$$

**Proposition 10.1** (Squared error loss.)**.** *If* $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$ *and* $\int \theta^2 p(\theta \mid \mathbf{x}) \, d\theta < \infty$*, then* $\hat{\theta}_{\text{Bayes}}$ *is the posterior mean, i.e.*

$$\hat{\theta}_{\text{Bayes}} = \int_{\Theta} \theta p(\theta \mid \mathbf{x}) \, d\theta.$$

*Proof.*

□

*Remark* 10.1. If we consider squared error loss, then the posterior mean minimizes the risk and the minimized risk is

$$R(\theta, \hat{\theta}_{\text{Bayes}}) = \int (\theta - \hat{\theta}_{\text{Bayes}})^2 p(\theta \mid \mathbf{x})\, d\theta,$$

which is the posterior variance.

There are naturally a large number of possible Bayes estimates, depending on the choice of loss function. For example,

1. If $L(\theta, \hat{\theta}) = |\theta - \hat{\theta}|$, then the Bayes estimate is the posterior median.
2. If $\theta$ is discrete and $L(\theta, \hat{\theta}) = \mathbb{I}(\theta \neq \hat{\theta})$, then the Bayes estimate is the posterior mode.

*Remark* 10.2. As usual, we can still think of a Bayes estimator as a random variable. In the case of the posterior mean, this would be

$$\hat{\theta}_{\text{Bayes}}(\mathbf{X}) = \int_{\Theta} \theta p(\theta \mid \mathbf{X})\, d\theta,$$

where the posterior density is now a random function since $\mathbf{X}$ is random. This is beyond the scope of this unit. However, in some situations where there is a closed form expression for the posterior mean as a function of the data $\mathbf{x}$, $\hat{\theta}_{\text{Bayes}}(\mathbf{X})$ is somewhat simpler to think about.

## 10.2 Credible intervals

**Definition 10.2** (Symmetric $(1-\alpha)$ credible interval.)**.** In the scalar case, one finds $\theta_1$ and $\theta_2 \in \Theta$ such that

$$\int_{-\infty}^{\theta_1} p(\theta|\mathbf{x}) d\theta = \alpha/2 \quad \text{and} \quad \int_{\theta_2}^{+\infty} p(\theta|\mathbf{x}) d\theta = \alpha/2,$$

and therefore such that

$$\mathbb{P}(\theta \in [\theta_1, \theta_2] | \mathbf{x}) = \int_{\theta_1}^{\theta_2} p(\theta|\mathbf{x}) d\theta = 1 - \alpha.$$

The interval $(\theta_1, \theta_2)$ is called a *symmetric* $(1-\alpha)$ *credible interval.*

These sets are relatively easy to obtain in practice, as this corresponds to finding two quantiles of the posterior distribution. Here is an example where the posterior is a Beta density.

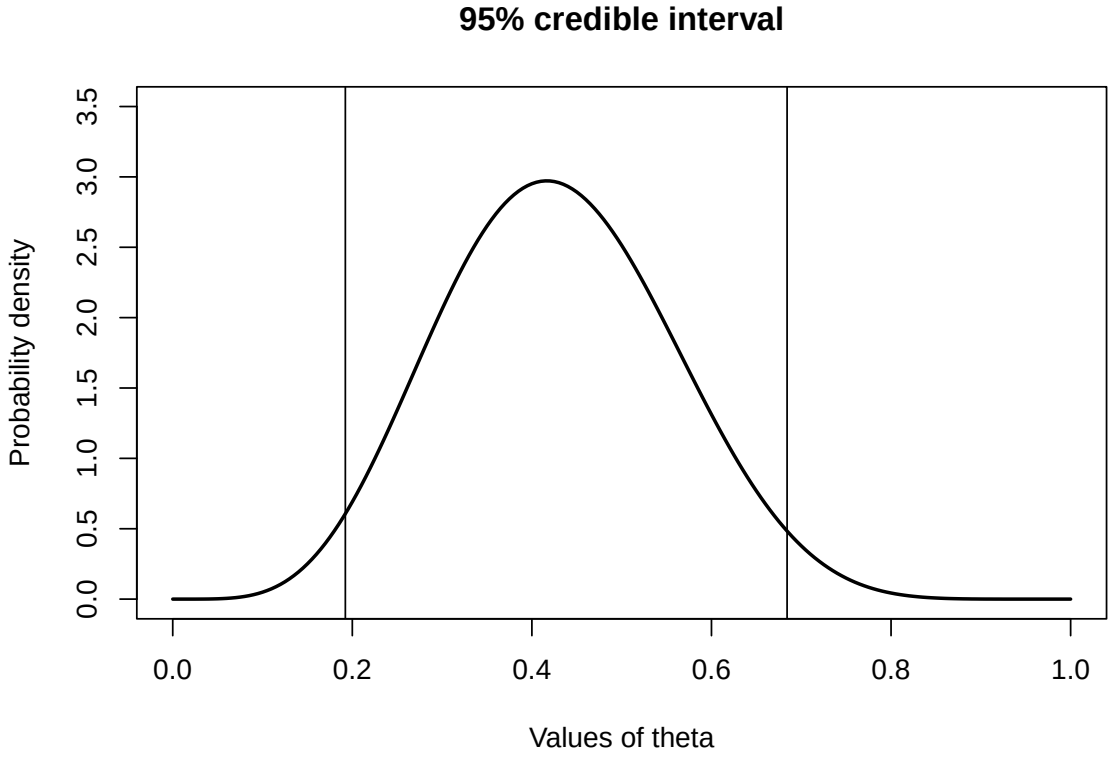
```
thetas <- seq(from = 0, to = 1, by = 0.005)
alpha <- 2
beta <- 2
n <- 10
x <- 4
Pposterior <- dbeta(thetas, x+alpha, beta+n-x)
plot(thetas, Pposterior, ylim=c(0,3.5), type = "l", lty = 1, lwd = 2,
main = "95% credible interval", xlab = "Values of theta", ylab = "Probability density")
aa <- 0.05
theta1 <- qbeta(aa/2, x+alpha, n-x+beta); theta1

## [1] 0.1922324

theta2 <- qbeta(1-aa/2, x+alpha, n-x+beta); theta2

## [1] 0.6842224

abline(v=theta1)
abline(v=theta2)
```




This is difficult to generalize to the multidimensional scenario. One can also consider the following type of credible intervals.

**Definition 10.3** (High posterior density set)**.** The high posterior density ($\mathcal{HPD}(1-\alpha)$) set, is the level set

$$\mathcal{HPD}_v := \{\theta \in \Theta : p(\theta|\mathbf{x}) \geq v\},$$

where $v$ is chosen such that

$$\mathbb{P}\left(\theta \in \mathcal{HPD}_v|\mathbf{x}\right) = \int_{\mathcal{HPD}_v} p(\theta|\mathbf{x})d\theta = 1-\alpha.$$

These confidence interval/sets have a very nice property.

**Theorem 10.1.** *The $\mathcal{HPD}(1-\alpha)$ credible set is the smallest subset of $\Theta$ containing exactly $1-\alpha$ of the total probability.*

These type of sets are more difficult to determine without a computer, and an example of code is provided in the last question of Homework sheet 9. However in some simple scenarios this can be determined analytically.

**Example 10.2.** Assume that the posterior density is a Pareto$(\theta_m, \beta)$ for $\beta, \theta_m > 0$, i.e. $p(\theta \mid \mathbf{x}) = \beta\theta_m^\beta/\theta^{\beta+1}\mathbb{I}\{\theta \geq \theta_m\}$, then the posterior density is non-increasing and we know that a high posterior density set is going to be of the form $[\theta_m, z]$ where $z$ must be chosen such that

$$\int_{\theta_m}^{z} p(\theta \mid \mathbf{x})\mathrm{d}\mathbf{x} = 1-\alpha.$$

In order to find $z$ one needs an expression for the integral, which is possible in this case

$$1-\alpha=\beta\theta_m^\beta\int_{\theta_m}^{z}\theta^{-(\beta+1)}\mathrm{d}\theta=1-\left(\frac{\theta_m}{z}\right)^\beta,$$

which gives

$$z=\frac{\theta_m}{\alpha^{1/\beta}}.$$

In general, one can use the quantile function or numerical approximations in R if the integral is not tractable.

Intuitively, increasing the number of observations should increase our certainty about likely values of $\theta$, meaning that one should expect the posterior density to concentrate on a set of values as $n$ becomes large. Example 10.1 illustrates this. In fact, provided some conditions are satisfied, the posterior density will often look similar to that of a normal distribution as $n\to\infty$: this result is known as the Bernstein–von Mises phenomenon, and its precise statement and proof are beyond the scope of this course (for example one needs to introduce a distance between probability distributions in order to define what is meant by "looks similar"). One can observe one aspect of this phenomenon by investigating the behaviour of, say, the posterior mean and variance.

**Example 10.3.** We have established that for the Beta/Bernoulli model, with statistical parameter $\theta$, the posterior for $n$ observations is a $\mathrm{Beta}(n\bar{x}+\alpha, n(1-\bar{x})+\beta)$ distribution, whose mean and variance are

$$\frac{n\bar{x}+\alpha}{n+\alpha+\beta}=\frac{\bar{x}+\alpha/n}{1+(\alpha+\beta)/n}$$

and

$$\frac{(n\bar{x}+\alpha)[n(1-\bar{x})+\beta]}{(n+\alpha+\beta)^2(n+\alpha+\beta+1)}=\frac{(\bar{x}+\alpha/n)[(1-\bar{x})+\beta/n]}{[1+(\alpha+\beta)/n]^2[n+\alpha+\beta+1]}.$$

*Remark* 10.3. In this example it is easier to see that the Bayes estimator associated with squared error loss is

$$\hat{\theta}_{\mathrm{Bayes}}(\mathbf{X})=\frac{n\bar{X}+\alpha}{n+\alpha+\beta},$$

and hence that $\hat{\theta}_{\mathrm{Bayes}}(\mathbf{X})\to_{\mathbb{P}(\cdot;\theta)}\theta$ by the WLLN and the fact that $\alpha/n$ and $(\alpha+\beta)/n$ both converge to 0 as $n\to\infty$. Similarly, the posterior variance,

$$\frac{(\bar{X}+\alpha/n)[(1-\bar{X})+\beta/n]}{[1+(\alpha+\beta)/n]^2[n+\alpha+\beta+1]}\to_{\mathbb{P}(\cdot;\theta)}0,$$

since $\bar{X}\to_{\mathbb{P}(\cdot;\theta)}\theta$. This is a rudimentary frequentist analysis of a Bayes estimator.

# 11 Bayesian hypothesis testing

Consider a model and $\mathbf{x}$ leading to a posterior distribution/density $p(\theta|\mathbf{x})$ and assume that we would like to test whether

$$H_0 : \theta \in \Theta_0 \qquad \text{vs.} \qquad H_1 : \theta \in \Theta_1.$$

The important difference here to the frequentist setting is that we can calculate

$$\mathbb{P}(\vartheta \in \Theta_0 \mid \mathbf{X} = \mathbf{x}) = \int_{\Theta_0} p(\theta \mid \mathbf{x}) d\theta,$$

and

$$\mathbb{P}(\vartheta \in \Theta_1 \mid \mathbf{X} = \mathbf{x}) = \int_{\Theta_1} p(\theta \mid \mathbf{x}) d\theta.$$

We might, e.g., decide to choose the hypothesis with the largest probability.

We will consider in detail here only the two simple hypotheses setting

$$H_0 : \theta = \theta_0, \qquad \text{vs.} \qquad H_1 : \theta = \theta_1,$$

to contrast it with the Neyman–Pearson approach. In a Bayesian framework one can in fact take into account the cost of making an error, and base our decision on the minimization of this cost. We can construct a table of losses:

| Truth\Decision | $\theta_0$ | $\theta_1$ |
|---|---|---|
| $\theta_0$ | $L_{00}$ | $L_{10}$ |
| $\theta_1$ | $L_{01}$ | $L_{11}$ |

That is, for $i, j \in \{0, 1\}$ we let $L_{ij}$ be the loss for choosing $H_i$ when $\theta = \theta_j$. We assume $L_{ij} > L_{jj}$ for $i \neq j$, that is it is always more costly to make the wrong decision. We then aim to choose the hypothesis with the smallest posterior expected cost.

When we choose $H_0$ we risk the following loss

$$R_0 = L_{00} \times p(\theta_0|\mathbf{x}) + L_{01} \times p(\theta_1|\mathbf{x})$$

and when we choose $H_1$ we risk the loss

$$R_1 = L_{10} \times p(\theta_0|\mathbf{x}) + L_{11} \times p(\theta_1|\mathbf{x}).$$

We will choose the hypothesis that minimizes our expected posterior loss. Hence, we choose $H_1$ (reject $H_0$) if $R_1 < R_0$, i.e.

$$L_{10} \times p(\theta_0|\mathbf{x}) + L_{11} \times p(\theta_1|\mathbf{x}) < L_{00} \times p(\theta_0|\mathbf{x}) + L_{01} \times p(\theta_1|\mathbf{x}),$$

which holds if and only if

$$p(\theta_0 \mid \mathbf{x}) [L_{10} - L_{00}] < p(\theta_1 \mid \mathbf{x}) [L_{01} - L_{11}],$$

and hence if and only if

$$\frac{p(\theta_1 \mid \mathbf{x})}{p(\theta_0 \mid \mathbf{x})} > \frac{L_{10} - L_{00}}{L_{01} - L_{11}}.$$

We can recast this as

$$\frac{p(\mathbf{x} \mid \theta_1) p(\theta_1)}{p(\mathbf{x} \mid \theta_0) p(\theta_0)} > \frac{L_{10} - L_{00}}{L_{01} - L_{11}},$$

or

$$\frac{L(\theta_1;\mathbf{x})}{L(\theta_0;\mathbf{x})} > \frac{p(\theta_0)\,[L_{10}-L_{00}]}{p(\theta_1)\,[L_{01}-L_{11}]}.$$

This is exactly of the same form as the Neyman–Pearson test, except that the "critical value" is chosen according to our prior and our assessment of the risk of taking the wrong decision.

In particular,

1. the greater the cost of a Type I error ($L_{10}$), the higher the threshold,
2. the greater the cost of a Type II error ($L_{01}$) the smaller the threshold,
3. the greater the prior probability of $H_0$, $p(\theta_0)$, the higher the threshold,
4. the greater the prior probability of $H_1$, $p(\theta_1)$, the smaller the threshold.

# 12 Case Studies

Since statistics is more than just deriving estimators, confidence intervals, credible intervals and hypthosis tests on paper, here are a number of case studies in R where we can explore the content of the course concepts in more detail. RMkdown source files for these case studies can be found on Blackboard.

1. Hypothesis testing
2. Assessing hypothesis tests
3. Generalised Likelihood ratio tests
4. Goodness-of-fit tests
5. Bayesian Inference

# Case study 6: Hypothesis testing

Juliette Unwin

## Recap of the main terms involved in hypothesis testing for a parameter $\theta \in \mathbb{R}$

### The two hypotheses

We have realizations $(x_1, \ldots, x_n) := \mathbf{x}$ of $(X_1, \ldots, X_n) := \mathbf{X} \sim f(\cdot; \theta)$. We want to assess which of two statements (*hypotheses*) about the parameter $\theta$ is more consistent with these realizations. The first hypothesis is the *null* hypothesis, $H_0$ and we look at its simplest possible form:

$$H_0 : \theta = \theta_0.$$

The second hypothesis is the *alternative* hypothesis and we assume that it has one of the following forms

$$H_1 : \theta > \theta_0 \quad \text{or} \quad H_1 : \theta < \theta_0 \quad \text{or} \quad H_1 : \theta \neq \theta_0 \quad \text{or} \quad H_1 : \theta = \theta_1.$$

The choice of the alternative depends on the problem we have to study.

### The test statistic

Once we formed the test, we need to choose an appropriate test statistic, $T(\mathbf{X})$. In Statistics 2 we will introduce specific criteria for a good test statistic. For now, by appropriate we refer to a test statistic:

(1) whose distribution under the null hypothesis is known or can be approximated and

(2) an 'extreme' value of the observed test statistic, $T(\mathbf{x}) := t_{obs}$, suggests that $H_0$ is unlikely to be true and $H_1$ is more consistent with these realizations.

### Type I and Type II errors

When we perform a hypothesis test and we come up with a decision we may commit one of the following errors:

**Type I error**: Reject $H_0$ while it is in fact true

**Type II error**: Not reject $H_0$ while it is in fact false.

Obviously our goal is to minimize both of them but as there is an unavoidable trade-off between the two, we need to decide which of the two is the least acceptable and make sure that we control it. This is Type I error (why?).

### The critical region

The critical (or rejection) region, $R$, are those values of $\mathbf{x}$ that would lead in rejecting the $H_0$, i.e.

$$R : \{\mathbf{x} \in \mathbb{R}^n : T(\mathbf{x}) \text{ is 'extreme' if } H_0 \text{ is true } \}.$$

To control Type I error we determine $R$ so that

$$\mathbb{P}[T(\mathbf{X}) \in R | H_0 \text{ is true}] = \alpha,$$

where $\alpha \in (0, 1)$ is the significance level (usually it takes values 0.05 or 0.01 but in general its value is strongly related to the cost that comes with wrongly rejecting the $H_0$.)

The rejection region is now the tool we have for deciding between the two hypothesis. So, we now have a complete test.

### p-value

Having determined the critical value we can then compute the *p-value* of the test which is

$$p - value := \mathbb{P}[T(\mathbf{X}) \text{ is at least as extreme as } T(\mathbf{x}) | H_0 \text{ is true}].$$

### Power of the test

The power of the test is a measure for the ability of the test to identify when the $H_0$ is not true. It is defined as:

$$\text{power} = \mathbb{P}[T(\mathbf{X} \in R | H_1 \text{ is true})] = 1 - \mathbb{P}(\text{Type II error})$$

It is important to know the power of a test, as the way we formed the test we make it harder to reject $H_0$ than retaining it. So, we should know how much this (biased) method affects the test in wrongly deciding in favor of the $H_0$.

## Practical Application

After a first year exam, a student sends an email to the lecturer of the unit complaining about the exam paper. The student claimed that the exam was much harder compared to previous years. The lecturer has not started marking the 300 exam scripts yet, so he decides to perform a hypothesis test in order to check the student's claim. To do so, he randomly selects 20 scripts, marks them and records the marks in R:

```
marks<-c(54, 58, 84, 75, 59, 33, 52, 78, 51, 56, 80, 37, 67, 61, 30, 71, 55, 68, 36, 89)
```

A reasonable model for these data $(x_1, \ldots, x_{20})$ is the normal distribution, i.e. the lecturer assumes that $(X_1, \ldots, X_n) \sim \mathcal{N}(\mu, \sigma^2)$. He also knows that for the previous 5 years the average mark was 64 and hence we wants to perform the following test:

$$H_0 : \mu = 64 \quad \text{Vs} \quad H_1 : \mu < 64.$$

If there is no information for the variance, a test statistic that satisfies the criteria mentioned above is ,

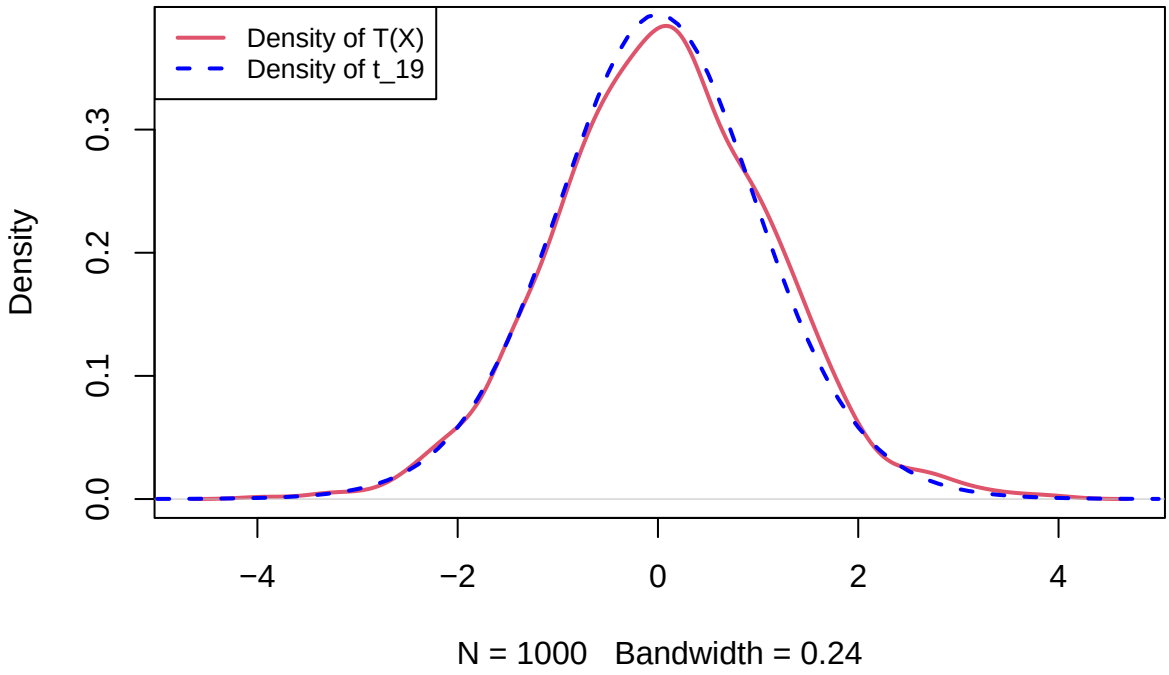
$$T(\mathbf{X}) = \frac{\sqrt{n}(\overline{X} - 64)}{S},$$

whose distribution is $t_{n-1}$ when $H_0$ holds.

We can verify the distribution of $T(\mathbf{X})$ by simulations (repeated experiments) and plotting its density against the density of $t_{n-1}$:

```
set.seed(56382765)
N=1000
n=20
tstat<-rep(NA,N)
for(i in 1:N){
```

```
  x<-rnorm(n,64,15) #assume that sigma=15
  s=sqrt(1/(n-1)*(sum((x-mean(x))^2)) )
  tstat[i]=sqrt(n)*(mean(x)-64)/s
}
plot(density(tstat),col=2,lwd=2,main="")
lines(seq(-5,5,by=0.1),dt(seq(-5,5,by=0.1),n-1),col='blue',lty=2,lwd=2)
#lines(density(rt(1000,n-1,col='blue',lty=2,lwd=2) can also
#be used for plotting the density of a sample form t_19
legend("topleft",legend=c("Density of T(X)", "Density of t_19"),
       cex=0.8,lty=c(1,2),lwd=c(2,2),col=c(2,'blue'))
```




Returning to the test, assume that $\alpha = 0.05$. The critical region will be of the form

$$R = \{\mathbf{x} : T(\mathbf{x}) < c\},$$

where $c$ will be chosen so that

$$\mathbb{P}[T(\mathbf{X}) < c; \mu = 64] = 0.05.$$

Note that here we reject for small values of $T(\mathbf{X})$ (as implied by $T(\mathbf{X}) < c$) as 'extreme' values for the $H_0$ are those who will be more consistent with $H_1$.

We can compute the critical value, $c$, as follows:

```
n<-20
crit<-qt(0.05,df=n-1)
crit
```

```
## [1] -1.729133
```

Hence, we will reject when the observed value of the test statistic is less than $-1.729$, i.e.

$$R = \{\mathbf{x} : T(\mathbf{x}) < -1.729\}.$$

We compute the observed test statistic:

```r
s<-sqrt(1/(n-1)*(sum((marks-mean(marks))^2)) )
tobs<-sqrt(n)*(mean(marks)-64)/s
tobs
```

```
## [1] -1.124126
```

This value does not fall in the rejection region, hence we do not have enough evidence from the data to reject the null hypothesis. This means the student's claim of a hard exam is not supported by the data at significance level 0.05.

In order to find the minimum value of $\alpha$ for which we would reject the $H_0$, we need to compute the p-value:

```r
pvalue<-pt(tobs,n-1)
pvalue
```

```
## [1] 0.1374809
```

This is relatively high, which indicates that we would not reject $H_0$ easily.

Now, we will compute the power for the following test:

$$H_0 : \mu = 64 \quad \text{Vs} \quad H_1 : \mu = 62.$$

By definition,

$$\text{power} = \mathbb{P}[T(\mathbf{X}) < -1.729 | H_1 \text{ is true}] = \mathbb{P}[T(\mathbf{X}) < -1.729 | \mu = 62].$$

The exact way to compute this probability is to use the non-central student t distribution, $t_\nu(\delta)$, which is defined as:

If $\frac{Z}{\sqrt{\mathcal{X}^2_\nu/\nu}} \sim t_\nu$ then the random variable $\frac{Z+\delta}{\sqrt{\mathcal{X}^2_\nu/\nu}} \sim t_\nu(\delta)$

Under $H_1$, we have that $\frac{\overline{X}-62}{S/\sqrt{n}} \sim t_{n-1}$, so

$$\mathbb{P}[T(X) < -1.729 | \mu = 62] = \mathbb{P}\left[\frac{\overline{X} - 64}{S/\sqrt{n}} < -1.729 | \mu = 62\right] =$$

$$\mathbb{P}\left[\frac{\overline{X} - 62 - 2}{S/\sqrt{n}} < -1.729 | \mu = 62\right] = \mathbb{P}\left[T' < -1.729\right],$$

where $T' \sim t_{n-1}(-2)$.

This can be computed directly in R as follows,

```r
power<-pt(crit,df=n-1,ncp=-2*sqrt(n)/s) #ncp is the non-centrality parameter
power
```

```
## [1] 0.1270542
```

Alternatively, we could assume that $S \approx s$ and simply approximate the power using the central t-distribution (or standard normal):

$$\text{power} = \mathbb{P}\left[\frac{\overline{X} - 64}{S/\sqrt{n}} < -1.729 | \mu = 62\right] = \mathbb{P}\left[\frac{\overline{X} - 62}{S/\sqrt{n}} < -1.729 + \frac{2}{S/\sqrt{n}} | \mu = 62\right] \approx \mathbb{P}\left[\tilde{T} < -1.729 + \frac{2}{S/\sqrt{n}}\right],$$

where $\tilde{T} \sim t_{n-1}$.

```r
power<-pt(crit+2*sqrt(n)/s,df=n-1)
power
```

```
## [1] 0.1212573
```

which is close to the exact result.

Let's see now how the sample size would modify the result. Assume that we have each mark recorded twice:

```
marks2<-c(marks,marks)
n=40
s<-sqrt(1/(n-1)*(sum((marks-mean(marks))^2)) )
tobs<-sqrt(n)*(mean(marks)-64)/s
paste("tobs=", round(tobs,4))
```

```
## [1] "tobs= -2.2776"
```

```
pvalue<-pt(tobs,df=n-1)
paste("pvalue=",round(pvalue,4))
```

```
## [1] "pvalue= 0.0142"
```

```
power<-pt(crit,df=n-1,ncp=-2*sqrt(n)/s)
paste("power=",round(power,4))
```

```
## [1] "power= 0.259"
```

# Case study 7: Assessing hypothesis tests

## Signifigance level and power

Juliette Unwin

In this case study we will investigate a hypothesis test for the parameter $p$ of observations from Geometric($p$).

Let $\mathbf{x} = (x_1, .., x_n)$ be realizations of i.i.d. random variables $X_1, ..., X_n$ where $X_i \sim$Geometric($p$). We want to perform the following test:

$$H_0 : p = \frac{1}{2} \quad \text{Vs} \quad H_1 : p = p_1, \quad \text{for some } p_1 < \frac{1}{2}.$$

**Test statistic**

We know that if we want to form a hypothesis test of a specified significance level (say $\alpha = 0.05$), the Neyman-Pearson test statistic,

$$T_{NP}(\mathbf{x}) = \frac{L(p_1; \mathbf{x})}{L(p_0; \mathbf{x})} = \left(\frac{1-p_1}{1-p_0}\right)^{(\sum_{i=1}^n x_i)-n} \left(\frac{p_1}{p_0}\right)^n,$$

is optimal, in the sense that it is the most powerful among all tests of level $\alpha$.

In section 19 of the notes, we show that $Y = \sum_{i=1}^{n} X_i$ is equivalent to $T_{NP}$. In addition, it has the advantage that it has a Negative Binomial distribution with parameters $n$ and $p$, in contrast to $T_{NP}$ whose distribution is not known. $Y$ counts the number of Bernoulli trials until we obtain $n$ successes and its pmf is given by

$$p_Y(y) = \binom{y-1}{n-1} p^n (1-p)^{y-n}, \ y \in \{n, n+1, ...\}$$

We write $Y \sim \text{NegBin}(n, p)$. Hence, under the null hypothesis the distribution of $Y$ is $\text{NegBin}\left(n, \frac{1}{2}\right)$.

**Type I error - Significance level**

The rejection region is of the form

$$R = \left\{ \mathbf{x} : \sum_{i=1}^{n} x_i \geq c \right\}.$$

If we choose the significance level to be $\alpha = 0.05$, we compute $c$ so that

$$\mathbb{P}\left(Y \geq c; p = \frac{1}{2}\right) \leq 0.05,$$

or equivalently

$$\mathbb{P}\left(Y \leq c - 1; p = \frac{1}{2}\right) \geq 0.95, \text{ for } c \in \mathbb{Z}^+.$$

First, we create the probability mass function (*dNegBin*), the probability distribution function (*pNegBin*) and the quantile function (*qNegBin*) for the Negative Binomial in R (alternatively we could use dnbinom, pnbinom, qnbinom but we need to be careful with the choice of the parameters as it is defined differently compared to the way it is described here.)

```r
dNegBin<-function(y,n,p){
  stopifnot(all(p>=0 & p<=1 & y>=n & as.integer(n) & as.integer(y) & n>0))
  choose(y-1,n-1)*p^n*(1-p)^(y-n)
}

pNegBin<-function(y,n,p){
  prob<-0
  for(i in n:floor(y)){
    prob=prob+dNegBin(i,n,p)
  }
  return(prob)
}

#The quantile function is not perfectly coded here but it works
#for the needs of this case study
qNegBin<-function(q,n,p){
  x<-seq(n,n+500,by=1)
  probx<-sapply(x, function(u) pNegBin(u,n,p))
  min(x[probx>=q])
}
```

To compute the critical value ($crit$) for $\alpha = 0.05$ we use qNegBin. Assume we have a sample of size $n = 25$.

```r
crit<-qNegBin(q=1-0.05,n=25,p=0.5) +1
pcrit<-round(pNegBin(crit-1,25,0.5),3) #P(Y<=crit-1) rounded to 3 s.f.
 c(c=crit,  "P(Y>=c)"=1-pcrit)
```

```
##       c P(Y>=c)
##  63.000   0.049
```

Thus, we reject the null hypothesis when $\sum_{i=1}^{25} x_i \geq 63$.

### Power function and power of the test

Now we want to compute the power of the test for a specific value $p_1$. First we write the code for the power function,

$$\pi(p; Y, c) = \mathbb{P}(Y \geq c; p)$$

```r
powerf<-function(c,n,p){
  1-pNegBin(c-1,n,p)
}
```

We can confirm that for $p = 0.5$, this is (approximately) the significance level.

```r
powerf(crit,25,0.5)
```

```
## [1] 0.04897722
```

If the alternative is $H_1 : p = \frac{1}{4}$, the power of the test is

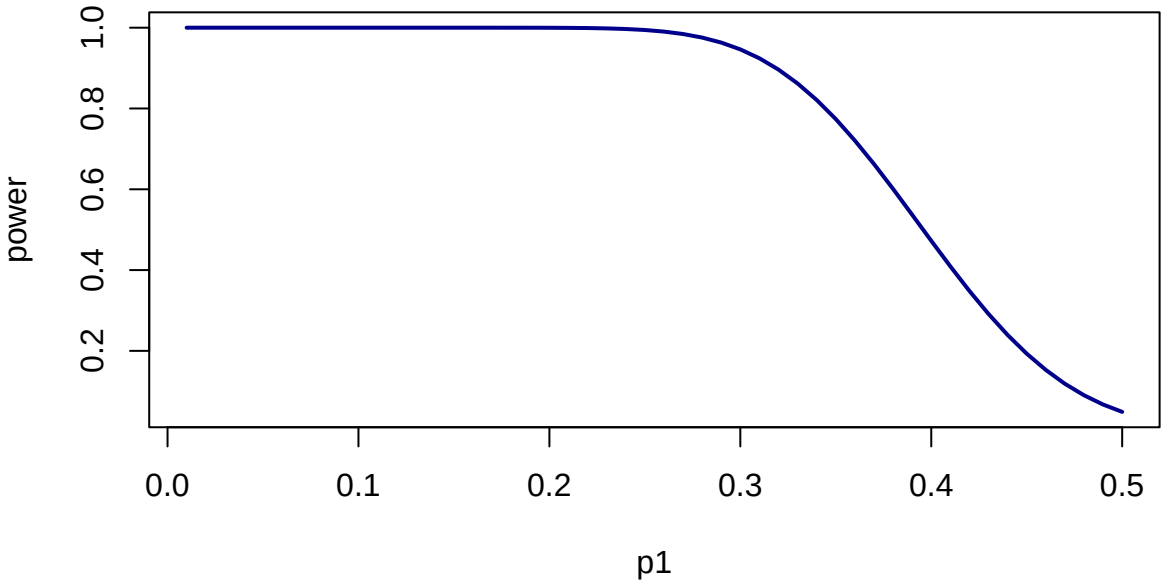
```r
powerf(crit,25,1/4)
```

```
## [1] 0.9943004
```

### Plotting the power in terms of $p_1$ and $n$

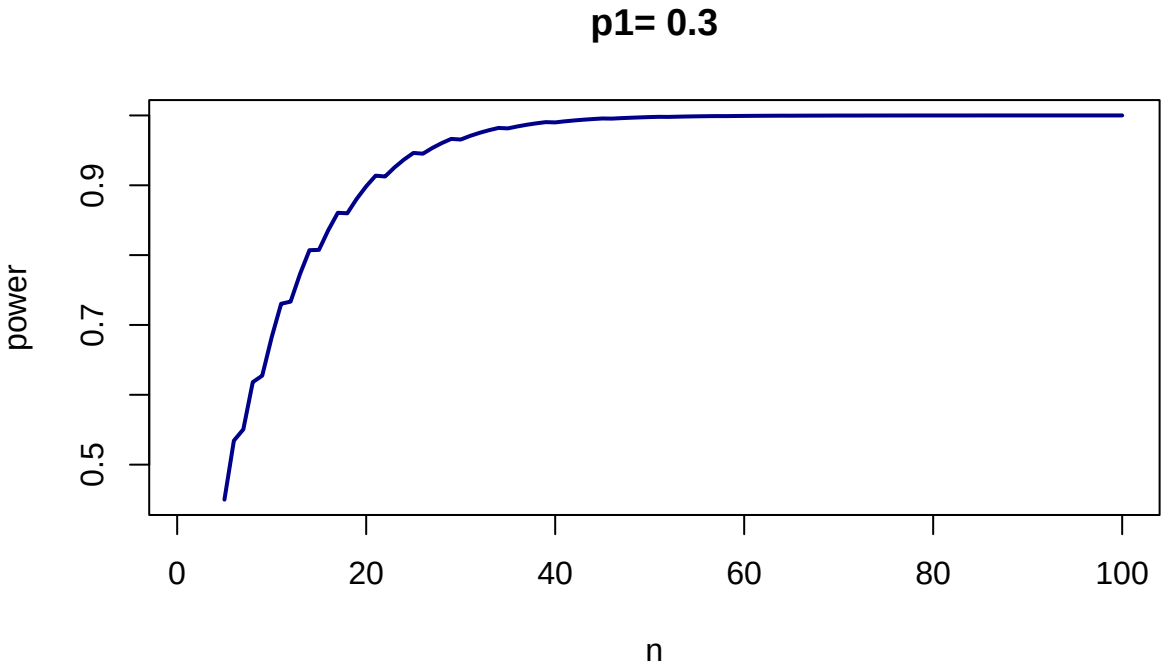
We plot the power function for various values of $p_1$ to see how the value of the parameter under the alternative hypothesis affects the power of the test.

```
p1<-seq(0.01,0.5,0.01)
plot(p1,powerf(crit,n=25,p1),type='l',col='darkblue',lwd=2,ylab='power')
```




We can also illustrate graphically the effect of the sample size on the power of the test:

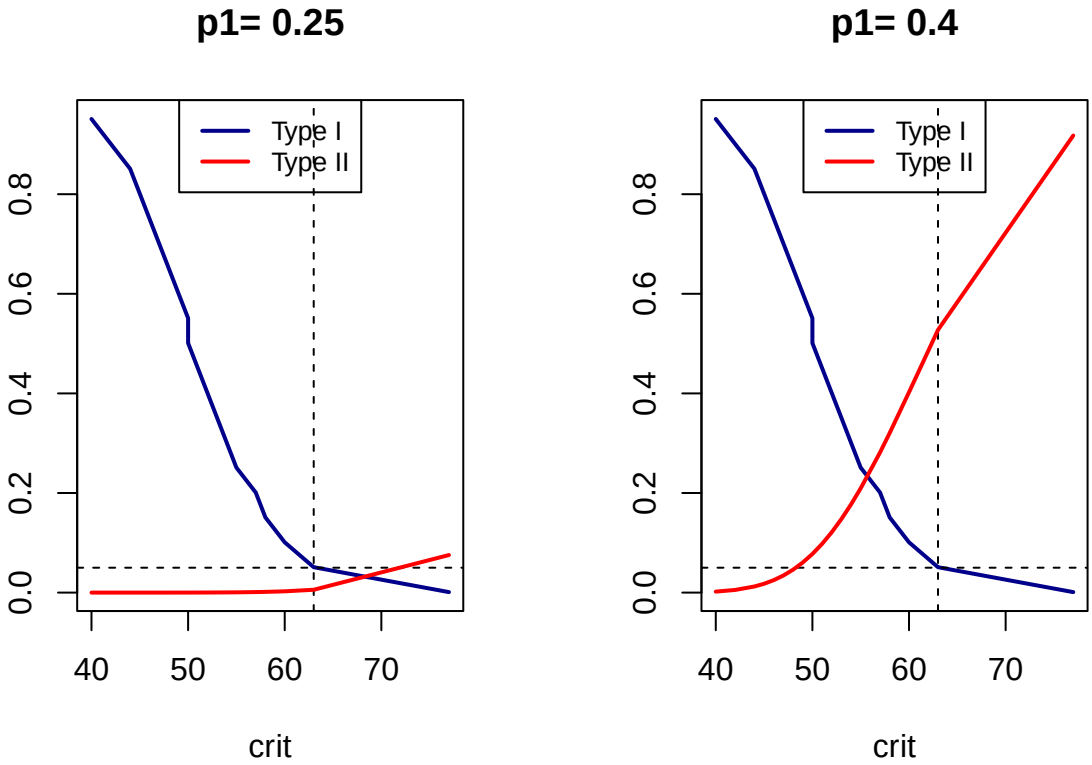
```
p1=0.3
crit<-rep(NA,100)
power<-rep(NA,100)
for (n in 5:100){
crit[n]<-qNegBin(q=1-0.05,n,p=0.5) +1
power[n]<-powerf(crit[n],n,p1)
}
plot(1:100,power,type='l',lwd=2,col='darkblue',xlab="n", main=paste("p1=",p1))
```




The sample size plays an important role when designing tests as it is the main tool we have to increase the power of a test and hence reduce Type II error.

**Trade-off between type I and type II errors**

```
par(mfrow=c(1,2))
n=25
p0=0.5
p1=c(0.25, 0.4)
for(i in p1){
alpha<-seq(0.001,1,by=0.05)
crit<-sapply(alpha, function(a) qNegBin(q=1-a,n=25,p=p0) +1)
power<-sapply(crit, function(c) powerf(c,n,i))
typeI<-alpha
typeII<-1-power
plot(crit, typeI, type='l',col='darkblue',lwd=2,main=paste("p1=",i), ylab="")
abline(h=0.05,lty=2)
abline(v=63,lty=2)
lines(crit, typeII,type='l',col='red',lwd=2)
legend("top",legend=c("Type I","Type II"),lwd=c(2,2), col=c("darkblue", "red"),cex=0.8)
}
```




It is obvious from these two graphs that there is a trade-off between the two errors. This explains why we usually fix Type I error and look for tests that would minimize Type II error. Also we see that the closer $p_1$ is to $p_0$, the hardest it gets for the test to identify when $H_1$ is true as the type II error increases and hence the power(=1-type II error) decreases.

# Case study 8: Generalised Likelihood ratio tests

Juliette Unwin

## Linear Regression Model

Consider that we aim to model a set of observations $Y_1, ..., Y_n$ of random variable $Y$ as follows:

$$Y_i = \beta_o + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + \epsilon_i, \quad i = 1, 2, ...n,$$

where

- $x_{ij}$ is the $i-th$ observation of a random variable (covariate) $X_j$
- $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$ and
- $\beta_0, \ldots, \beta_k$ are the unknown parameters of the model.

This model is called a multiple regression model. $Y$ is the response variable and $X_1, .., X_k$ are the predictors. We assume that $Y_i$ are independent given the observed values of $X_1, .., X_k$. The model can alternatively be written as

$$\mathbb{E}(Y_i | x_{i1}, ..., x_{ik}) = \beta_o + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} = \sum_{j=0}^{k} \beta_j x_{ij},$$

where we assume that $x_{i0} = 1 \ \forall i = 1, .., n$.

It is easy to check that $Y_i \sim \mathcal{N}(\sum_{j=0}^{k} \beta_j x_{ij}, \sigma^2)$. Hence, the likelihood of $\boldsymbol{\theta} = (\beta_0, ..., \beta_k, \sigma^2)$ given $\mathbf{y} := (y_1, ..., y_n)$ is

$$L(\boldsymbol{\theta}; \mathbf{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta_1 x_{i1} - ... - \beta_k x_{ik})^2 \right\}$$

We can rewrite the model in matrix form,

$$Y = X\boldsymbol{\beta} + \epsilon,$$

where,

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} x_{10} & x_{11} & \ldots & x_{1k} \\ x_{20} & x_{21} & \ldots & x_{2k} \\ & \vdots & & \\ x_{n0} & x_{n1} & \ldots & x_{nk} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \text{and} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

The likelihood then becomes

$$L(\boldsymbol{\theta}; \mathbf{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left\{ -\frac{1}{2\sigma^2} (Y - X\boldsymbol{\beta})^T (Y - X\boldsymbol{\beta}) \right\}.$$

## The MLE estimators

Denote by $\hat{\boldsymbol{\beta}} := (\hat{\boldsymbol{\beta}}_0, ..., \hat{\beta}_k)$ the maximum likelihood estimator (MLE) of $\boldsymbol{\beta}$ and by $\hat{\sigma}^2$ the MLE of $\sigma^2$. We are given that

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y \text{ and } \hat{\sigma}^2 = \frac{S^2}{n}, \tag{1}$$

where $X^T$ is the transpose of $X$ and $S^2 = (Y - X\hat{\boldsymbol{\beta}})^T (Y - X\hat{\boldsymbol{\beta}})$.

In this case study we will focus in testing whether some of the parameters, $\beta_j$, are equal to 0 or not. Note that if a coefficient, $\beta_j$=0, the the corresponding predictor $X_j$ does not contribute to the model and hence it is an indication that it is not a helpful variable to explain the variability of $Y$. The test is of the form:

$$H_0 : \beta_{i_1} = ... = \beta_{i_l} = 0 \quad \text{Vs} \quad H_1 : \text{at least one of } \beta_{i_1}, ..., \beta_{i_l} \neq 0.$$

We will use the Generalised Likelihood Ratio (GLR) test statistic. This is given by

$$\Lambda_n(\mathbf{x}) := \frac{\sup_{H_0} f_n(\mathbf{y}; \boldsymbol{\beta}, \sigma^2)}{\sup_{\boldsymbol{\beta} \in \mathbb{R}^{k+1}} f_n(\mathbf{y}; \boldsymbol{\beta}, \sigma^2)} = \frac{L(\hat{\boldsymbol{\beta}}_0, \hat{\sigma}_0^2; \mathbf{y})}{L(\hat{\boldsymbol{\beta}}_{MLE}, \hat{\sigma}_{MLE}^2; \mathbf{y})},$$

where

- $\hat{\boldsymbol{\beta}}_0$ and $\hat{\sigma}_0^2$ are the MLE of $\boldsymbol{\beta}$ and $\sigma^2$ under the null hypothesis and
- $\hat{\boldsymbol{\beta}}_{MLE}$ and $\hat{\sigma}_{MLE}^2$ are the MLE of $\boldsymbol{\beta}$ and $\sigma^2$ for the full model (no restrictions on $\beta_i$).

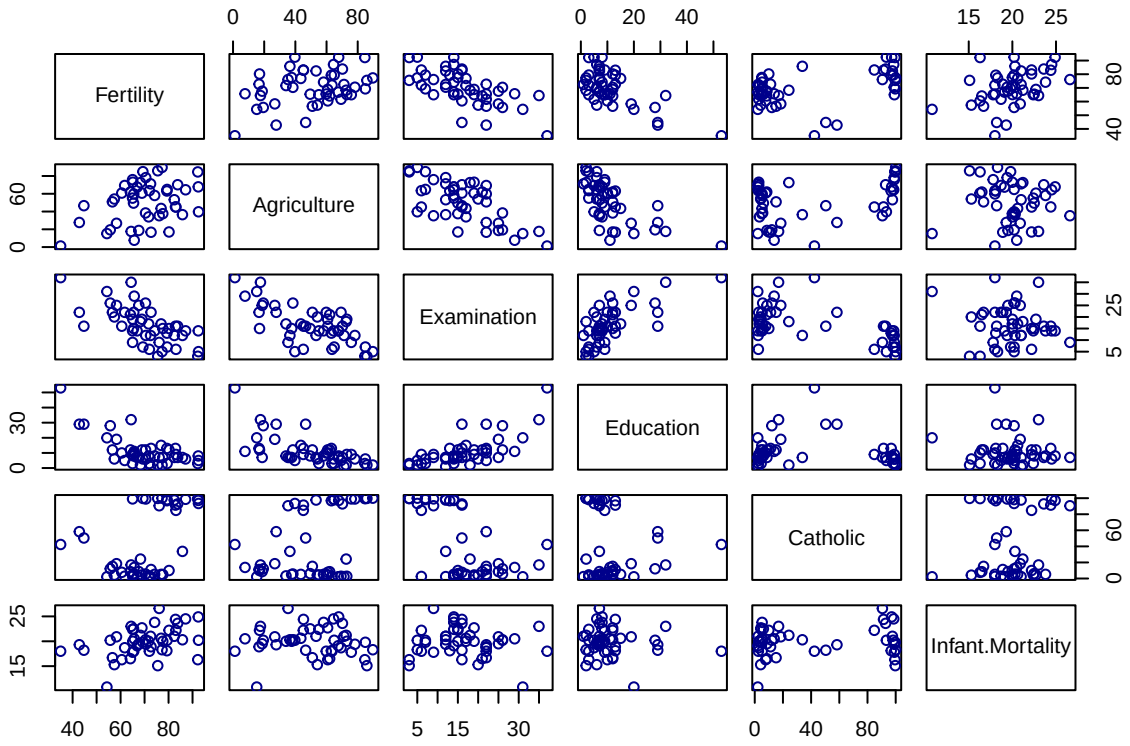
We know that $T = -2 \log \Lambda_n$ has (approximately) a $\mathcal{X}_d^2$ distribution, where the degrees of freedom are $d = \dim(\Theta) - \dim(\Theta_0) = (k + 2) - (k + 2 - l) = l$.

## Data application

Let's see how this works in practice. We will use the *swiss* dataset which contain the following measurements for 47 French-speaking "provinces" in Switzerland at about 1888:

- $Y$: Fertility Ig, 'common standardized fertility measure'
- $X_1$: Agriculture % of males involved in agriculture as occupation
- $X_2$: Examination % draftees receiving highest mark on army examination
- $X_3$: Education % education beyond primary school for draftees.
- $X_4$: Catholic % 'catholic' (as opposed to 'protestant').
- $X_5$: Infant.Mortality live births who live less than 1 year.

```
#View and explore data
#View(swiss)
#?swiss
plot(swiss,col='darkblue')
```




From the graphs we see that some of the predictors are not uncorrelated which would normally require further investigation as it may cause colinearity problems. For this case study we will ignore this fact.

Consider the full model:

$$Y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_5 x_{i5} + \epsilon_i.$$

Assume that we want to test whether variables Argiculture and Examination are statistically significant for the response, $Y$. Hence, we want to perform the test with null hypothesis

$$H_0 : \beta_1 = \beta_2 = 0.$$

We first need to compute the MLE for $\boldsymbol{\beta}$ and $\sigma^2$ under the restricted model (H_0) and for the full model. We start with the full model and use the MLE estimators given in (1).

```
Xfull<-cbind(1,as.matrix(swiss[,2:6])) # we form the matrix X. Note that we add a
                                        #column of 1 to include the intercept term b_0
Y<-swiss[,1]
betaMLE<-solve(t(Xfull)%*%Xfull) %*% t(Xfull) %*%Y
betaMLE
```

```
##                          [,1]
##                  66.9151817
## Agriculture      -0.1721140
## Examination      -0.2580082
## Education        -0.8709401
## Catholic          0.1041153
## Infant.Mortality  1.0770481
```

Now, under the null hypothesis the parameters $\beta_1$ and $\beta_2$ are 0. Hence, we have the restricted model

$$Y_i = Y_i = \beta_0 + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \epsilon_i.$$

So, we can compute the MLE for this model now by adjusting the $X$ matrix.

```
Xnull<-cbind(1,as.matrix(swiss[,4:6])) #exclude the 2 variables which have
                                       #coefficients equal to 0 under H_0
Y<-swiss[,1]
betaMLE0<-solve(t(Xnull)%*%Xnull) %*% t(Xnull) %*%Y
betaMLE0
```

```
##                          [,1]
##                   48.67707330
## Education         -0.75924577
## Catholic           0.09606607
## Infant.Mortality   1.29614813
```

We can now compute the test statistic.

```
ell<-function(Y,X,beta) {
  n<-length(Y)
  S2=t(Y-X%*%beta)%*%(Y-X%*%beta)
sigma2<-S2/n
  1/(2*pi*sigma2)^(n/2)*exp(-1/(2*sigma2)*S2)
}
 Lambda<-ell(Y,Xnull,betaMLE0)/ell(Y,Xfull,betaMLE)
 T=-2*log(Lambda)
 pval<-pchisq(T,df=2,lower.tail=FALSE)
 c(T=T,pvalue=pval)
```

```
##          T     pvalue
## 6.59687492 0.03694084
```

The p-value is smaller than 0.05 so at this significance level we reject the null hypothesis, which means that at least for one of the two variables, Agriculture and/or Examination, we do not have enough evidence to support that they are not statistically important to explain the variability of Fertility. We could repeat the process to test each of the variables separately.

Note that in R we can perform the test using the *lrtest* function from the lmtest package. Lets see if we can confirm the result we obtained above:

```
model1<-lm(Fertility~.,data=swiss)
model2<-lm(Fertility~Education+Catholic+Infant.Mortality,data=swiss)
library(lmtest)
lrtest(model1,model2)
```

```
## Likelihood ratio test
##
## Model 1: Fertility ~ Agriculture + Examination + Education + Catholic +
##     Infant.Mortality
## Model 2: Fertility ~ Education + Catholic + Infant.Mortality
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   7 -156.04
## 2   5 -159.33 -2 6.5969    0.03694 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# Case study 9: Goodness-of-fit tests

Juliette Unwin

## 1. Goodness-of-fit for Exponential distribution

(Make sure you have read the relevant document and covered the material for week 9.)

We will show how to perform the Pearson goodness-of-fit test for a continuous random variable. Here we simulate observations from Exponential(0.5) and we will test whether the Pearson test statistic will be able to confirm this.

Recall that the Pearson test statistic is given by

$$T_{Pearson}(\mathbf{X}) = \sum_{i=1}^{m} \frac{(O_i - E_i)^2}{E_i},$$

where $m$ is the number of intervals used to quantize the observed values, $O_i$ is the number of the observed values in category (interval) $i$ and $E_i$ is the expected number of observations in category $i$, when the null hypothesis is true. Note that we choose $m$ so that all $E_i \geq 5$.

Under the null hypothesis, $T_{Pearson}$ converges in distribution to $\mathcal{X}^2_{m-1}$.

```
set.seed(4687681)
n=100
data<-rexp(n,0.5)
summary(data)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.001439 0.749395 1.463424 2.236118 3.345293 9.651472
```

First trial to quantize data:

```
breaks<-c(-Inf,seq(0,9,by=1),Inf)
Obs<-table(cut(data,breaks))
Obs
```

```
##
## (-Inf,0]    (0,1]    (1,2]    (2,3]    (3,4]    (4,5]    (5,6]    (6,7]
##        0       39       17       13       14        7        6        1
##    (7,8]    (8,9] (9, Inf]
##        1        0        2
```

```
mle<-1/mean(data) # we use the mle (instead of the optimal value) for simplicity
prop<-pexp(breaks[-1],mle)-pexp(breaks[-length(breaks)],mle)
Exp<-n*prop
round(cbind(Obs,Exp),1)
```

```
##          Obs  Exp
## (-Inf,0]   0  0.0
## (0,1]     39 36.1
## (1,2]     17 23.1
```

```
## (2,3]     13 14.7
## (3,4]     14  9.4
## (4,5]      7  6.0
## (5,6]      6  3.9
## (6,7]      1  2.5
## (7,8]      1  1.6
## (8,9]      0  1.0
## (9, Inf]   2  1.8
```

We observe that we have some $E_i < 5$. So, we need to try use different intervals as our classes.

```
breaks<-c(-Inf,seq(0.5,3,by=0.5),4,5,Inf)
Obs<-table(cut(data,breaks))
Obs
```

```
##
## (-Inf,0.5]    (0.5,1]    (1,1.5]    (1.5,2]    (2,2.5]    (2.5,3]      (3,4]
##         20         19         11          6          5          8         14
##      (4,5]   (5, Inf]
##          7         10
```

```
mle<-1/mean(data)
prop<-pexp(breaks[-1],mle)-pexp(breaks[-length(breaks)],mle)
Exp<-n*prop
round(cbind(Obs,Exp),1)
```

```
##            Obs  Exp
## (-Inf,0.5]  20 20.0
## (0.5,1]     19 16.0
## (1,1.5]     11 12.8
## (1.5,2]      6 10.2
## (2,2.5]      5  8.2
## (2.5,3]      8  6.6
## (3,4]       14  9.4
## (4,5]        7  6.0
## (5, Inf]    10 10.7
```

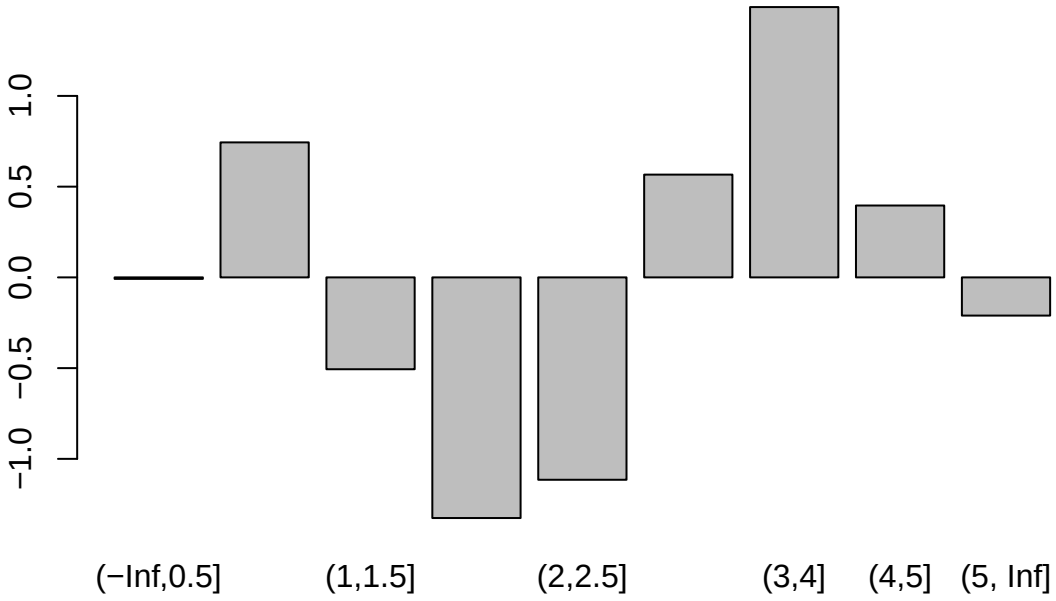
Now all $E_i >= 5$, so we keep this classification and hence $m = 9$. We can now compute the test statistic and the p-value range (range as we used the MLE for $\lambda$ instead of $\lambda_{opt}$):

```
Test<-sum((Obs-Exp)^2/Exp)
pval_up<-1-pchisq(Test,df=length(breaks)-1-1) #d.f=m-1
pval_low<-1-pchisq(Test,df=length(breaks)-1-1-1) #d.f=m-1-1 as we use mle instead of optimal value
c(pval_low=pval_low,pval_up=pval_up)
```

```
## pval_low  pval_up
## 0.476854 0.585535
```

If we want to observe where biggest differences occur we can use *barplot*:

```
barplot((Obs-Exp)/sqrt(Exp),main="Hanging Chi-gram")
```




## 2. Confirming that the p-value is uniformly distributed in the interval (0,1)

In the notes (see Remark 20.2) we mention that for a simple null hypothesis, i.e. $H_0 : \theta = \theta_0$, and assuming $T(\mathbf{X})$ is a continuous random variable with $X \sim f(.;\theta_0)$, the distribution of $p(\mathbf{X})$ is uniform under the null hypothesis. We will confirm this for the Normal distribution when $\sigma^2 = 4$ is known. Consider the test:

$$H_0 : \mu = 0 \quad \text{Vs} \quad H_1 : \mu > 0.$$

The optimal test statistic (in the Neyman-Pearson sense) is $T(\mathbf{X}) = \bar{X}$ and the p-value is

$$p(\mathbf{x}) = \mathbb{P}(\bar{X} \geq \bar{x}; \mu = 0),$$

where $\bar{x}$ is the observed test statistic.

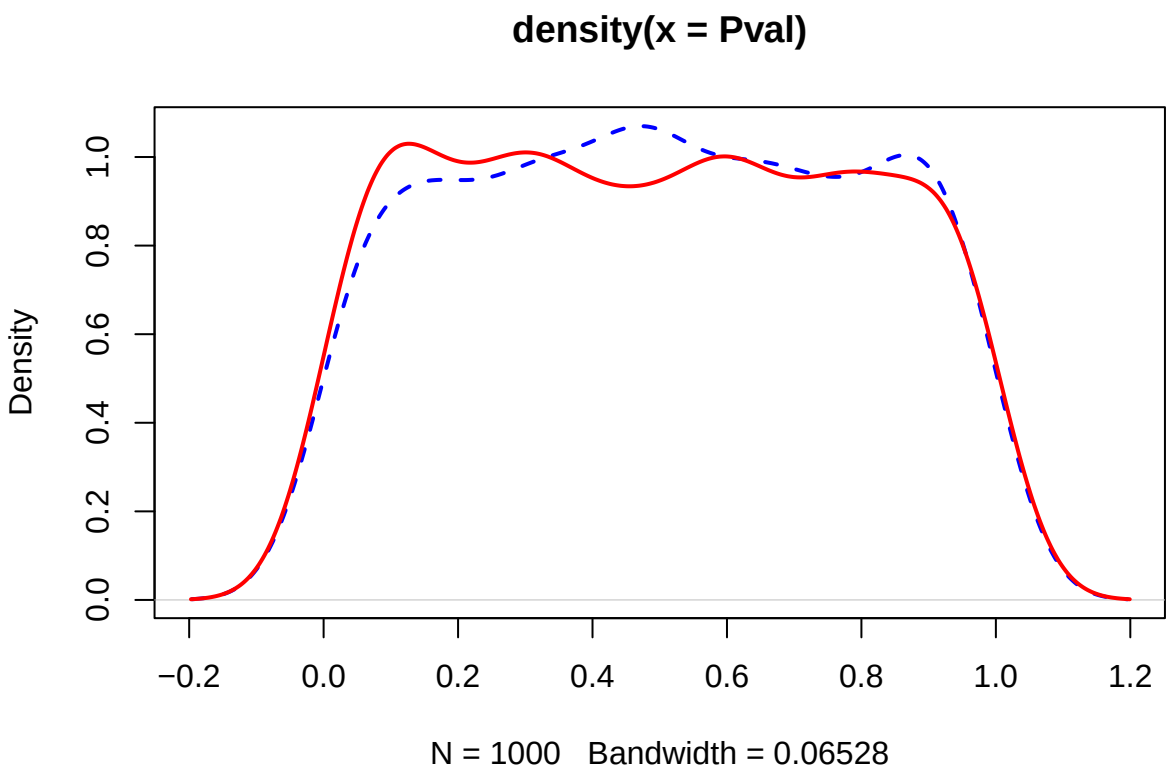
We want to confirm that $p(\mathbf{X}) \sim U(0,1)$. First we write the code to simulate one value for $p(\mathbf{X})$.

```
set.seed(Sys.time()) #remove seed as we will need several samples
pvalue<-function(mu0,sigma,n,N) {
count<-0
xbar<-mean(rnorm(n,mu0,sigma))
for(i in 1:N) {
Xbar<-mean(rnorm(n,mu0,sigma))
 if(Xbar>=xbar) count=count+1
}
pval=count/N
return(count/N)
}
```

Now we can use the above function (pvalue) to simulate as many values we want:

```r
simul<-1000 #number of simulations
Pval<-rep(NA,simul)
for(i in 1:simul) {
  Pval[i]=pvalue(mu0=0,sigma=2,n=20,N=100)
}

plot(density(Pval),col='blue',lty=2,lwd=2)
lines(density(runif(1000)),col='red',lwd=2)
```




We are now ready to perform the Goodness of fit test.

```r
n<-length(Pval) # which is equal to simul
breaks<-c(-Inf, seq(0.1,0.9,by=0.1),Inf)
Obs <- table(cut(Pval, breaks))
Obs
```

```
##
## (-Inf,0.1]  (0.1,0.2]  (0.2,0.3]  (0.3,0.4]  (0.4,0.5]  (0.5,0.6]  (0.6,0.7]
##        105         97         89        102        114         94        101
##  (0.7,0.8]  (0.8,0.9] (0.9, Inf]
##         95        105         98
```

```r
prop<-punif(breaks[-1])-punif(breaks[-length(breaks)])
Exp<-n*prop
round(cbind(Obs,Exp),1)
```

```
##            Obs Exp
## (-Inf,0.1] 105 100
## (0.1,0.2]   97 100
## (0.2,0.3]   89 100
## (0.3,0.4]  102 100
```

```
## (0.4,0.5]  114 100
## (0.5,0.6]   94 100
## (0.6,0.7]  101 100
## (0.7,0.8]   95 100
## (0.8,0.9]  105 100
## (0.9, Inf]  98 100
```

```
Test<-sum((Obs-Exp)^2/Exp)
test_pval<-1-pchisq(Test,df=length(breaks)-1-1) #d.f=m-1 (note that breaks has dimension m+1)
test_pval
```

```
## [1] 0.8786182
```

## 3. Exercise:

Confirm that for $X_i$ i.i.d Geometric$(p)$, the $\sum_{i=1}^{n} X_i \sim NegBin(n,p)$. You may want to use the function *pNegBin* from week's 8 case study.

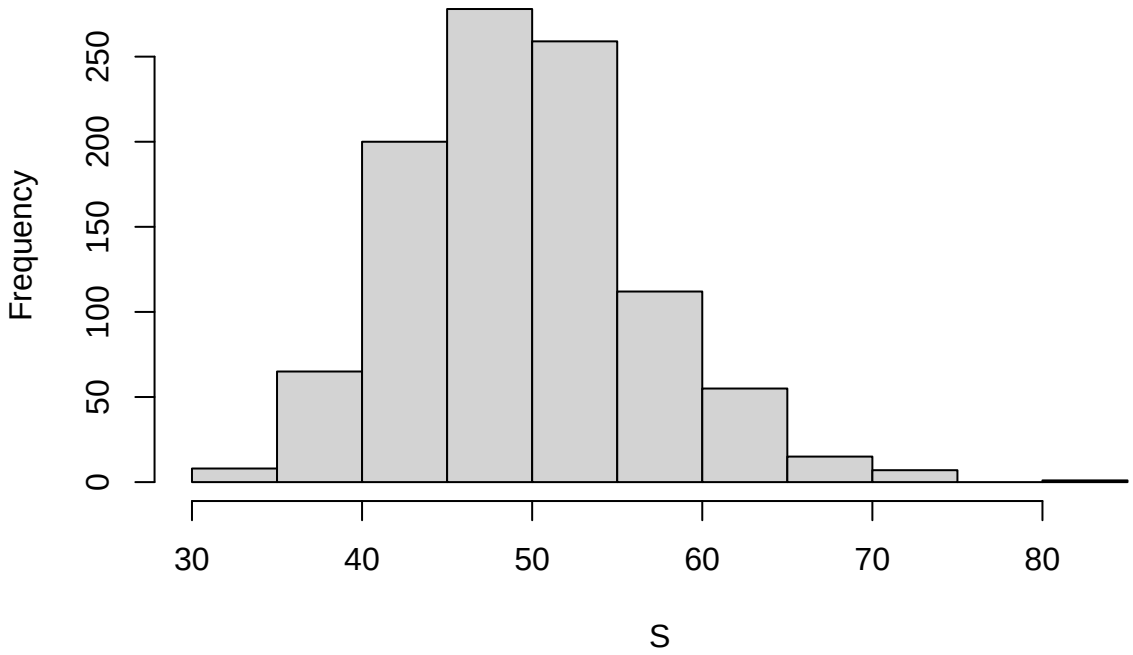
Solution: We will simulate samples from Geometric with probability $p$ and then apply the goodness-of-fit test for the sums of the samples to see if Negative Binomial is a good fit.

We first copy the code for Case Study of week 8 for the pmf and probability distribution function of Negative Binomial:

```
dNegBin<-function(y,n,p){
  stopifnot(all(p>=0 & p<=1 & y>=n & as.integer(n) & as.integer(y) & n>0))
  choose(y-1,n-1)*p^n*(1-p)^(y-n)
}

pNegBin<-function(y,n,p){
  prob<-0
  for(i in n:floor(y)){
    prob=prob+dNegBin(i,n,p)
  }
  return(prob)
}
```

```
n<-25 #sample size
N<-1000 #number of simulations
p<-0.5
S<-rep(NA,N)
for(i in 1:N){
S[i]<-sum(rgeom(n,p)+1) # we add 1 as the geometric in R counts the number of failures
                        #until the first success, instead of number of trials
}
hist(S)
```




```
breaks<-c(25,35,seq(40,60,by=5),100)
Obs <- table(cut(S, breaks))
Obs
```

```
##
##  (25,35]  (35,40]  (40,45]  (45,50]  (50,55]  (55,60] (60,100]
##        8       65      200      278      259      112       78
```

```
y<-c(1:(length(breaks)-1))
prop<-sapply(y,function(i) pNegBin(breaks[i+1],n,p)-pNegBin(breaks[i],n,p))
Exp<-N*prop
round(cbind(Obs,Exp),1)
```

```
##          Obs   Exp
## (25,35]    8   8.3
## (35,40]   65  68.6
## (40,45]  200 198.8
## (45,50]  278 280.4
## (50,55]  259 234.5
## (55,60]  112 131.9
## (60,100]  78  77.5
```

```
Test<-sum((Obs-Exp)^2/Exp)
test_pval<-1-pchisq(Test,df=length(breaks)-1-1)
test_pval
```

```
## [1] 0.4469786
```

# Case study 10: Bayesian Inference

Juliette Unwin

## The posterior for Normal likelihood and Normal prior

Suppose that we have a single observation $x$ from $N(\theta, \sigma^2)$, where $\sigma^2$ is known and we use a normal prior for $\theta$, i.e. $\theta \sim N(\mu_{pr}, \sigma^2_{pr})$, where $\mu_{pr}$ and $\sigma^2_{pr}$ are known. We will show that the prior distribution of $\theta$ is a conjugate, i.e. the posterior distribution is also Normal.

We know that

$$p(\theta|x) \propto L(\theta; x)p(\theta),$$

where,

- $p(\theta)$ is the prior of $\theta$
- $L(\theta; x)$ is the likelihood of $\theta$ having observed $x$ and
- $p(\theta|x)$ is the posterior of $\theta$ given the observed value $x$

Hence,

$$\begin{aligned}
p(\theta|x) &\propto \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\theta)^2\right\} \times \frac{1}{\sqrt{2\pi\sigma^2_{pr}}} \exp\left\{-\frac{1}{2\sigma^2_{pr}}(\theta-\mu_{pr})^2\right\} \\
&\propto \exp\left\{-\frac{1}{2\sigma^2}(x-\theta)^2 - \frac{1}{2\sigma^2_{pr}}(\theta-\mu_{pr})^2\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}(x^2-2\theta x+\theta^2) - \frac{1}{2\sigma^2_{pr}}(\theta^2-2\mu_{pr}\theta+\mu^2_{pr})\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left[\theta^2\left(\frac{1}{\sigma^2}+\frac{1}{\sigma^2_{pr}}\right) - 2\theta\left(\frac{x}{\sigma^2}+\frac{\mu_{pr}}{\sigma^2_{pr}}\right)\right]\right\} \\
&= \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{\sigma^2_{pr}}\right)\left[\theta^2 - 2\theta\underbrace{\left(\frac{x}{\sigma^2}+\frac{\mu_{pr}}{\sigma^2_{pr}}\right)\left(\frac{\sigma^2\sigma^2_{pr}}{\sigma^2+\sigma^2_{pr}}\right)}_{\mu_{post}}\right]\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{\sigma^2_{pr}}\right)(\theta-\mu_{post})^2\right\} \quad \text{(by completing the square)} \\
&= \exp\left\{-\frac{1}{2\sigma^2_{post}}(\theta-\mu_{post})^2\right\},
\end{aligned}$$

where,

$$\frac{1}{\sigma^2_{post}} = \frac{1}{\sigma^2} + \frac{1}{\sigma^2_{pr}} \tag{1}$$

and

$$\begin{aligned} \mu_{post} &= \left(\frac{x}{\sigma^2} + \frac{\mu_{pr}}{\sigma^2_{pr}}\right)\left(\frac{\sigma^2\sigma^2_{pr}}{\sigma^2+\sigma^2_{pr}}\right) \\ &= \sigma^2_{post}\left(\frac{x}{\sigma^2} + \frac{\mu_{pr}}{\sigma^2_{pr}}\right) \end{aligned} \tag{2}$$

If we set $a = \frac{1}{\sigma^2}$ and $b = \frac{1}{\sigma^2_{pr}}$ we can write the posterior parameters as follows:

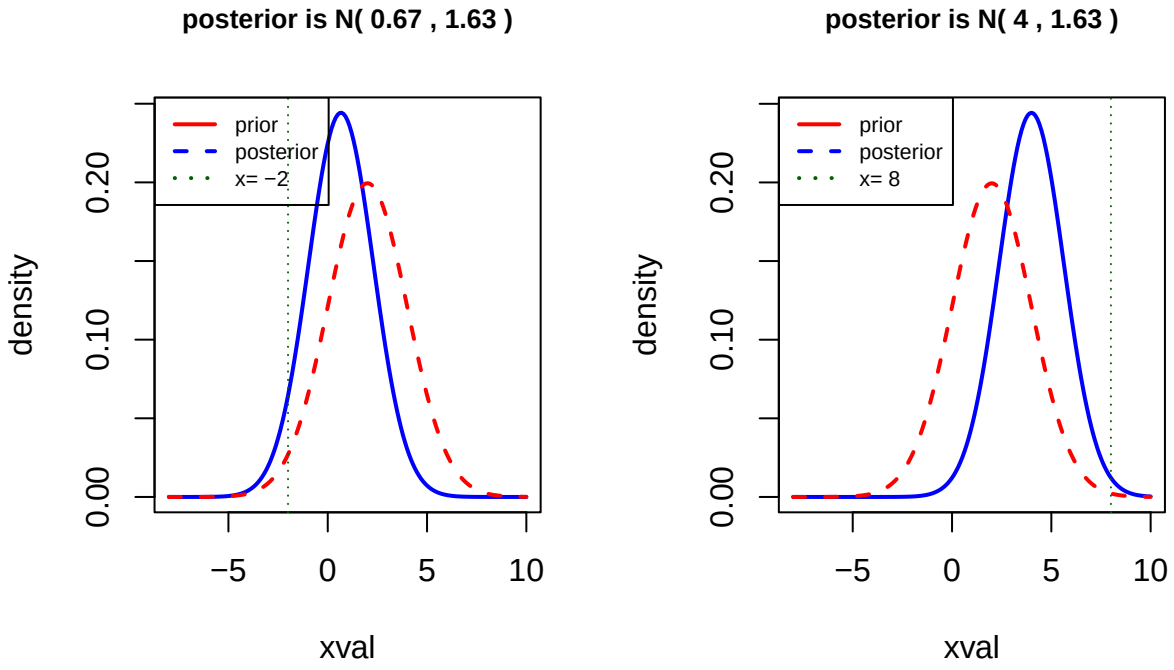
$$\sigma^2_{post} = \frac{1}{a+b} \quad \text{and} \quad \mu_{post} = \frac{a}{a+b}x + \frac{b}{a+b}\mu_{pr}.$$

Hence, we showed that $\theta|x \sim N(\mu_{post}, \sigma^2_{post})$.

From (1), it is easy to verify that $\sigma^2_{post} < \sigma^2_{pr}$, which means that the variance (uncertainty) of $\theta$ is reduced in the light of new information ($x$).

Lets see how the observed value, $x$, affects the distribution of the posterior. Assume that the prior parameters are $\mu_{pr} = 2, \sigma^2_{pr} = 4$ and let $\sigma^2 = 8$. We will derive and plot the posterior when $x = -2$ and $x = 8$.

```
# we define the known parameters
mu_pr<-2 #prior mean
sigma_pr<-2 #prior s.d.
sigma<-sqrt(8) #s.d. for likelihood
a<-1/sigma_pr^2
b<-1/sigma^2
# We will plot the prior and posterior for (1) x=-2 and (2) x=8
par(mfrow=c(1,2))
for(x in c(-2,8)) {
sigma_post<-1/sqrt(a+b)
mu_post<-a/(a+b) *mu_pr+b/(a+b)*x
xval<-seq(-8,10,0.1)
plot(xval,dnorm(xval,mu_post,sigma_post),ylab='density',type='l',col='blue',lwd=2,
      main=paste("posterior is N(", round(mu_post,2), ",", round(sigma_post,2), ")"),
     cex.main=0.8)
lines(xval,dnorm(xval,mu_pr,sigma_pr),type='l',col='red',lwd=2,lty=2)
abline(v=x,col='darkgreen',lty=3)
legend('topleft', legend=c('prior','posterior',paste("x=", x)),lwd=c(2,2),lty=c(1,2,3),
       col=c('red','blue','darkgreen'), cex=0.7)
}
```




## Updating the posterior when new data are observed

The posterior in the previous section was derived for a single observation. What happens if we have more observations available, say $x_1, x_2, .., x_n$ which are realizations of i.i.d $X_1, ..., X_n$ from $N(\theta, \sigma^2)$ ? One way is to compute the posterior in a similar way as as with one observation. Alternatively, we can avoid the complicated computations if we observe that for $n = 2$,

$$\begin{aligned} p(\theta|x_1, x_2) &\propto L(\theta; x_1, x_2) \times p(\theta) \\ &= L(\theta; x_1) \times L(\theta; x_2) \times p(\theta) \\ &\propto L(\theta; x_2) \times p(\theta|x_1). \end{aligned}$$

This means that we can think of $p(\theta|x_1)$ as the new prior for $\theta|(x_1, x_2)$ and hence, the new prior parameters will be the posterior parameters we computed for one observation. These are $\mu_{post}$ and $\sigma^2_{post}$ given by (2) and (1) respectively (simply replace $x$ with $x_1$).

By iterations, we can keep updating the posterior until all $x_1, x_2, ...x_n$ are used. In R it is easy to apply this method.

```
mu_pr<-2
sigma_pr<-2
sigma<-sqrt(8)
a<-1/sigma_pr^2
b<-1/sigma^2

n<-20 #sample size
set.seed(6734671)
x<-rnorm(n,5,sigma)  # assume sample is from N(5,4) (i.e. theta=5)
Mean<-rep(NA,n+1)
Var<-rep(NA,n+1)
Mean[1]<-mu_pr
Var[1]<-sigma_pr^2
for(i in 2:(n+1)) {
  a=1/ Var[i-1]
```

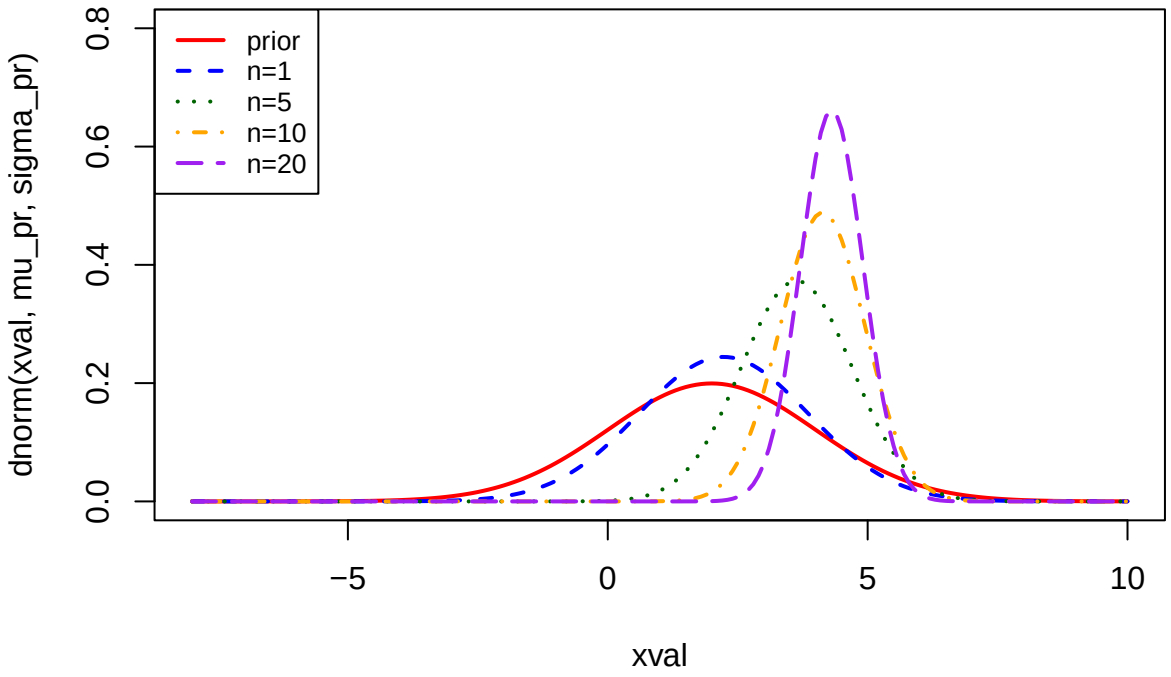
```
Var[i]=1/(a+b)
Mean[i]=a/(a+b) *Mean[i-1]+b/(a+b)*x[i-1]
}

parameters<-matrix(NA,ncol=3,nrow=n+1)
colnames(parameters)=c("iter","mean","variance")
parameters[,1]<-seq(0,n,1)
parameters[,2]<-Mean
parameters[,3]<-Var
parameters
```

```
##       iter     mean  variance
##  [1,]    0 2.000000 4.0000000
##  [2,]    1 2.227743 2.6666667
##  [3,]    2 2.939601 2.0000000
##  [4,]    3 2.616503 1.6000000
##  [5,]    4 3.693490 1.3333333
##  [6,]    5 3.634240 1.1428571
##  [7,]    6 4.098852 1.0000000
##  [8,]    7 4.095851 0.8888889
##  [9,]    8 3.916585 0.8000000
## [10,]    9 4.201533 0.7272727
## [11,]   10 4.134149 0.6666667
## [12,]   11 4.486747 0.6153846
## [13,]   12 4.212795 0.5714286
## [14,]   13 4.107826 0.5333333
## [15,]   14 3.860847 0.5000000
## [16,]   15 3.848807 0.4705882
## [17,]   16 3.924167 0.4444444
## [18,]   17 4.072405 0.4210526
## [19,]   18 4.240440 0.4000000
## [20,]   19 4.180994 0.3809524
## [21,]   20 4.307666 0.3636364
```

We can plot the densities of the posteriors after 1, 5, 10 and 20 iterations to see how the distribution is updated as the sample size increases:

```
plot(xval,dnorm(xval,mu_pr,sigma_pr),type='l',col='red',lwd=2,ylim=c(0,0.8))
  lines(xval,dnorm(xval,Mean[2],sqrt(Var[2])),lty=2,col='blue',lwd=2)
  lines(xval,dnorm(xval,Mean[6],sqrt(Var[6])),lty=3,col='darkgreen',lwd=2)
  lines(xval,dnorm(xval,Mean[11],sqrt(Var[11])),lty=4,col='orange',lwd=2)
  lines(xval,dnorm(xval,Mean[21],sqrt(Var[21])),lty=5, col='purple',lwd=2)
  legend("topleft",legend=c("prior","n=1", "n=5", "n=10", "n=20"),lwd=2,lty=1:5,
         col=c("red","blue","darkgreen","orange","purple"),cex=0.8)
```




Remark 1: The theoretical values of the (final) posterior parameters for a sample $x_1, ...., x_n$ are given by

$$\frac{1}{\sigma^2_{post}} = \frac{n}{\sigma^2} + \frac{1}{\sigma^2_{pr}} \quad \text{and} \quad \mu_{post} = \sigma^2_{post}\left(\frac{\sum_{i=1}^{n} x_i}{\sigma^2} + \frac{\mu_{pr}}{\sigma^2_{pr}}\right).$$

Remark 2: Since the posterior has a normal distribution, it is unimodal and symmetric around the posterior mean. Hence, we know that the Bayes estimate for any of the three loss functions covered in the notes (see p.118,119), it will be the posterior mean. It is also easy to compute credible intervals as it only requires to compute the quantiles of a normally distributed random variable.